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Embedded System Design

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AEZG512/AELZG512/DMZG512
Embedded System Design
Remote Lab Contact Session-03

Contents

- Expt.3. Implementation of Delay Generation Using Timer LPTMR with Interrupt on the S32K144 Microcontroller
- Expt.4. Generation of Pulse Width Modulation Signal Using FTM Module on the S32K144 Microcontroller
- Expt.5. Analog Signal Acquisition Using ADC Module on the S32K144 Microcontroller with PWM Output Connected to ADC Input.



Expt.3. Implementation of Delay Generation Using Timer LPTMR with Interrupt on the S32K144 Microcontroller

- **Objective**

- ✓ Develop an interrupt-based timer delay application using the LPTMR module of the S32K144 microcontroller with S32 SDK.

- **Problem Description**

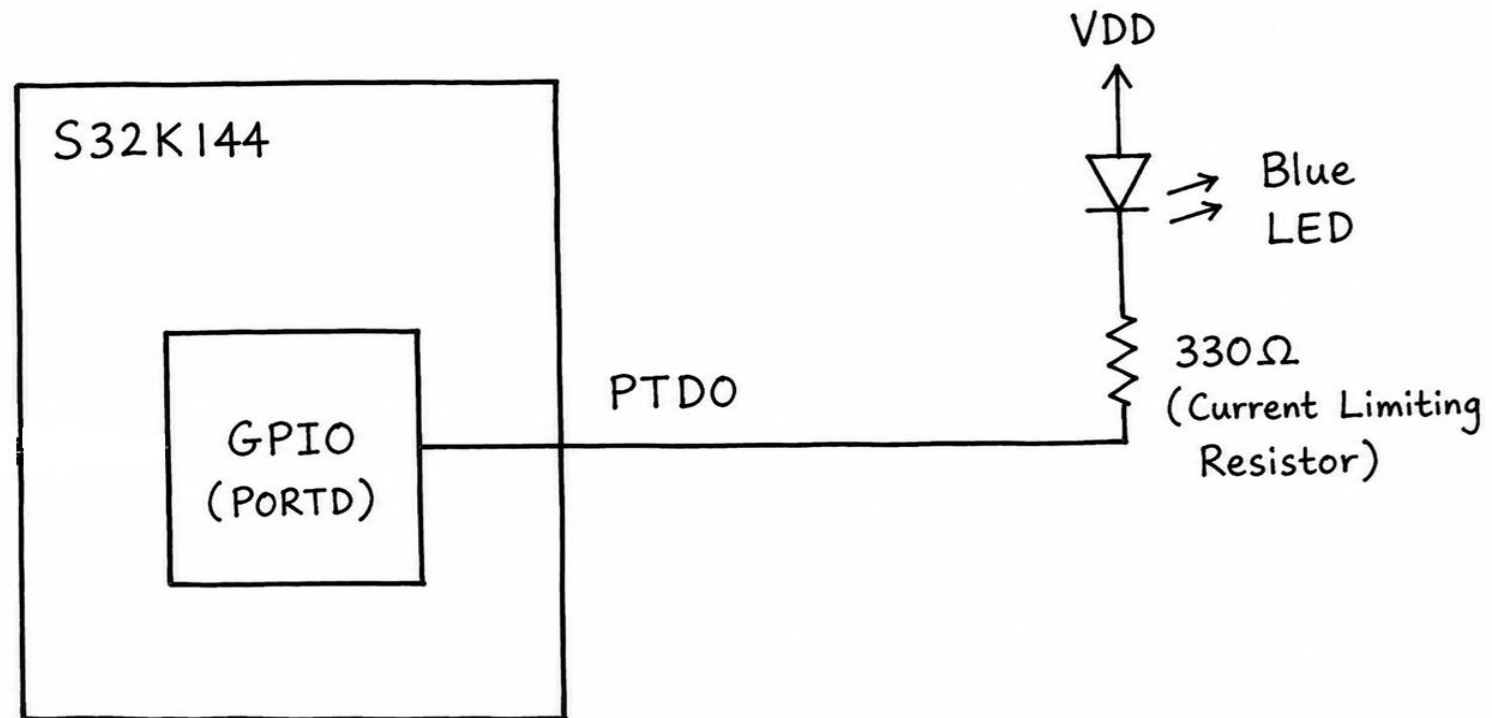
- ✓ The timer should generate a periodic interrupt after a fixed time interval. The CPU should not wait continuously in a polling loop. Instead, the timer interrupt service routine should toggle a GPIO output (PTD0) by the main program.

- **Functional Requirements**

- Configure system clock and timer peripheral clock.
- Configure LPTMR to generate a periodic interrupt.
- Enable the corresponding interrupt in the interrupt controller.
- Write or configure the timer callback / interrupt handler.
- Toggle an LED or GPIO output (PTD0) inside the interrupt callback or based on an interrupt flag.
- Ensure that the timer interrupt flag is cleared properly.

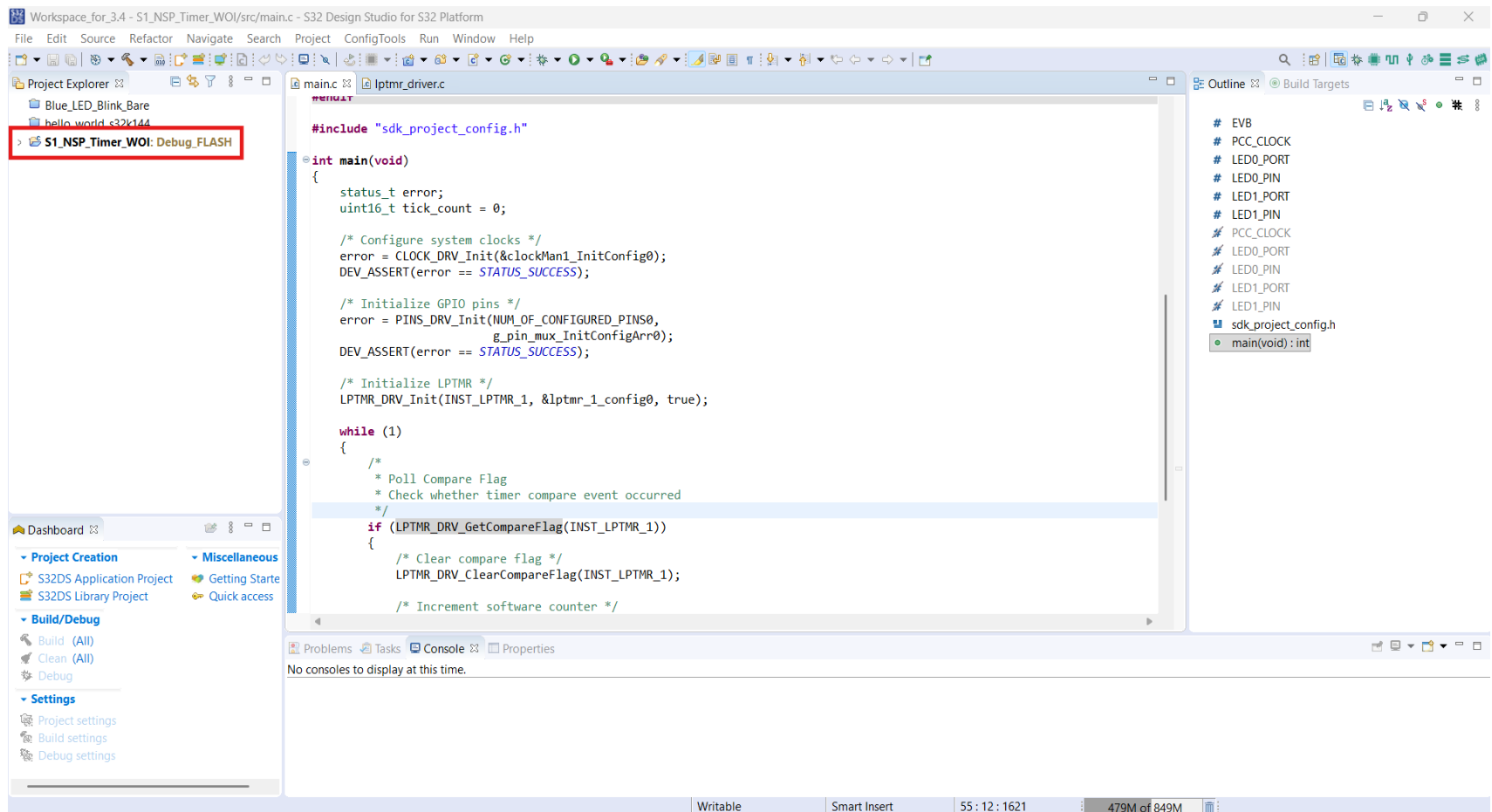
- **Expected Output**

- The LED or GPIO output should toggle periodically based on timer interrupts without using a blocking software delay.

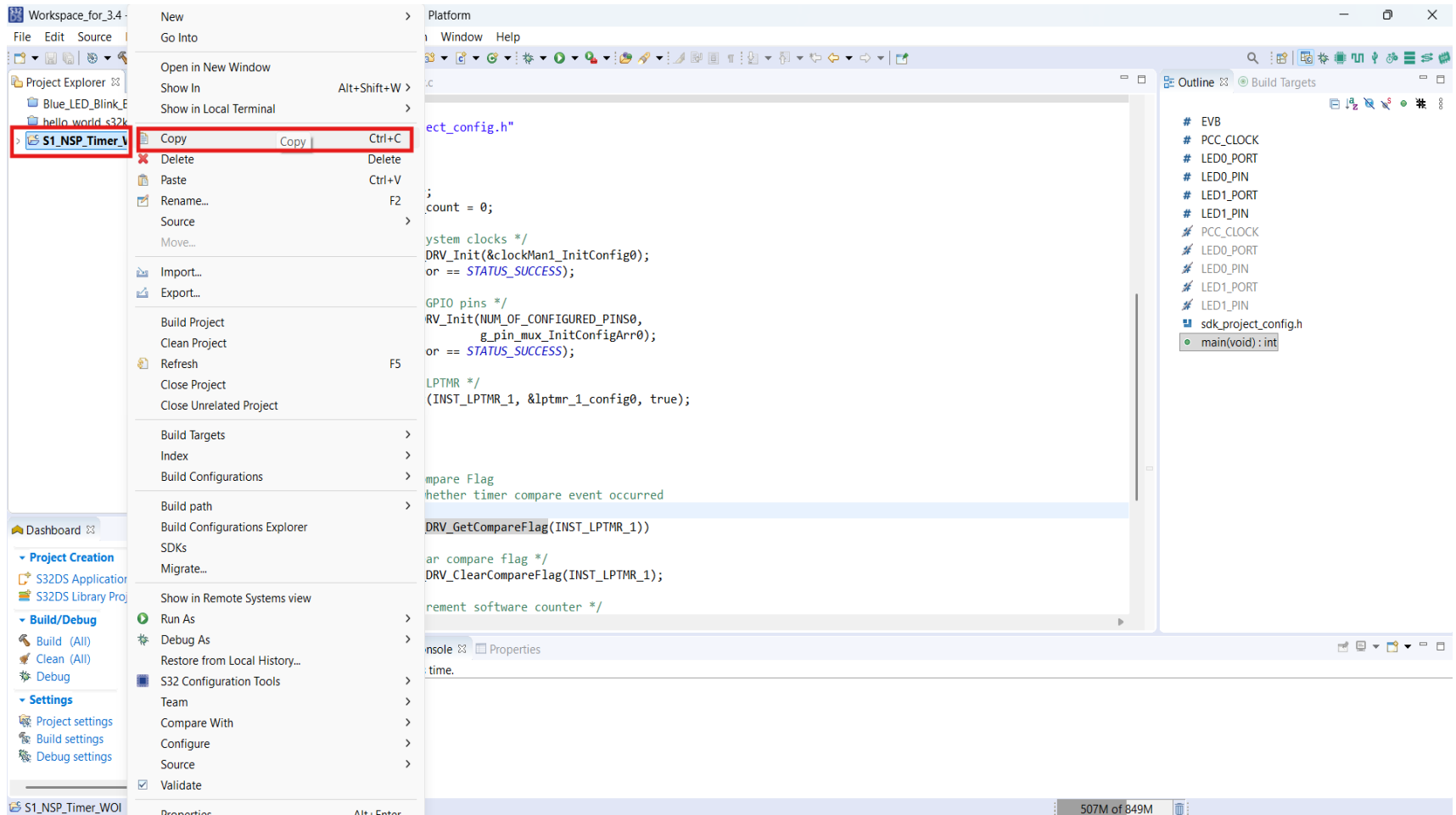


Refer Expt-2 project creation from S1_26_AE_AEL_DM_ZG512_Lab_CS_02.pdf
for Timer without interrupt.

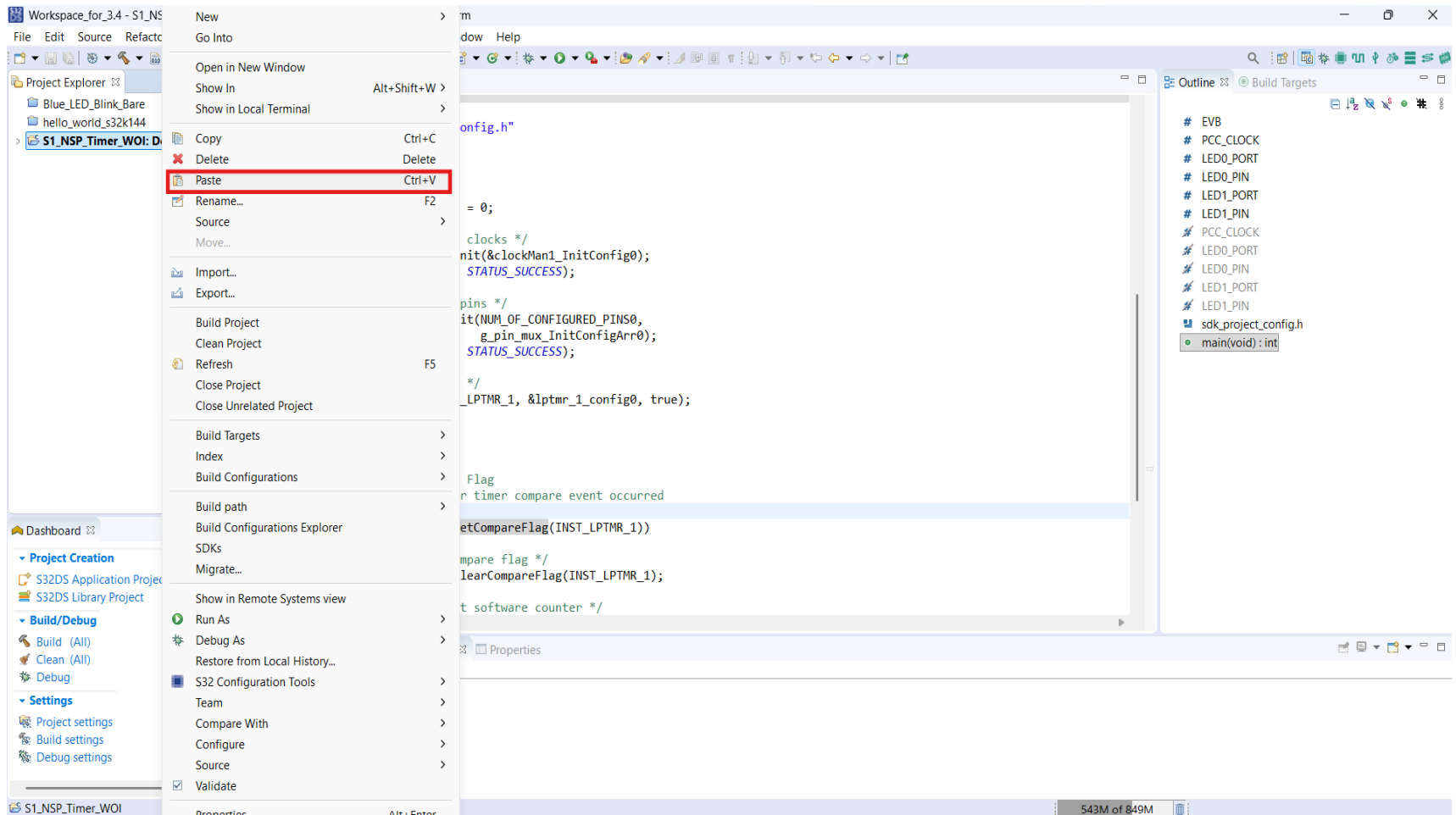
- S1_NSP_Timer_WOI Project is available in Project Explorer



- Select S1_NSP_Timer_WOI project and right click and select **Copy**



- Now click on **Paste**



The screenshot shows an IDE interface with a context menu open over a file named 'S1_NSP_Timer_WOI'. The 'Paste' option is highlighted with a red rectangle. The menu also shows 'Copy' (Ctrl+C), 'Delete' (Delete), 'Rename...' (F2), 'Source', 'Move...', 'Import...', 'Export...', 'Build Project', 'Clean Project', 'Refresh' (F5), 'Close Project', 'Close Unrelated Project', 'Build Targets', 'Index', 'Build Configurations', 'Build path', 'Build Configurations Explorer', 'SDKs', 'Migrate...', 'Show in Remote Systems view', 'Run As', 'Debug As', 'Restore from Local History...', 'S32 Configuration Tools', 'Team', 'Compare With', 'Configure', 'Source', and 'Validate' (Alt+Enter).

The background code in the editor is as follows:

```
onfig.h"

= 0;

clocks */
nit(&clockMan1_InitConfig0);
STATUS_SUCCESS);

pins */
it(NUM_OF_CONFIGURED_PINS0,
g_pin_mux_InitConfigArr0);
STATUS_SUCCESS);

*/
_LPTMR_1, &lptmr_1_config0, true);

Flag
r timer compare event occurred

etCompareFlag(INST_LPTMR_1))

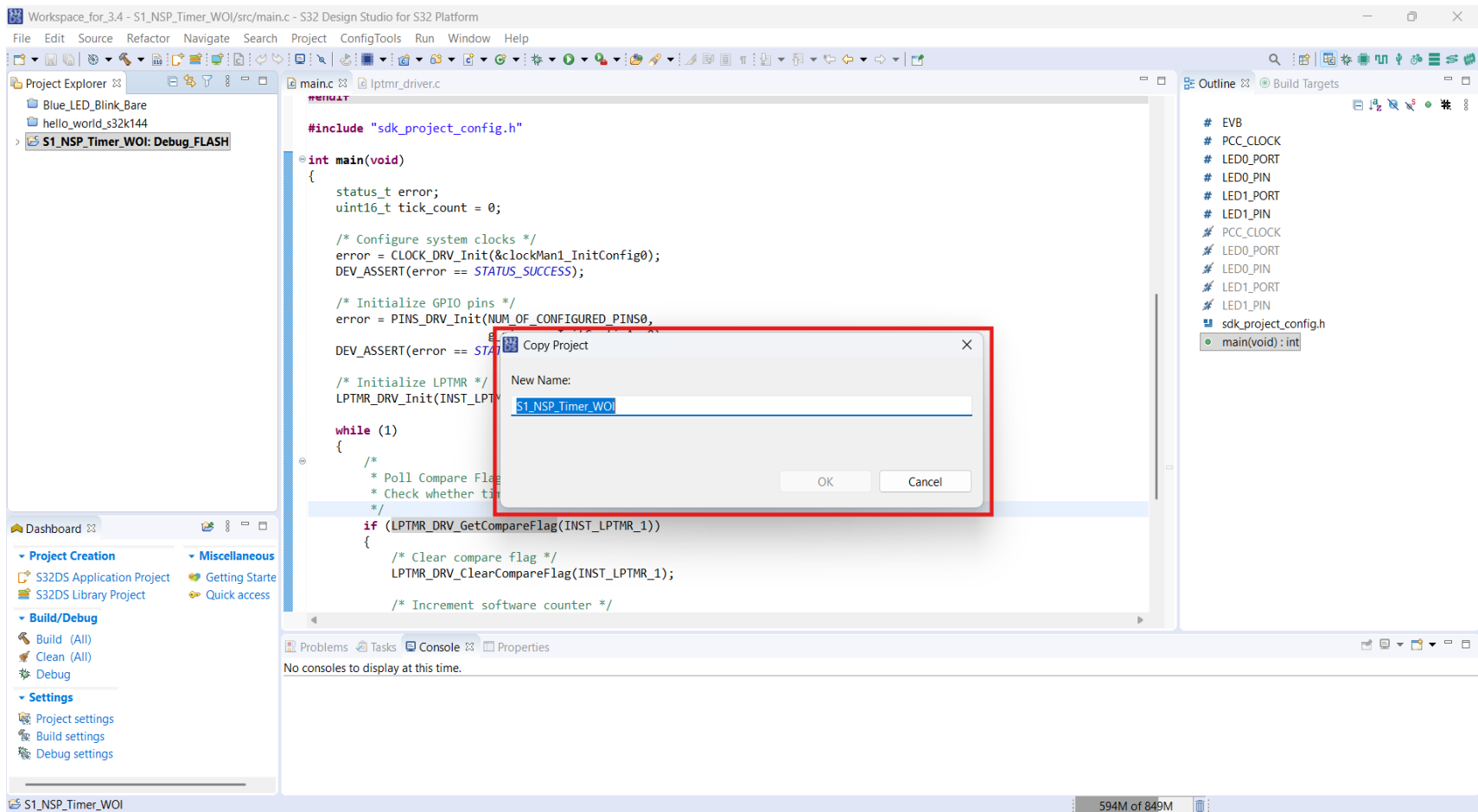
mpare flag */
learCompareFlag(INST_LPTMR_1);

t software counter */
```

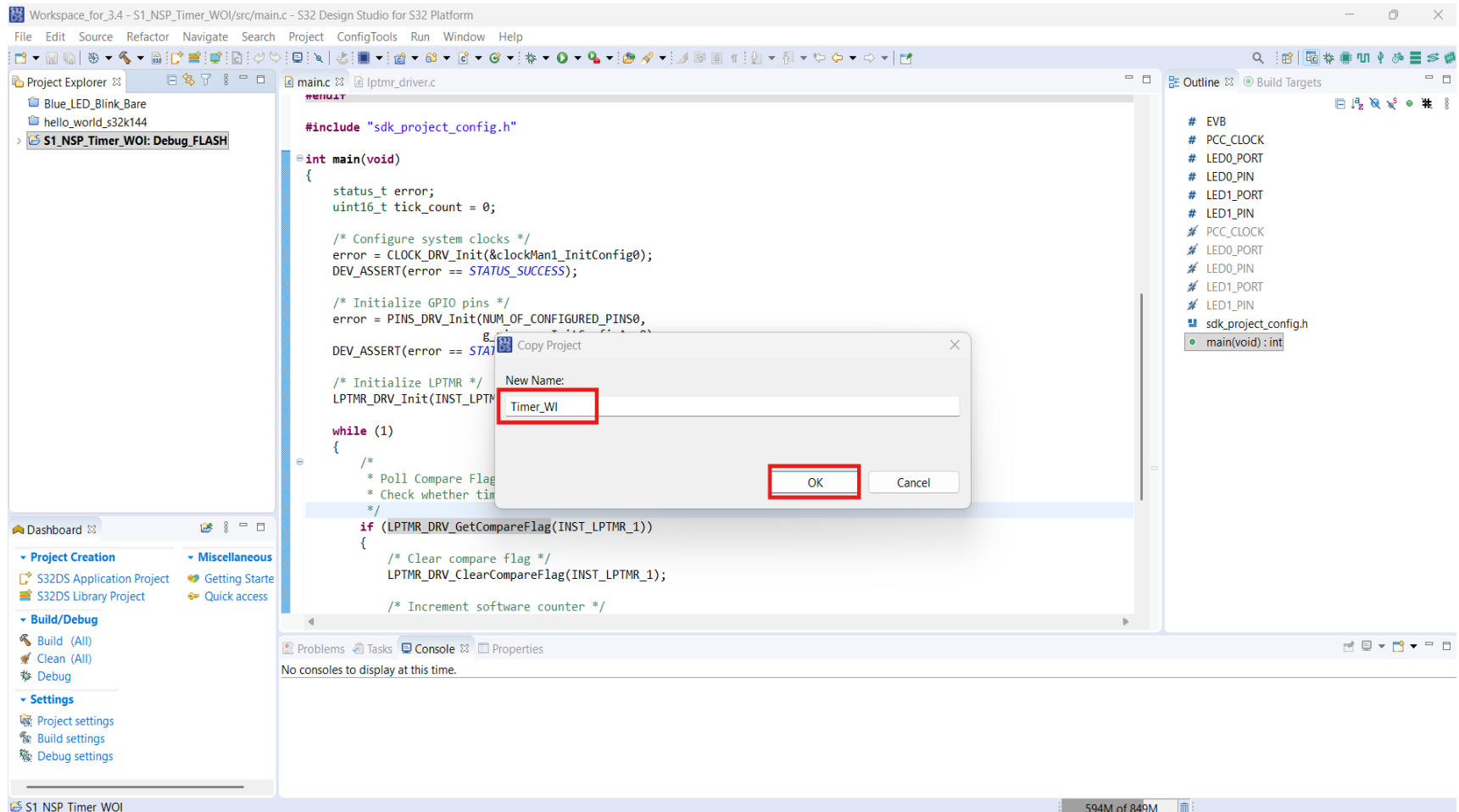
The right sidebar shows the 'Outline' view with the following contents:

```
# EVB
# PCC_CLOCK
# LED0_PORT
# LED0_PIN
# LED1_PORT
# LED1_PIN
# PCC_CLOCK
# LED0_PORT
# LED0_PIN
# LED1_PORT
# LED1_PIN
sdk_project_config.h
main(void): int
```

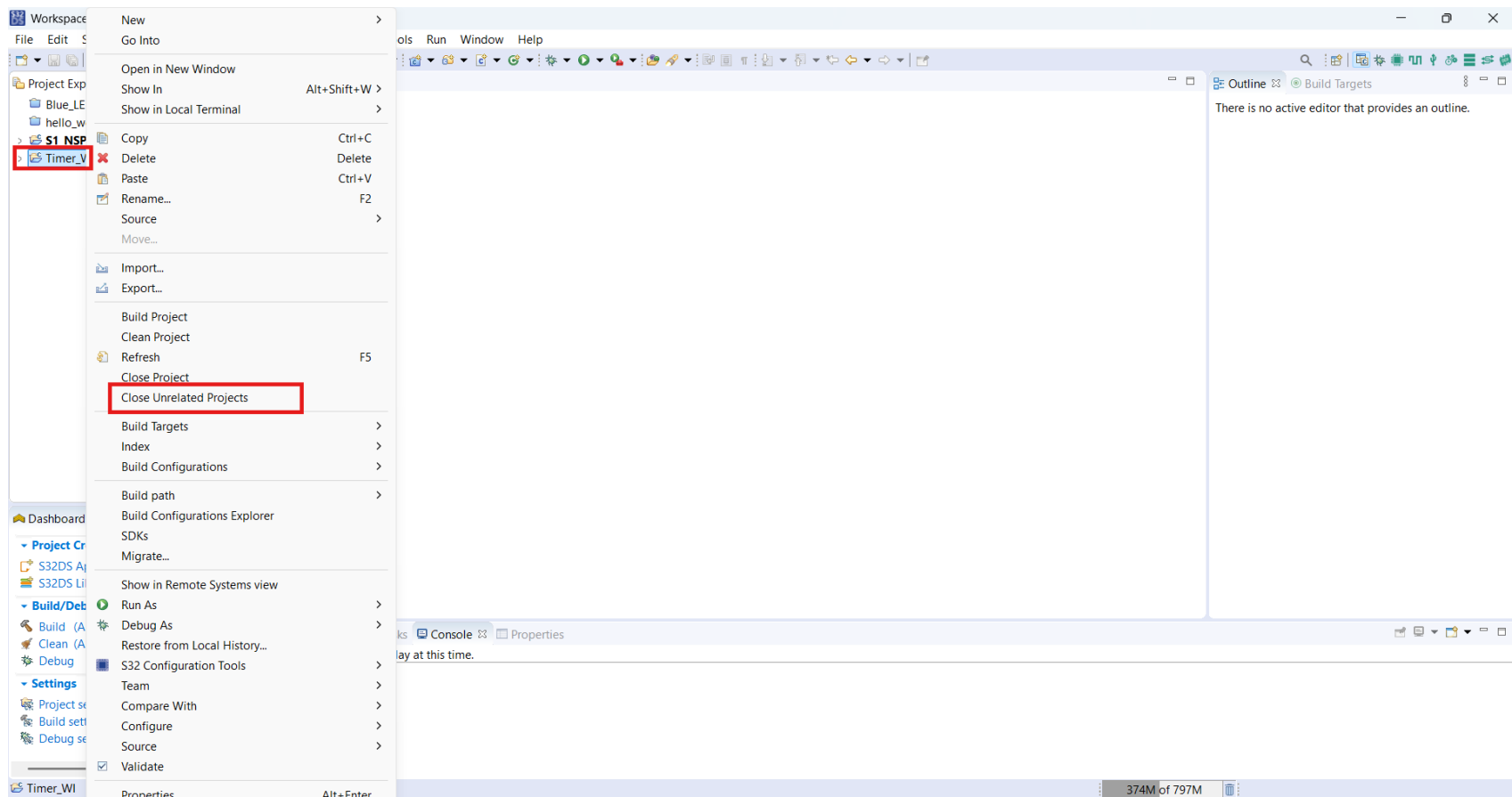
- Pop up window opens



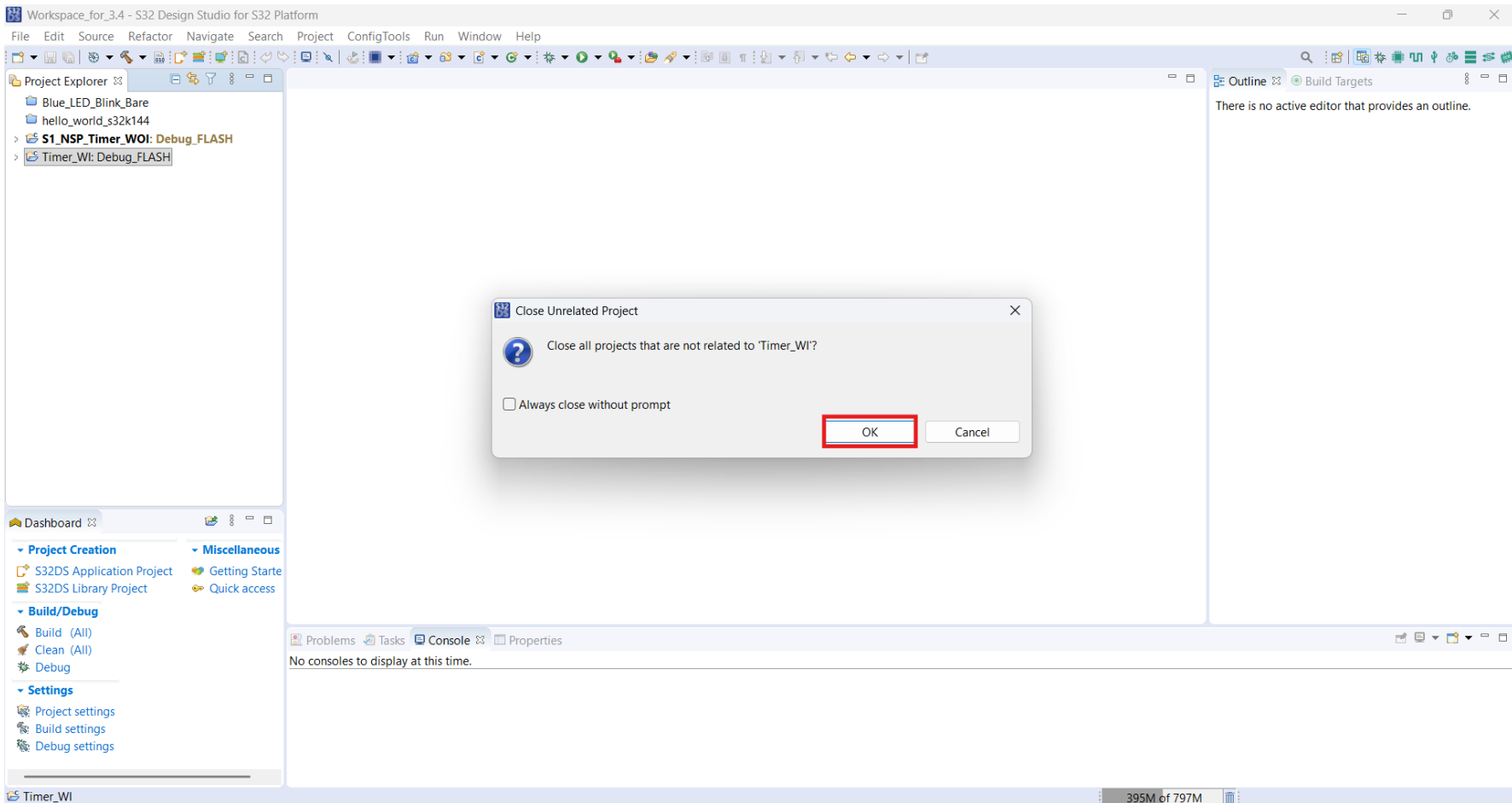
- Give name to project as Timer_WI and Click on OK

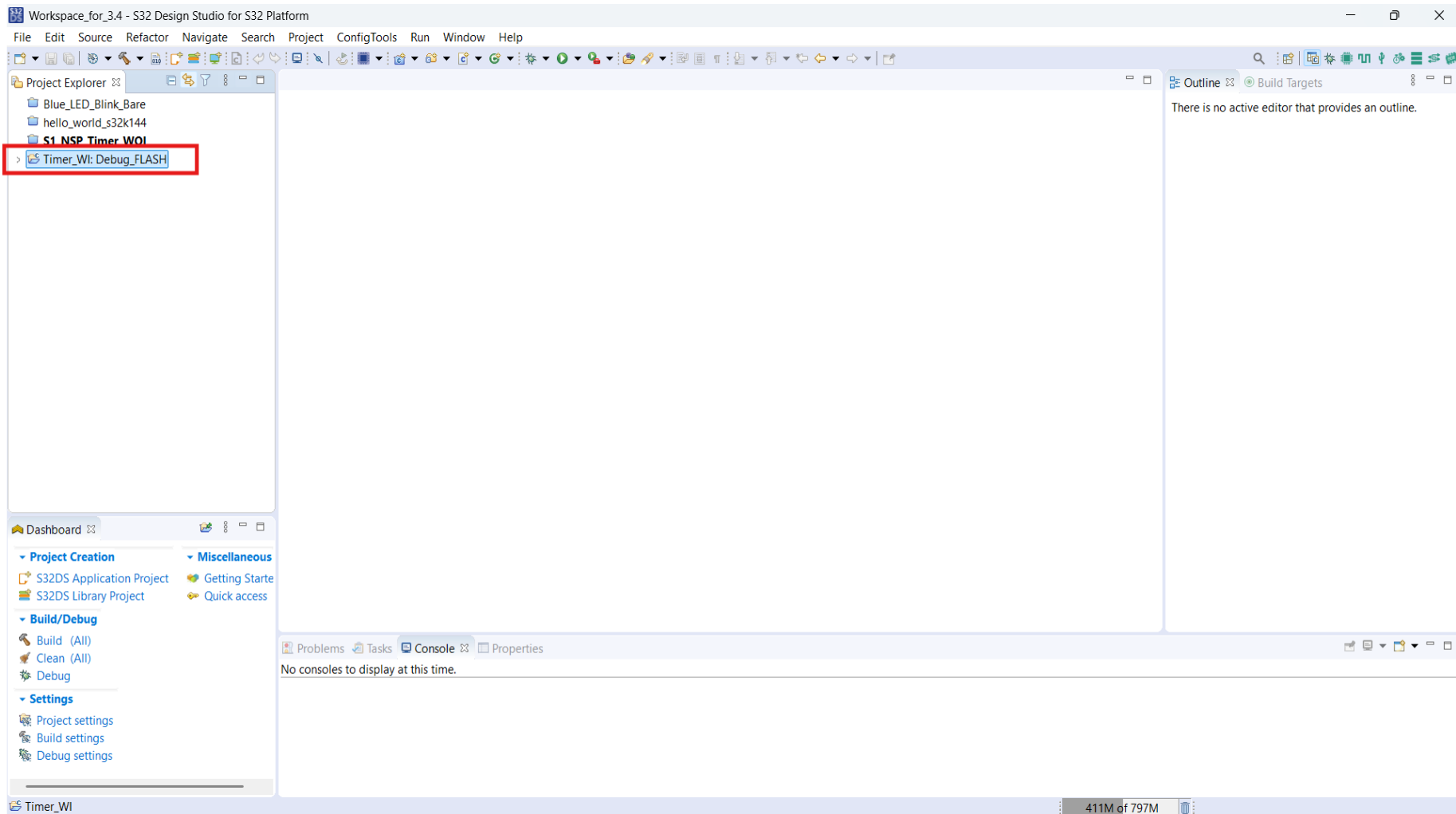


- Select Timer_WI project and right click and click on **Close Unrelated Projects**

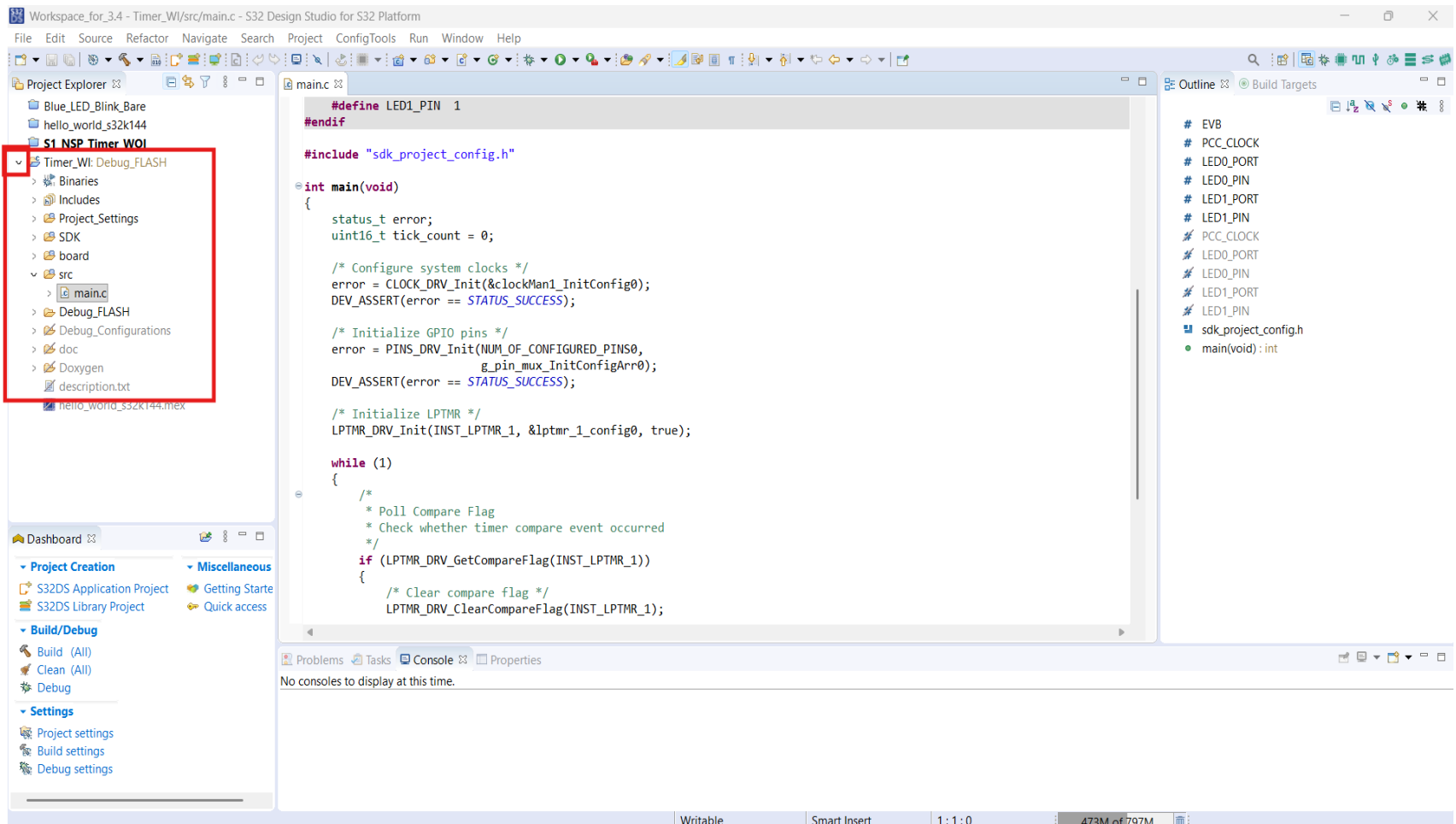


- Click on OK





- Expand the project and open src→main.c file



- Keep the Pin Routing as per previous Timer_WOI project.

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

Timer_WOI Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin name Label Identifier PORT FTM

1	PTE16		PTE16	FTM2_CH7
2	PTE15		PTE15	FTM2_CH6
3	PTD1		PTD1	FTM0_CH3[...]
4	PTD0	RGB_LED_BLUE	RGB_LED_BLUE	FTM0_CH2[...]
5	PTE11		PTE11	FTM2_CH5
6	PTE10		PTE10	FTM2_CH4
7	PTE13		PTE13	FTM2_FLT0
8	PTE5		PTE5	TCLK2[...]
9	PTE4		PTE4	FTM2_QD_P...
10	VDD_10			
11	VDDA			
12	VREFH			
13	VREFL			
14	VSS_14			
15	PTB7		PTB7	
16	PTB6		PTB6	
17	PTE14		PTE14	FTM0_FLT1[...]
18	PTE3		PTE3	FTM0_FLT0[...]
19	PTE12		PTE12	FTM0_FLT3
20	PTD17		PTD17	FTM0_FLT2

Routing Details

Routing Details for BOARD... 5

Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Select	Digital Filter	Drive Strength	Passive Filter	Initial Value
->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	n/a	n/a	Low
->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High
->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High
->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Down	Disabled	Low	n/a	High

S32K144_LQFP100 - LQFP 100 package

Configuration - C

Configuration - F

Processor: S32K1

Part number: S32K1

Core: Cortex

SDK Version: s32sdl

Project

Pins

Config electric and ru

Generated code

Update code ena

board\pin_mux.

board\pin_mux.

Functional group

BOARD_InitPins

Other tools

Problems

type filter text

Level Resource

- For Peripheral Configuration: Select ConfigTools→Peripherals

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project **ConfigTools** Pins Run Window Help

Timer_WOI BOARD_InitPins

Pins Peripheral... Power Gro... Package

ConfigTools menu:

- Pins
- Clocks
- Peripherals**
- IVT
- DCD
- QuadSPI
- DDR
- Config Tools Overview
- Update Code
- Configuration Preferences
- Data Manager

Pin list:

Pin	Pin name	Label	Identifier	PO
1	PTE16			PT16
2	PTE15			PT15
3	PTD1			PTD1
4	PTD0	RGB_LE...	RGB_LED_...	PTD0
5	PTE11			PT11
6	PTE10			PT10
7	PTE13			PT13
8	PTE5			PT5
9	PTE4			PT4
10	VDD_10			
11	VDDA			
12	VREFH			
13	VREFL			
14	VSS_14			
15	PTB7			PTB7
16	PTB6			PTB6
17	PTE14			PT14
18	PTE3			PT3
19	PTE12			PT12

Routing Details for BOARD...

Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do

Overview

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

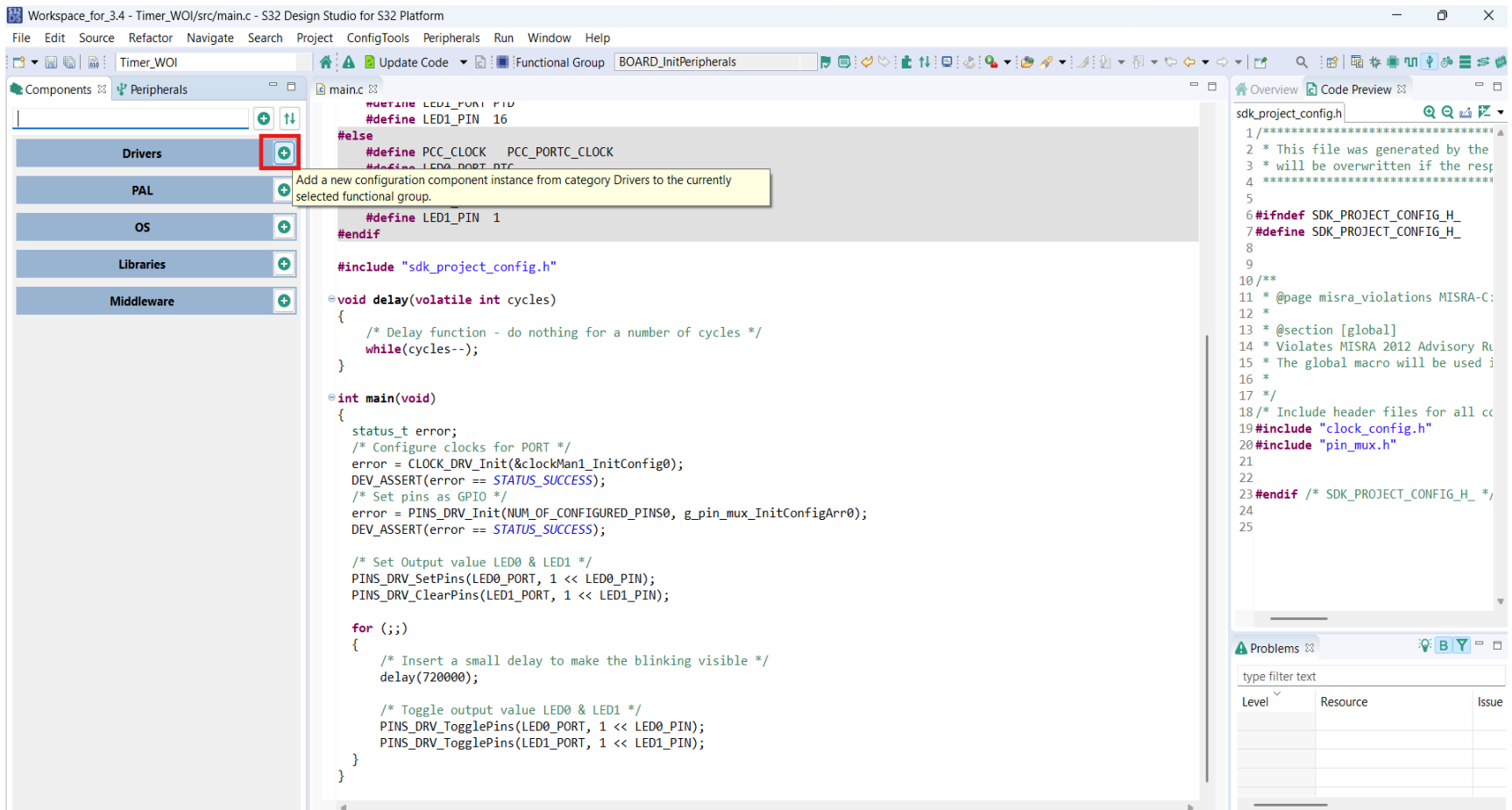
Other tools

Problems

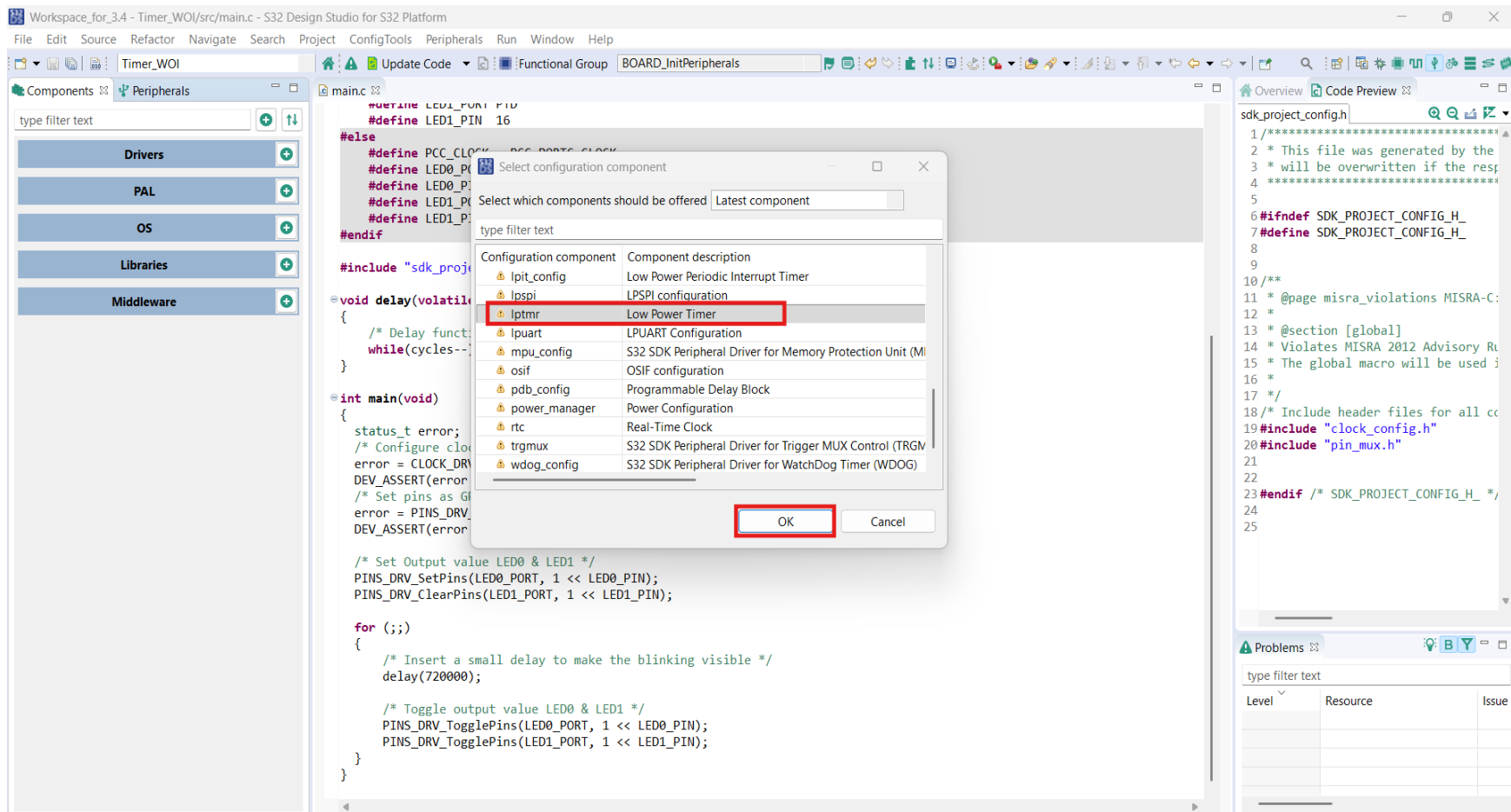
type filter text

Level	Resource	Issue	Origin

- Click on Add Drivers, new pop-up window opens



- Select **lptmr** and Click on OK



- Check settings as below for Low Power Timer

Workspace_for_3.4 - Timer_WI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Peripherals Run Window Help

Timer_WI

Components Peripherals

Drivers

lptmr_1

PAL

OS

Libraries

Middleware

main.c

```
status_t error;
uint16_t tick_count;

/* Configure system clock
error = CLOCK_DRV_INIT(CLOCK_DRV_INIT_FLAGS_NONE);
DEV_ASSERT(error == 0);

/* Initialize GPIO
error = PINS_DRV_INIT(PINS_DRV_INIT_FLAGS_NONE);
DEV_ASSERT(error == 0);

/* Initialize LPTMR
LPTMR_DRV_Init(1, &tick_count);

while (1)
{
    /* Poll Component
    /* Check whether timer has expired
    /* If timer has expired, increment tick count
    if (LPTMR_DRV_IsExpired(&tick_count))
    {
        /* Clear LPTMR interrupt
        LPTMR_DRV_ClearInterrupt(&tick_count);

        /* Increment tick count
        tick_count++;

        /* Toggle LED
        /* If tick count is odd, toggle LED
        if (tick_count % 2 == 1)
        {
            PINS_DRV_Write(PINS_DRV_PORT_LED, PINS_DRV_PIN_LED_TOGGLE);
        }
    }
}
```

Low Power Timer [Drivers]

Name lptmr_1

Mode General_Mode Peripheral LPTMR_0

General LPTMR Configuration

LPTMR Configurations

lptmr_1_config0	Name	lptmr_1_config0
	Read only	☒
	Work Mode	Timer
	DMA Request	☐
	Interrupt Enable	☒
	Free Run	☐
	Compare Value	1000
	Compare Value Units	Microseconds
	Input Clock	SIRC DIV2 clock
	Clock frequency	8 MHz
	Prescaler	Timer mode: prescaler 2, Glitch filter mode: invalid
	Bypass Prescaler	☐
	Pin Select (Pulse-Counter)	Count pulses from TRGMUX trigger
	Pin Polarity (Pulse-Counter)	Count pulse on rising edge

peripherals_lptmr_1.h

```
1 /* *****
2 * This file was generated by the S
3 * will be overwritten if the respo
4 *****
5
6 /* clang-format off */
7 /* TEXT BELOW IS USED AS SETTING FO
8 !!GlobalInfo
9 product: Peripherals v10.0
10 processor: S32K144
11 package_id: S32K144_LQFP100
12 mcu_data: s32sdk_s32k1xx_rtm_403
13 processor_version: 0.0.0
14 functionalGroups:
15 - name: BOARD_InitPeripherals
16   UUID: a6d43dcb-893a-49cd-9d9d-0bb
17   called_from_default_init: true
18   selectedCore: core0
19 * BE CAREFUL MODIFYING THIS COMMENT
20 /* clang-format on */
21
22 /* *****
23 * Included files
24 *****
25 #include "peripherals_lptmr_1.h"
26
27 /* *****
28 * lptmr_1 initialization code
29 *****
30 /* clang-format off */
```

Problems

Level	Resource	Issue
-------	----------	-------

- Click on **Update Code**

The screenshot shows the S32 Design Studio IDE interface. The top toolbar has the 'Update Code' button highlighted with a red box. The main workspace is divided into three panes:

- Left Pane (Components):** Shows a tree view with 'Drivers' selected, containing 'lptmr_1'.
- Center Pane (Low Power Timer [Drivers]):** Displays the configuration for 'lptmr_1'. The 'General LPTMR Configuration' section is expanded, showing 'LPTMR Configurations' with a table of settings for 'lptmr_1_config0':

Name	Value
Read only	☒
Work Mode	Timer
DMA Request	☐
Interrupt Enable	☒
Free Run	☐
Compare Value	1000
Compare Value Units	Microseconds
Input Clock	SIRC DIV2 clock
Clock frequency	8 MHz
Prescaler	Timer mode: prescaler 2, Glitch filter mode: invalid
Bypass Prescaler	☐
Pin Select (Pulse-Counter)	Count pulses from TRGMUX trigger
Pin Polarity (Pulse-Counter)	Count pulse on rising edge
- Right Pane (Code Preview):** Shows the source code for 'peripherals_lptmr_1.h', including comments and the '#include "peripherals_lptmr_1.h"' directive.

The bottom status bar shows 'Timer_WI' and '467M of 798M'.

- Click on OK

The screenshot displays the S32 Design Studio IDE interface. On the left, the 'Components' pane shows a tree view with 'Drivers', 'PAL', 'OS', 'Libraries', and 'Middleware'. The 'main.c' file is open in the editor, showing C code for a timer driver. In the center, the 'Low Power Timer [Drivers]' configuration window is open, displaying a table of generated files and their status. The 'OK' button is highlighted with a red rectangle. On the right, the 'Code Preview' pane shows the content of the 'peripherals_lptmr_1.h' header file. The 'Problems' pane at the bottom right is empty.

Generated file	Status
☒ Pins	
☒ board\pin_mux.c	no change
☒ pin_mux.h	no change
☒ Clocks	
☒ board\clock_config.c	no change
☒ clock_config.h	no change
☒ Peripherals	
☒ board\peripherals_lptmr_1.c	change
☒ peripherals_lptmr_1.h	no change
☒ sdk_project_config.h	no change
☐ DCD	warnings
☐ IVT	warnings
☐ QuadSPI	warnings
☐ DDR	warnings
☒ Configuration	
hello_world_s32k144.mex	change

Options
☒ Always show details before Update Code

OK Cancel

- After update click on C/C++ to change perspective

The screenshot displays the S32 Design Studio IDE interface. On the left, the 'Components' pane shows the 'Drivers' section with 'Lptmr_1' selected. The main editor shows the C source file 'main.c' with a timer initialization loop. The 'Low Power Timer [Drivers]' configuration window is open, showing the 'General LPTMR Configuration' for 'Lptmr_1'. A red arrow points to the 'C/C++' icon in the top toolbar, indicating the action to switch the perspective to C/C++.

Workspace: Workspace_for_3.4 - Timer_WI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Peripherals Run Window Help

Timer_WI

Components Peripherals

type filter text

Drivers

Lptmr_1

PAL

OS

Libraries

Middleware

main.c

```

status_t error;
uint16_t tick_count;

/* Configure system clock
error = CLOCK_DRV_INIT(CLOCK_DRV_INIT_FLAGS_NONE);
DEV_ASSERT(error == 0);

/* Initialize GPIO
error = PINS_DRV_INIT(PINS_DRV_INIT_FLAGS_NONE);
DEV_ASSERT(error == 0);

/* Initialize LPTMR
LPTMR_DRV_Init(LPTMR_DRV_INIT_FLAGS_NONE);

while (1)
{
    /* Poll Component
    /* Check whether timer is ready
    /*
    if (LPTMR_DRV_IS_READY())
    {
        /* Clear LPTMR
        LPTMR_DRV_CLEAR();

        /* Increment tick count
        tick_count++;

        /* Toggle LED
        /*
        if (tick_count % 1000 == 0)
        {
            PINS_DRV_WRITE(PINS_DRV_READ() ^ 0x01);
        }
    }
}

```

Lptmr_1

Name: Lptmr_1

Mode: General_Mode

Peripheral: LPTMR_0

General LPTMR Configuration

LPTMR Configurations

Lptmr_1_config0	Name	Value
	Read only	☒
	Work Mode	Timer
	DMA Request	☐
	Interrupt Enable	☒
	Free Run	☐
	Compare Value	1000
	Compare Value Units	Microseconds
	Input Clock	SIRC DIV2 clock
	Clock frequency	8 MHz
	Prescaler	Timer mode: prescaler 2, Glitch filter mode: invalid
	Bypass Prescaler	☐
	Pin Select (Pulse-Counter)	Count pulses from TRGMUX trigger
	Pin Polarity (Pulse-Counter)	Count pulse on rising edge

peripherals_lptmr_1.h

```

1 /* *****
2 * This file was generated by the S
3 * will be overwritten if the respe
4 *****
5
6 /* clang-format off */
7 /* TEXT BELOW IS USED AS SETTING FO
8 !!GlobalInfo
9 product: Peripherals v10.0
10 processor: S32K144
11 package_id: S32K144_LQFP100
12 mcu_data: s32sdk_s32k1xx_rtmm_403
13 processor_version: 0.0.0
14 functionalGroups:
15 - name: BOARD_InitPeripherals
16   UUID: a6d43dcb-893a-49cd-9d9d-0bb
17   called_from_default_init: true
18   selectedCore: core0
19 * BE CAREFUL MODIFYING THIS COMMENT
20 /* clang-format on */
21
22 /* *****
23 * Included files
24 *****
25 #include "peripherals_lptmr_1.h"
26
27 /* *****
28 * lptmr_1 initialization code
29 *****
30 /* clang-format off */

```

Problems

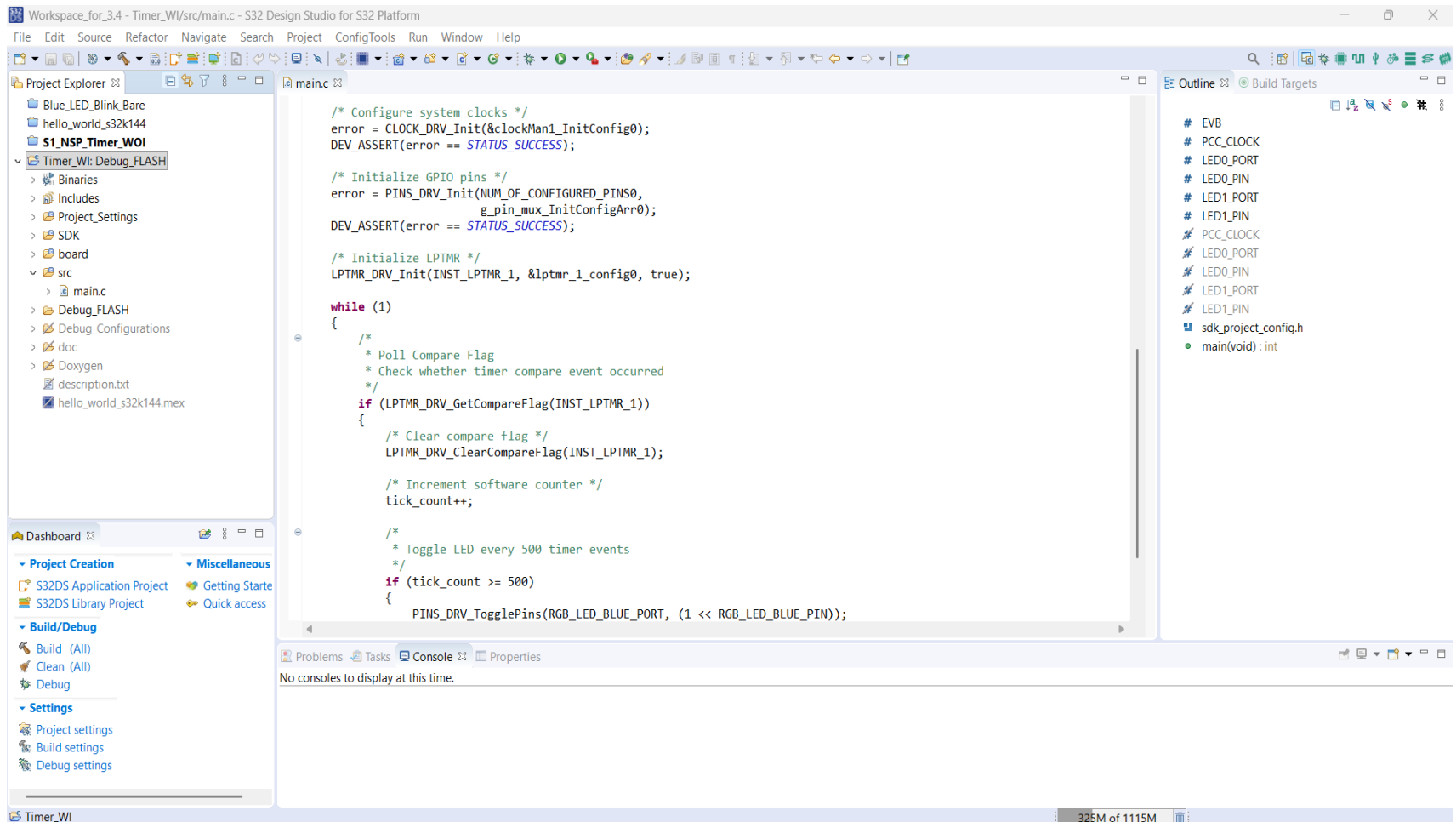
type filter text

Level	Resource	Issue

Timer_WI

308M of 1115M

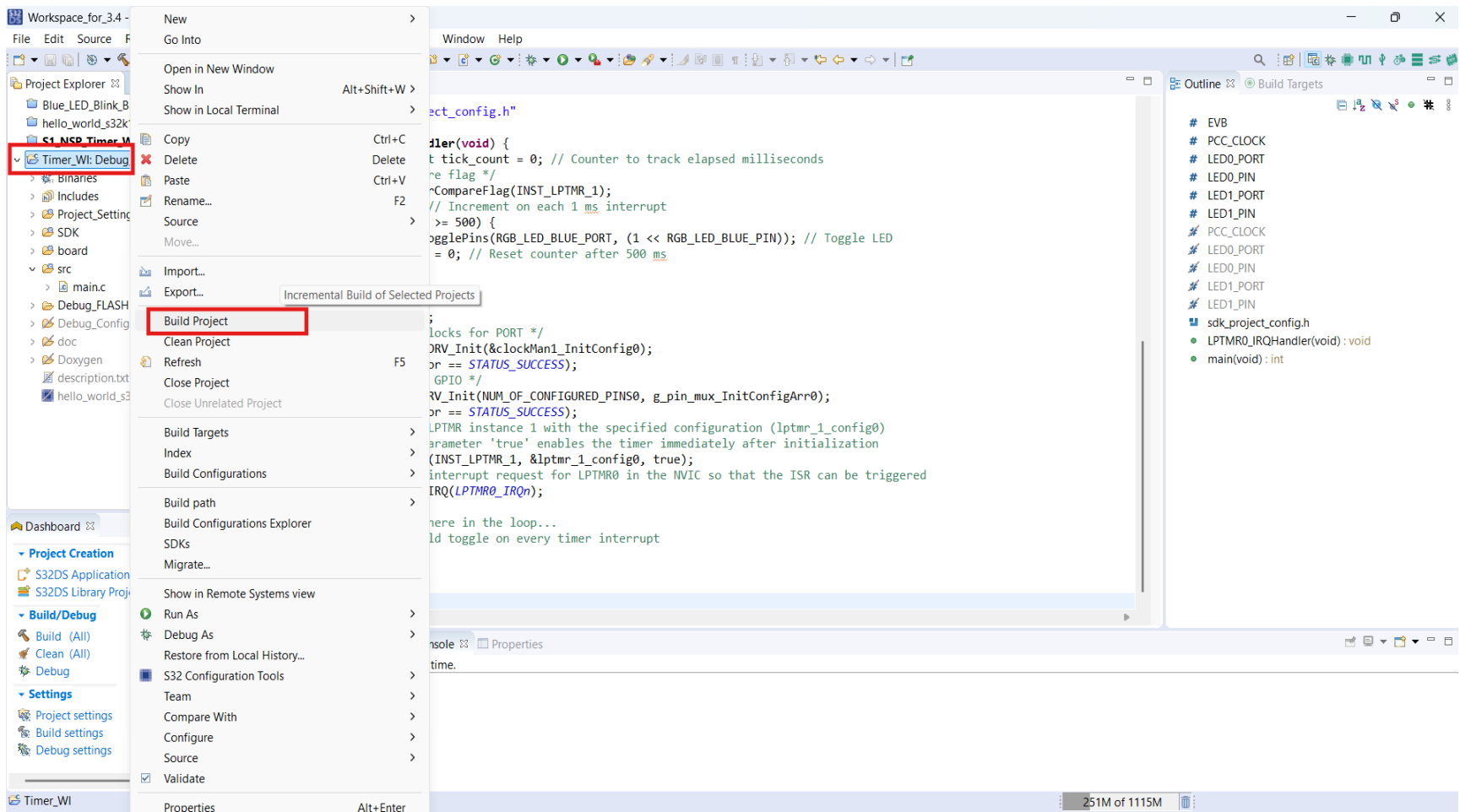
- C/C++ perspective



- Write C program as below and save it.

```
main.c ✕  
  
#include "sdk_project_config.h"  
  
void LPTMR0_IRQHandler(void) {  
    static uint16_t tick_count = 0; // Counter to track elapsed milliseconds  
    /* Clear compare flag */  
    LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);  
    tick_count++; // Increment on each 1 ms interrupt  
    if (tick_count >= 500) {  
        PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN)); // Toggle LED  
        tick_count = 0; // Reset counter after 500 ms  
    }  
}  
  
int main(void) {  
    status_t error;  
    /* Configure clocks for PORT */  
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);  
    DEV_ASSERT(error == STATUS_SUCCESS);  
    /* Set pins as GPIO */  
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);  
    DEV_ASSERT(error == STATUS_SUCCESS);  
    // Initialize LPTMR instance 1 with the specified configuration (lptmr_1_config0)  
    // The third parameter 'true' enables the timer immediately after initialization  
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);  
    // Enable the interrupt request for LPTMR0 in the NVIC so that the ISR can be triggered  
    INT_SYS_EnableIRQ(LPTMR0_IRQn);  
    for (;;) {  
        //Nothing here in the loop...  
        //LED should toggle on every timer interrupt  
    }  
}
```

- Right click on project and click on Build Project



- Check for any errors

The screenshot displays the S32 Design Studio IDE interface. The main editor shows the `main.c` file with the following code:

```
#include "sdk_project_config.h"

void LPTMR0_IRQHandler(void) {
    static uint16_t tick_count = 0; // Counter to track elapsed milliseconds
    /* Clear compare flag */
    LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
    tick_count++; // Increment on each 1 ms interrupt
    if (tick_count >= 500) {
        PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN)); // Toggle LED
        tick_count = 0; // Reset counter after 500 ms
    }
}

int main(void) {
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Initialize LPTMR instance 1 with the specified configuration (lptmr_1_config0)
    // The third parameter 'true' enables the timer immediately after initialization
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);
    // Enable the interrupt request for LPTMR0 in the NVIC so that the ISR can be triggered
    INT_SYS_EnableIRQ(LPTMR0_IRQn);
    for (;;) {
        //Nothing here in the loop...
        //LED should toggle on every timer interrupt
    }
}
```

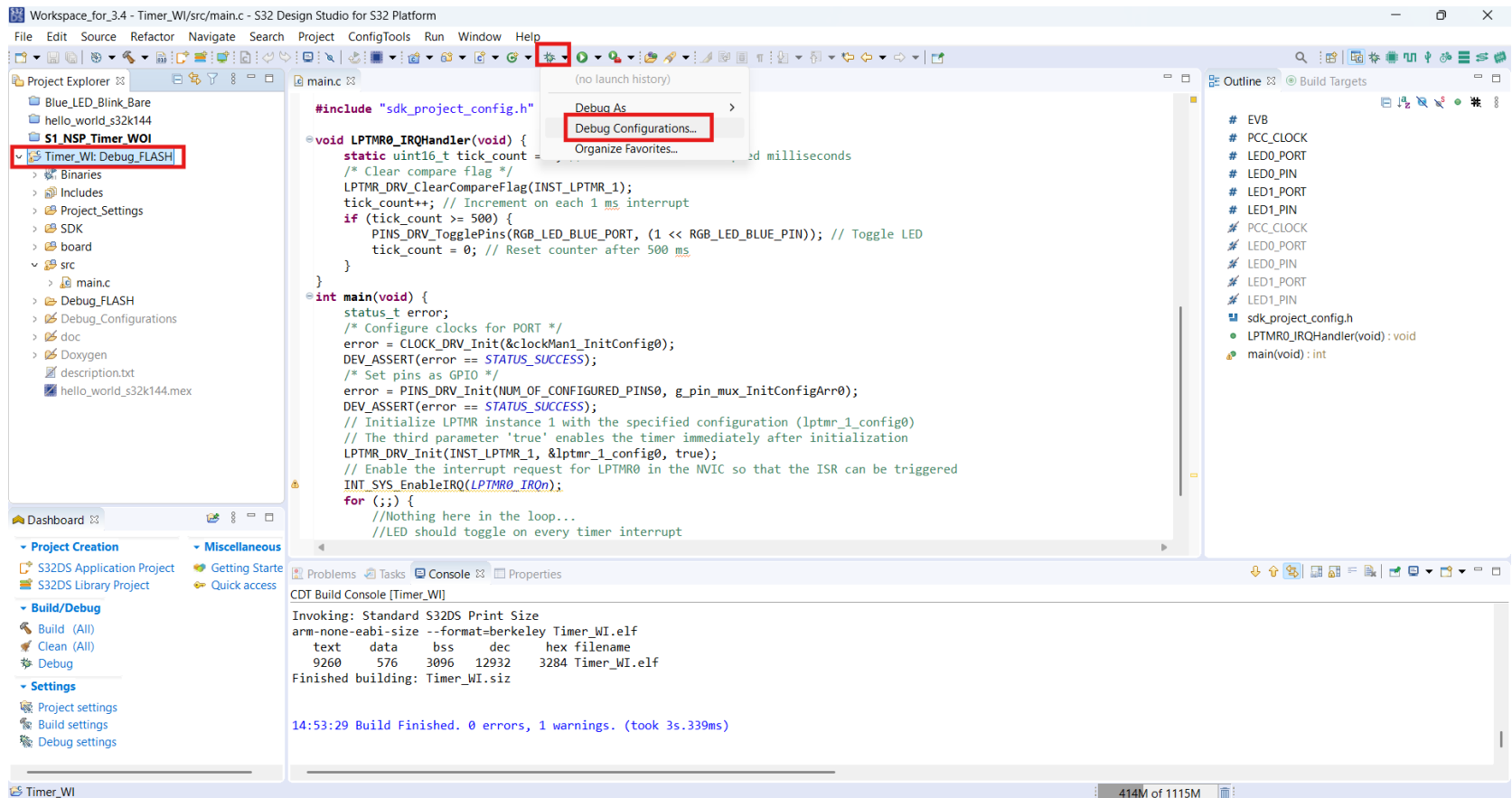
The left sidebar shows the Project Explorer with the project structure, including `Timer_WI: Debug_FLASH`. The bottom status bar indicates the build is finished with 0 errors and 1 warning.

The CDT Build Console [Timer_WI] shows the following output:

```
Invoking: Standard S32DS Print Size
arm-none-eabi-size --format=berkeley Timer_WI.elf
text  data  bss  dec  hex filename
9260   576   3096  12932  3284 Timer_WI.elf
Finished building: Timer_WI.siz

14:53:29 Build Finished. 0 errors, 1 warnings. (took 3s.339ms)
```

- Connect S32K144 development board to PC/Laptop
- Click on Debug → Debug Configurations



- Do configurations as below and Click on **Debug**

Workspace_for_3.4 - Timer_WI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help

Project Explorer

- Blue_LED_Blink_Bare
- hello_world_s32k144
- S1_NSP_Timer_WOI
- Timer_WI: Debug_FLASH
 - Binaries
 - Includes
 - Project_Settings
 - SDK
 - board
 - src
 - main.c
 - Debug_FLASH
 - Debug_Configurations
 - doc
 - Doxygen
 - description.txt
 - hello_world_s32k144.mex

Dashboard

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settings
 - Build settings
 - Debug settings

Debug Configurations

Create, manage, and run configurations

Name: Timer_WI_debug_flash_jlink

Debugger

J-Link GDB Server Setup

☒ Start the J-Link GDB server locally ☐ Connect to running target

Executable: \$(jlink_path)/\$(jlink_gdbserver) Browse... Variables...

Device name: S32K144 Supported device names

Endianness: ☒ Little ☐ Big

Connection: ☒ USB ☐ IP (USB serial or IP name/address)

Interface: ☐ SWD ☒ JTAG

Initial speed: ☐ Auto ☐ Adaptiv ☒ Fixed 1000 kHz

GDB port: 2331

SWO port: 2332 ☒ Verify downloads ☒ Initialize registers on start

Telnet port: 2333 ☒ Local host only ☐ Silent

Log file: Browse...

Other options: -singlerun -strict -timeout 0 -nogui

☒ Allocate console for the GDB server ☒ Allocate console for semihosting and SWO

GDB Client Setup

Executable: \$(S32DS_ARM32_TOOLCHAIN_DIR)/bin/\$(arm32_cross_prefix)gdb\$(arm32_cross_prefix) Browse... Variables...

Other options:

Commands: set mem inaccessible-by-default off

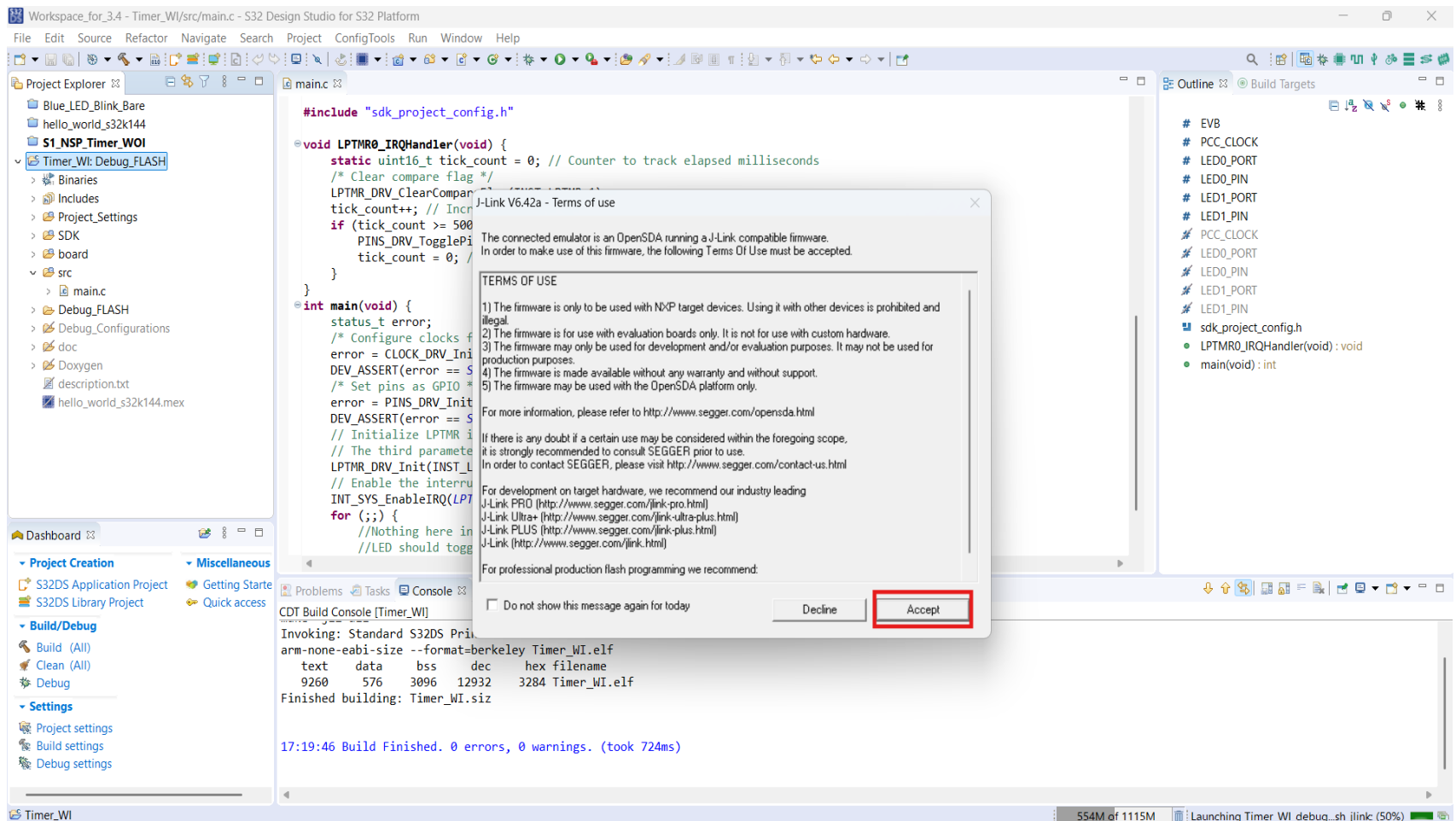
Filter matched 16 of 28 items

Revert Apply

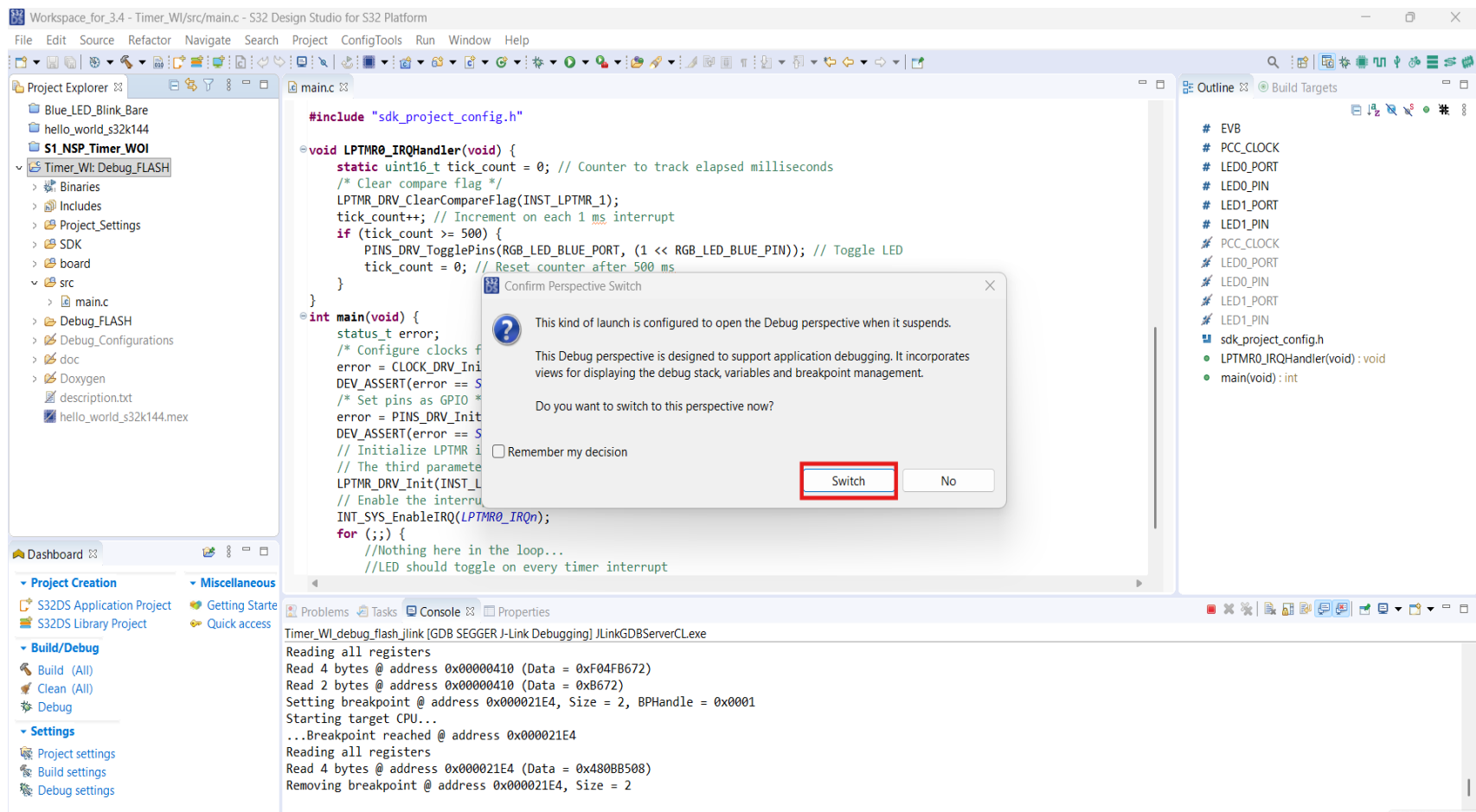
Debug Close

Timer_WI 449M of 1115M

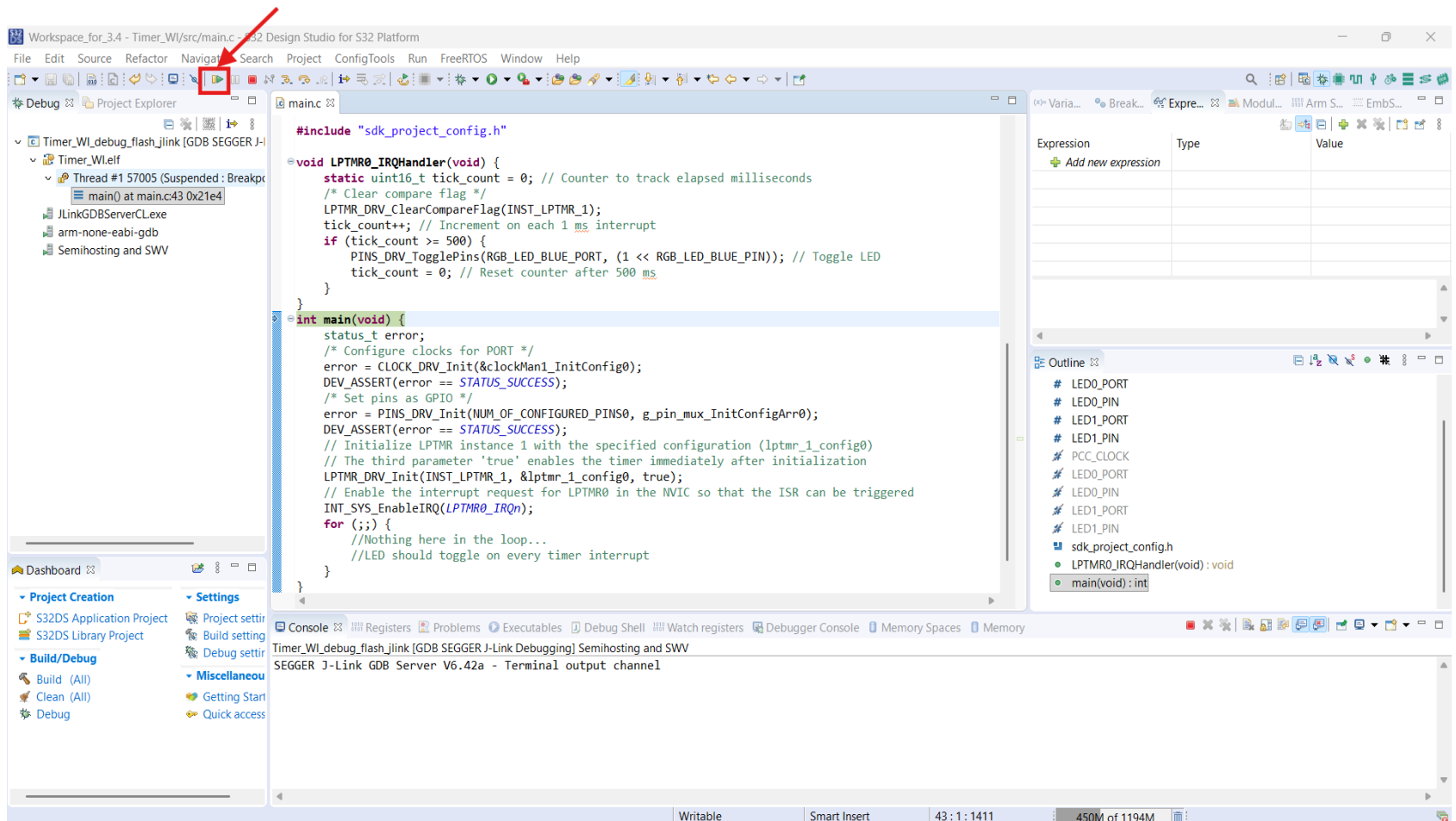
- Click on Accept



- Click on Switch, changes to Debug perspective



- To run code on board, click on Resume and see output on board



- Click on Terminate to stop debugging

The screenshot shows the S32 Design Studio IDE interface. The main window displays the source code for `main.c` in the `main` function. The `Debug` toolbar at the top has the `Terminate` button (represented by a red square icon) highlighted with a red box and an arrow. The `Project Explorer` on the left shows the project structure, including `Timer_Wl` and `Thread #1 57005 (Running: User Requ)`. The `Console` at the bottom shows the execution log, including messages about setting breakpoints and starting the target CPU.

```

static uint16_t tick_count = 0; // Counter to track elapsed milliseconds
/* Clear compare flag */
LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
tick_count++; // Increment on each 1 ms interrupt
if (tick_count >= 500) {
    PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN)); // Toggle LED
    tick_count = 0; // Reset counter after 500 ms
}

int main(void) {
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    // Initialize LPTMR instance 1 with the specified configuration (lptmr_1_config0)
    // The third parameter 'true' enables the timer immediately after initialization
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);
    // Enable the interrupt request for LPTMR0 in the NVIC so that the ISR can be triggered
    INT_SYS_EnableIRQ(LPTMR0_IRQn);
    for (;;) {
        //Nothing here in the loop...
        //LED should toggle on every timer interrupt
    }
}

```

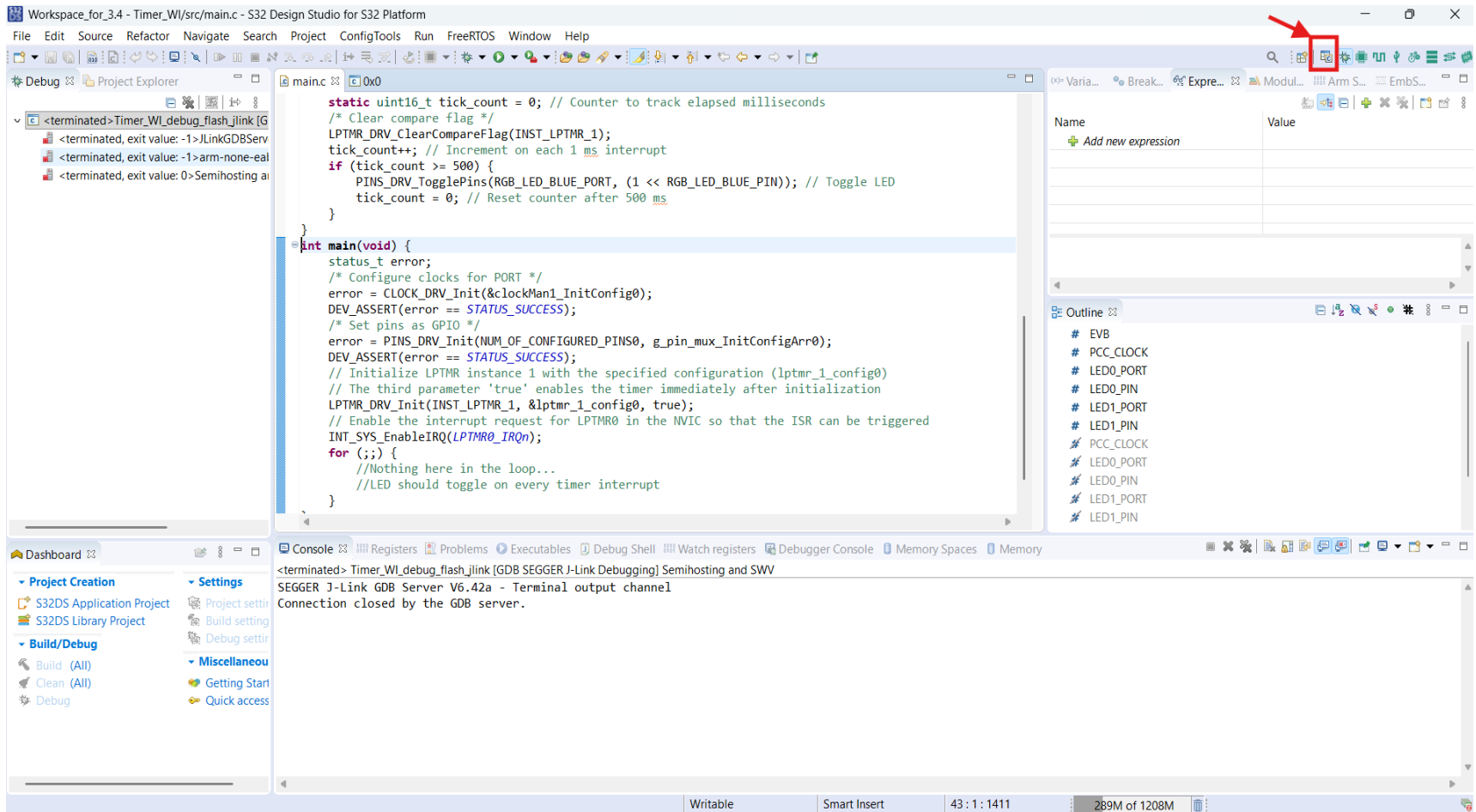
Console Output:

```

Timer_Wl_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCL.exe
CFBP 00000000, CONTROL 00, FAULTMASK 00, BASEPRI 00, PRIMASK 00
Reading all registers
Read 4 bytes @ address 0x00000410 (Data = 0xF04FB672)
Read 2 bytes @ address 0x00000410 (Data = 0xB672)
Setting breakpoint @ address 0x000021E4, Size = 2, BPHandle = 0x0001
Starting target CPU...
...Breakpoint reached @ address 0x000021E4
Reading all registers
Read 4 bytes @ address 0x000021E4 (Data = 0x480B508)
Removing breakpoint @ address 0x000021E4, Size = 2
Starting target CPU...

```

- Click on C/C++ to change perspective





Workspace_for_3.4 - Timer_WI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help

Project Explorer

- Blue_LED_Blink_Bare
- hello_world_s32k144
- S1_NSP_Timer_WOI
- Timer_WI: Debug_FLASH

main.c

```
static uint16_t tick_count = 0; // Counter to track elapsed milliseconds
/* Clear compare flag */
LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
tick_count++; // Increment on each 1 ms interrupt
if (tick_count >= 500) {
    PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN)); // Toggle LED
    tick_count = 0; // Reset counter after 500 ms
}

int main(void) {
    status_t error;
    /* Configure clocks for PORT */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    // Initialize LPTMR instance 1 with the specified configuration (lptmr_1_config0)
    // The third parameter 'true' enables the timer immediately after initialization
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);
    // Enable the interrupt request for LPTMR0 in the NVIC so that the ISR can be triggered
    INT_SYS_EnableIRQ(LPTMR0_IRQn);
    for (;;) {
        //Nothing here in the loop...
        //LED should toggle on every timer interrupt
    }
}
```

Outline

- # EVB
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN
- sdk_project_config.h
- LPTMR0_IRQHandler(void): void
- main(void): int

Dashboard

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Miscellaneous
 - Getting Started
 - Quick access
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settings
 - Build settings
 - Debug settings

Problems Tasks Console Properties

<terminated> Timer_WI_debug_flash.jlink [GDB SEGGER J-Link Debugging] Semihosting and SWV
SEGGER J-Link GDB Server V6.42a - Terminal output channel
Connection closed by the GDB server.

Writable Smart Insert 43 : 1 : 1411 305M of 1208M



Expt.4. Generation of Pulse Width Modulation Signal Using FTM Module on the S32K144 Microcontroller

- **Objective**

- ✓ Generate a PWM signal using the FTM module of the S32K144 microcontroller using S32 SDK.

- **Problem Description**

- ✓ The FlexTimer Module, FTM, shall be configured to generate a PWM signal on a selected output pin (PTD16). The PWM frequency and duty cycle shall be configurable in software. The generated PWM waveform shall be observed on PTD16 pin.

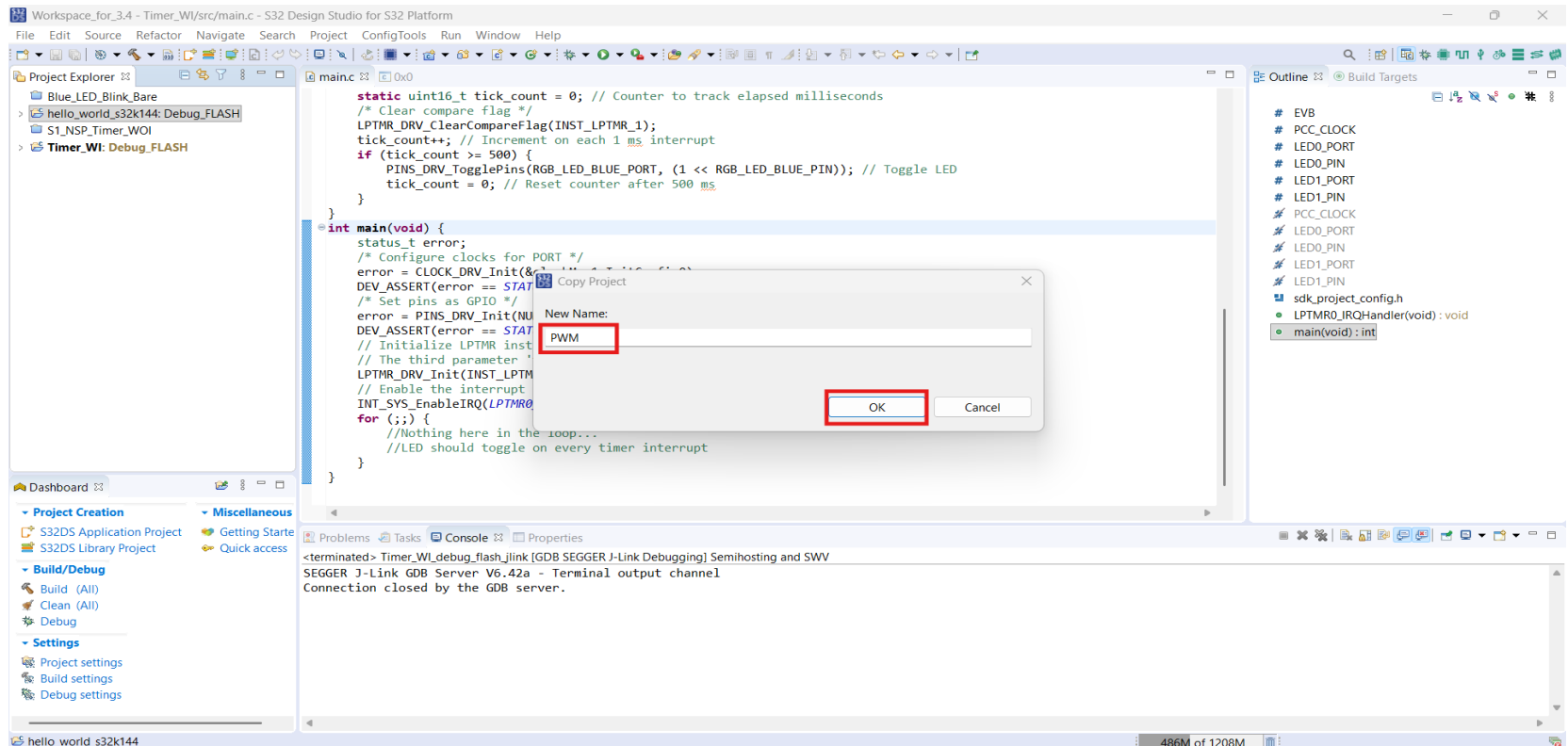
- **Functional Requirements**

- Configure system clock and FTM peripheral clock.
- Configure the selected FTM channel (FTM0_CH1) pin for PWM output using (PTD16) pin mux configuration.
- Configure PWM frequency, for example 1 kHz.
- Configure duty cycle, for example 25%, 50%, and 75%.
- Generate PWM signal continuously.
- Modify duty cycle during runtime or through different test cases.
- Verify PWM frequency and duty cycle using LED.

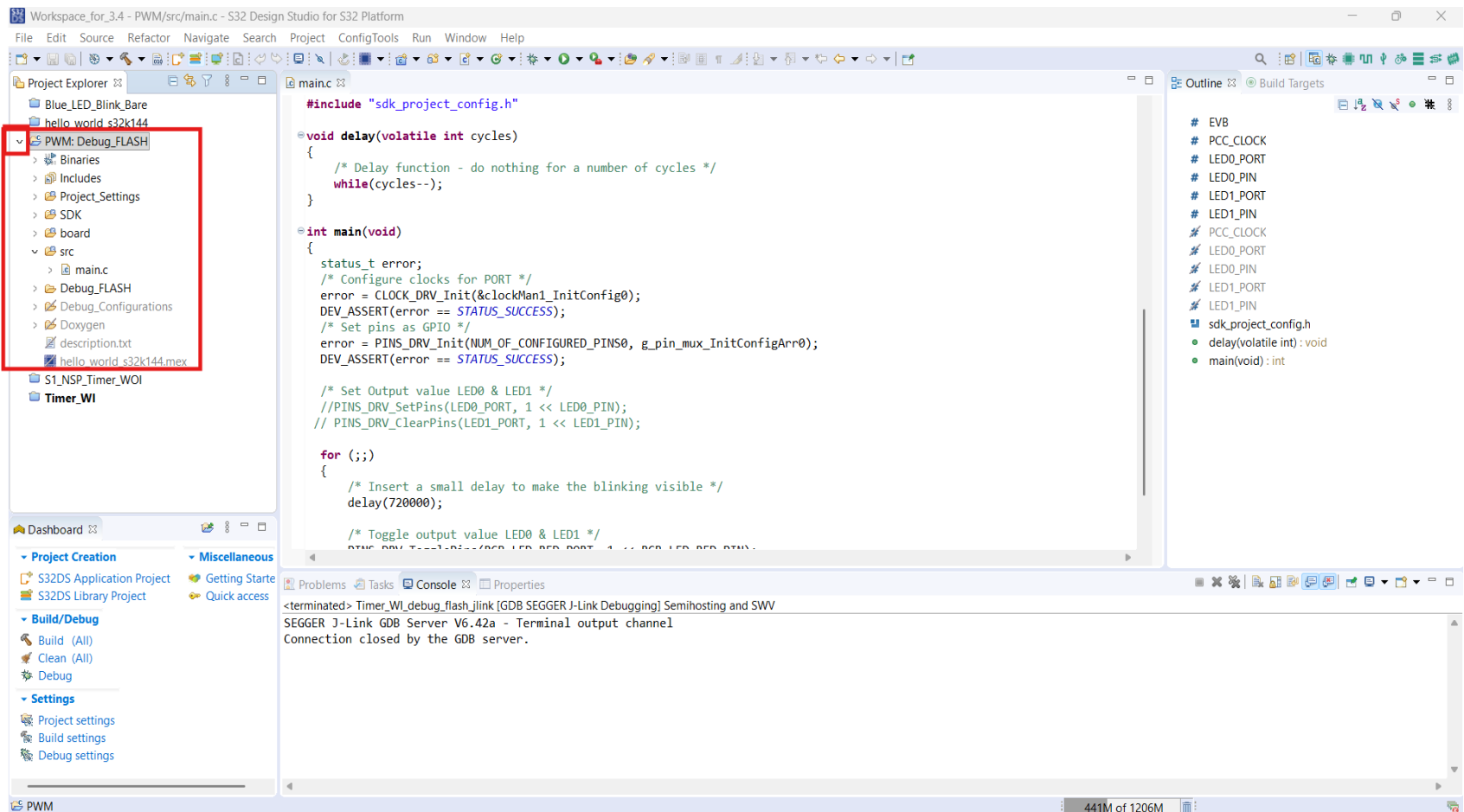
- **Expected Output**

- A PWM waveform should be generated on the selected FTM output pin PTD16 with the required frequency and duty cycle.

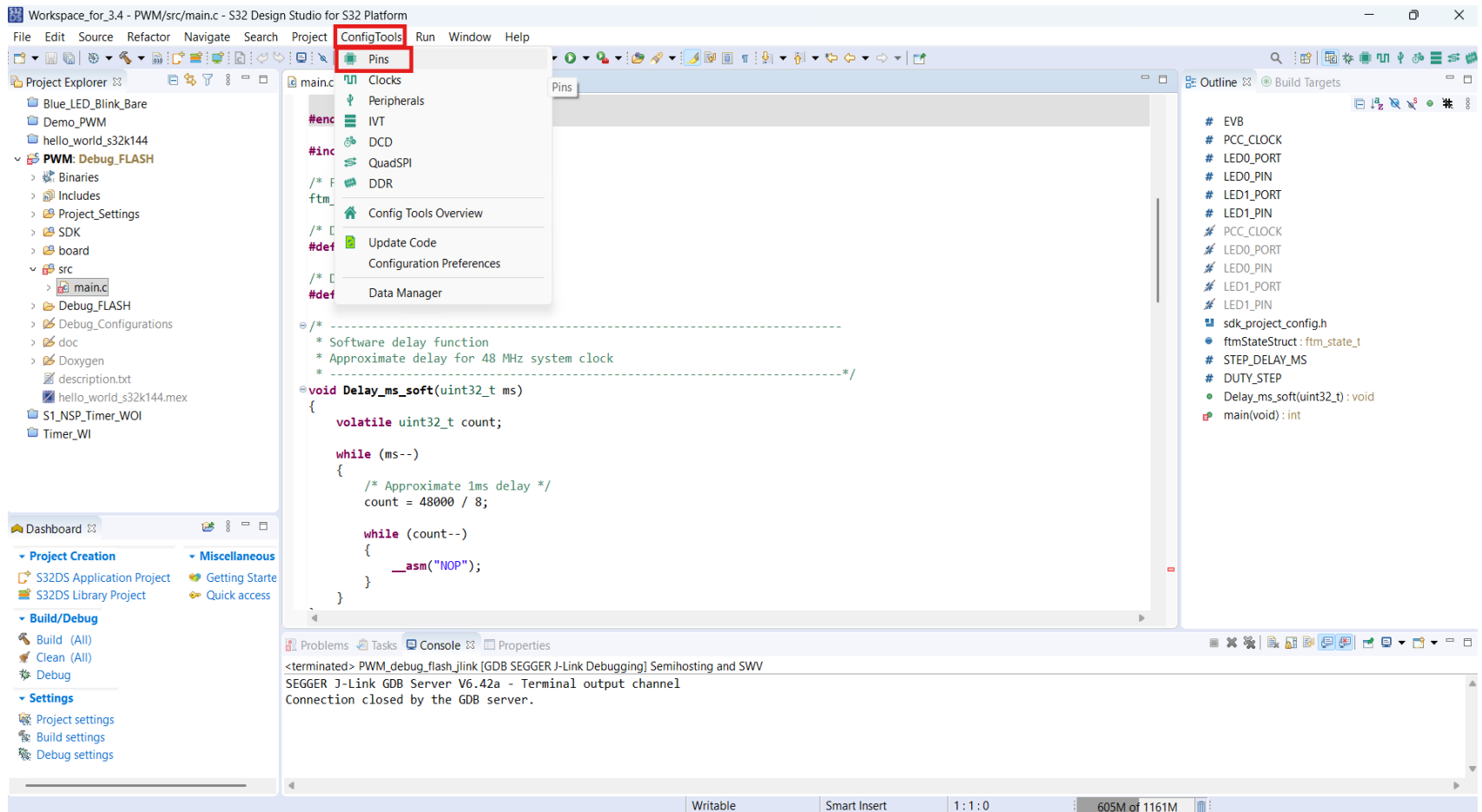
- Select hello_world_s32k144 project, copy and paste it and rename as PWM. Select PWM project and **Close Unrelated Projects**. Refer the S1_26_AE_AEL_DM_ZG512_Lab_CS_02.pdf



- Expand the project and open src→main.c file



- For Pin Routing Details: Click on ConfigTools→Pins



- Click on Pin 21, pop up window Pin[21] opens

Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi
☒	13	VREFL				
☒	14	VSS_14				
☐	15	PTB7		PTB7		
☐	16	PTB6		PTB6		
☐	17	PTE14		PTE14	FTM0_FLT1[...]	
☐	18	PTB3		PTB3	FTM0_FLT0[...]	
☐	19	PTB12		PTB12	FTM0_FLT3	
☐	20	PTD17		PTD17	FTM0_FLT2	
☐	21	PTD16	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPi0_SIN
☒	22	PTD15	RGB_LED_RED	PTD15	FTM0_CH0	LPSPi0_SCK
☐	23	PTE9		PTE9	FTM0_CH7	
☐	24	PTD14		PTD14	FTM2_CH5	
☐	25	PTD13		PTD13	FTM2_CH4	
☐	26	PTE8		PTE8	FTM0_CH6	
☐	27	PTB5		PTB5	FTM0_CH5	LPSPi0_PCS
☐	28	PTB4		PTB4	FTM0_CH4	LPSPi0_SOL
☐	29	PTC3		PTC3	FTM0_CH3	
☐	30	PTC2		PTC2	FTM0_CH2	
☐	31	PTD7		PTD7	FTM2_FLT3	
☐	32	PTD6		PTD6	FTM2_FLT2	

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked

Pin [21] window:

All signals on pin [21]:

- ☐ CMP0_RRT (CMP0,rrt)
- ☐ FTM0_CH1 (FTM0,ch,1)
- ☐ LPSPi0_SIN (LPSPi0,sin)
- ☐ PTD16 (PORTD,port,16)

Done

Overview Configuration - General Info

Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board/pin_mux.c
board/pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue	Origin

345M of 1164M

- Select FTM0_CH1, in new pop up window, select Output and click OK

Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM Update Code BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi
13	VREFL					
14	VSS_14					
15	PTB7			PTB7		
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPiO_SIN
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPiO_SCK
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTE8			PTE8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPiO_PCS
28	PTB4			PTB4	FTM0_CH4	LPSPiO_SOL
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTN6			PTN6	FTM2_FLT2	

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked

Direction required!

Select the routing direction for:
[Pin: PTD16 | Peripheral: FTM0 | Signal: ch, 1]

Input
Output

OK Cancel

Other tools

Problems

type filter text

Level	Resource	Issue	Origin
-------	----------	-------	--------

364M of 1164M

- Click on Done

Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin Pin name Label Identifier PORT FTM LPSPi

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi
13	VREFL					
14	VSS_14					
15	PTB7			PTB7		
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPi0_SIN
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPi0_SCK
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTE8			PTE8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPi0_PCS
28	PTB4			PTB4	FTM0_CH4	LPSPi0_SOL
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 5

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
21	FTM0	ch, 1	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked

Pin [21]

All signals on pin [21]:

- ☐ CMP0_RRT (CMP0,rrt)
- ☒ FTM0_CH1 (FTM0,ch,1)
- ☐ LPSPi0_SIN (LPSPi0,sin)
- ☐ PTD16 (PORTD,port,16)

Done

Overview Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

☒ Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue	Origin

440M of 1164M

Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package Overview Code Preview Registers

Pin name Label Identifier PORT FTM LPSPI

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPI
13	VREFL					
14	VSS_14					
15	PTB7			PTB7		
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPI0_SIN
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPI0_SCK
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTE8			PTE8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPI0_PCS
28	PTB4			PTB4	FTM0_CH4	LPSPI0_SOL
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 5

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked
21	FTM0	ch, 1	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue	Origin
-------	----------	-------	--------

470M of 1164M

- For Peripheral Configuration: Select ConfigTools→Peripherals

Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project **ConfigTools** Pins Run Window Help

Timer_WOI BOARD_InitPins

Pins Peripheral... Power Gro... Package

ConfigTools menu:

- Pins
- Clocks
- Peripherals**
- IVT
- DCD
- QuadSPI
- DDR
- Config Tools Overview
- Update Code
- Configuration Preferences
- Data Manager

Pin list:

Pin	Pin name	Label	Identifier	PO
1	PTE16			PT16
2	PTE15			PT15
3	PTD1			PTD1
4	PTD0	RGB_LE...	RGB_LED_...	PTD0
5	PTE11			PT11
6	PTE10			PT10
7	PTE13			PT13
8	PTE5			PT05
9	PTE4			PT04
10	VDD_10			
11	VDDA			
12	VREFH			
13	VREFL			
14	VSS_14			
15	PTB7			PTB7
16	PTB6			PTB6
17	PTE14			PT14
18	PTE3			PT03
19	PTE12			PT12

Routing Details for BOARD...

Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do
->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Do

Overview Configuration - General Info

Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c
board\pin_mux.h

Functional groups

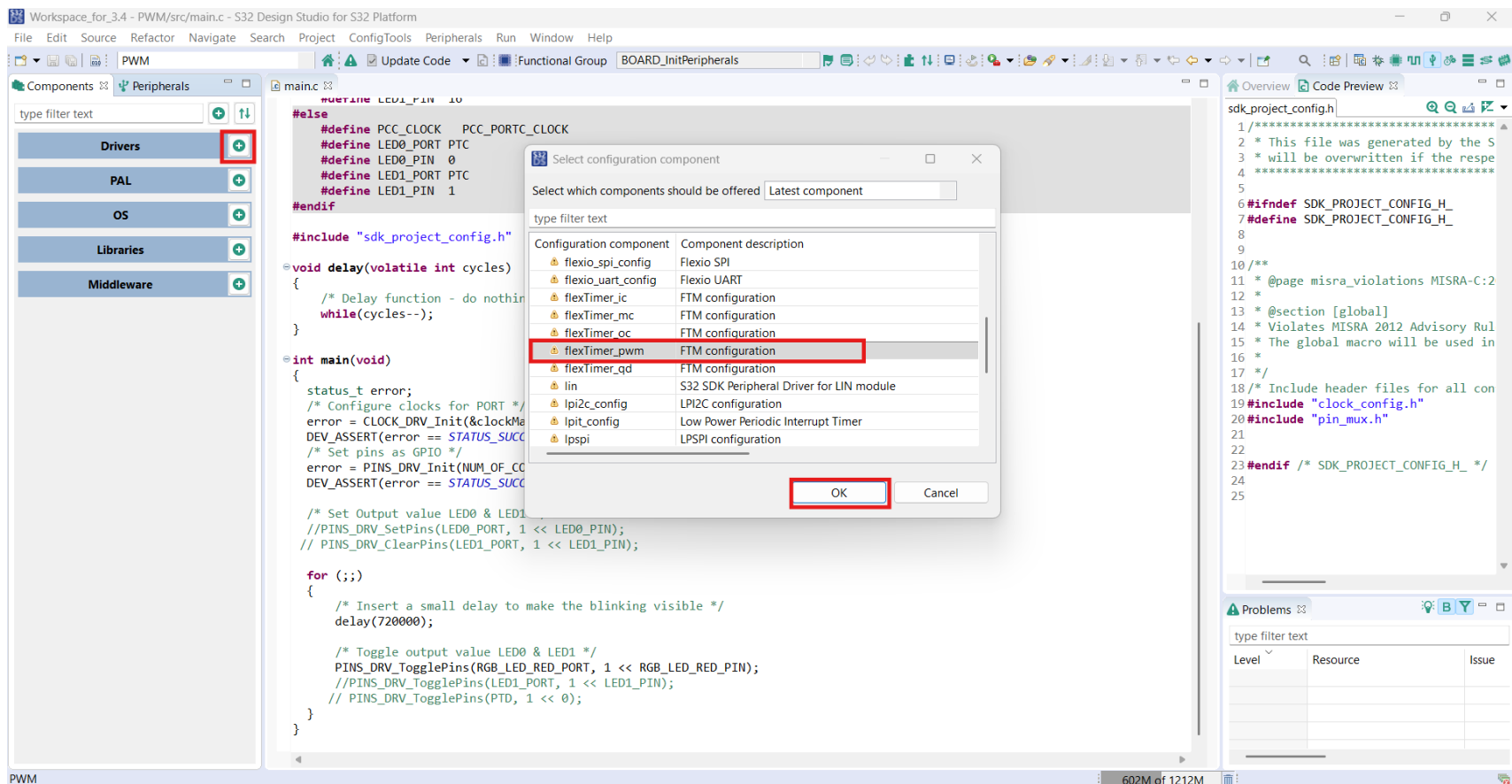
BOARD_InitPins

Other tools

Problems

Level	Resource	Issue	Origin

- Click on Add Drivers, new pop-up window opens. Select FlexTimer_pwm and Click on OK



- Check settings as below for FTM for Initialization tab

Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Peripherals Run Window Help

PWM

Components Peripherals

type filter text

Drivers

flexTimer_pwm_1

PAL

OS

Libraries

Middleware

main.c

```

}
}
/* Main Function
 *
 */
int main(void)
{
    status_t error;

    /* PWM duty variable
    uint16_t duty = 0;

    /* Direction flag
    bool increasing = false;

    /* Initialize Clock
    *
    error = CLOCK_DRV_INIT(CLOCK_DR
    DEV_ASSERT(error == 0);

    /* Initialize Pin
    *
    error = PINS_DRV_INIT(PIN_1,
    DEV_ASSERT(error == 0);

    /* Initialize FTM
    *
    FTM_DRV_Init(FTM0, &ftm0,
    &ftm0);

    /* Initialize PWM
    *
    FTM_DRV_InitPwm(FTM0, &ftm0,
    
```

flexTimer_pwm_1

FTM configuration [Drivers]

Name flexTimer_pwm_1

Mode General Mode Peripheral FTM_0

FTM Configuration Set

Initialization

FTM Clock Configuration

FTM Clock Source System clock

Frequency of source clock (Hz) 48000000

Clock Source Prescaler 32

Frequency of counter clock (Hz) 1500000

FTM Global Configuration

Mode Of Operation For FTM Edge Aligned PWM

Debug Mode Mode 3

Timer Overflow Interrupt ☐

Initialization Trigger ☐

FTM Register Sync Configuration

Software Trigger ☒

Hardware Trigger 1 ☐

Hardware Trigger 2 ☐

Hardware Trigger 3 ☐

Sync Point Immediate

Max Loading Point ☐

Min Loading Point ☐

Automatic Trigger Clear ☐

Inverting Control Synchronization PWM Synchronization

Software Output Control Synchronization PWM Synchronization

Output Mask Synchronization PWM Synchronization

peripherals_flexTimer_pwm_1.c

```

1 /* *****
2  * This file was generated by the
3  * will be overwritten if the resp
4  * *****
5
6 /* clang-format off */
7 /* TEXT BELOW IS USED AS SETTING F
8 !!GlobalInfo
9 product: Peripherals v10.0
10 processor: S32K144
11 package_id: S32K144_LQFP100
12 mcu_data: s32sdk_s32k1xx_rtm_403
13 processor_version: 0.0.0
14 functionalGroups:
15 - name: BOARD_InitPeripherals
16   UID: a6d43dcb-893a-49cd-9d9d-0b
17   called_from_default_init: true
18   selectedCore: core0
19   * BE CAREFUL MODIFYING THIS COMME
20 /* clang-format on */
21
22 /*****
23  * Included files
24  * *****
25 #include "peripherals_flexTimer_pwm_1
26
27 /*****
28  * flexTimer_pwm_1 initialization
29  * *****
30 /* clang-format off */

```

Problems

type filter text

Level	Resource	Issue

PWM

574M of 1062M

- Check settings as below for FTM, select Configuration and PWM Independent Configuration tabs

Workspace_for_3.4 - S32 Design Studio for S32 Platform

File Edit Navigate Search Project ConfigTools Peripherals Run Window Help

Components Peripherals

type filter text

Drivers

flexTimer_pwm_1

PAL

OS

Libraries

Middleware

flexTimer_pwm_1

FTM configuration (Drivers)

Name flexTimer_pwm_1

Mode General Mode Peripheral FTM_0

FTM Configuration Set

Initialization Configurations

PWM General Configuration

PWM Frequency[Hz] 1000

Dead Time Value In [Ticks] 0

Dead Time Pre-scaler Divided By 1

Fault Mode Initialization PWM Independent Configuration PWM Combined Configuration

Number Of Independent Channels 1

Independent Channels Configuration List

Index	Name	Channel hardware ID	Channel Polarity	Duty Cycle	External Trigger	Output Safe State	Enable Channel (n+1) Output	Channel Complementary Mode	Enable Deadtime
0	flexTimer_pwm_1_IndependentChannelConfig0	Channel 1	Active Low	3200	☐	High Level	☐	Invert Channel (n)	☐

Code Preview

```

1/******
2 * This file was generated by the
3 * will be overwritten if the resp
4 *****
5
6/* clang-format off */
7/* TEXT BELOW IS USED AS SETTING F
8!!GlobalInfo
9product: Peripherals v10.0
10processor: S32K144
11package_id: S32K144_LQFP100
12mcu_data: s32sdk_s32k1xx_rtm_403
13processor_version: 0.0.0
14functionalGroups:
15- name: BOARD_InitPeripherals
16  UUID: a6d43dcb-893a-49cd-9d9d-0b
17  called_from_default_init: true
18  selectedCore: core0
19  * BE CAREFUL MODIFYING THIS COMME
20/* clang-format on */
21
22/******
23 * Included files
24 *****
25#include "peripherals_flexTimer_pwm
26
27/******
28 * flexTimer_pwm_1 initialization
29 *****
30/* clang-format off */
  
```

Problems

type filter text

Level	Resource	Issue

PWM

584M of 1124M

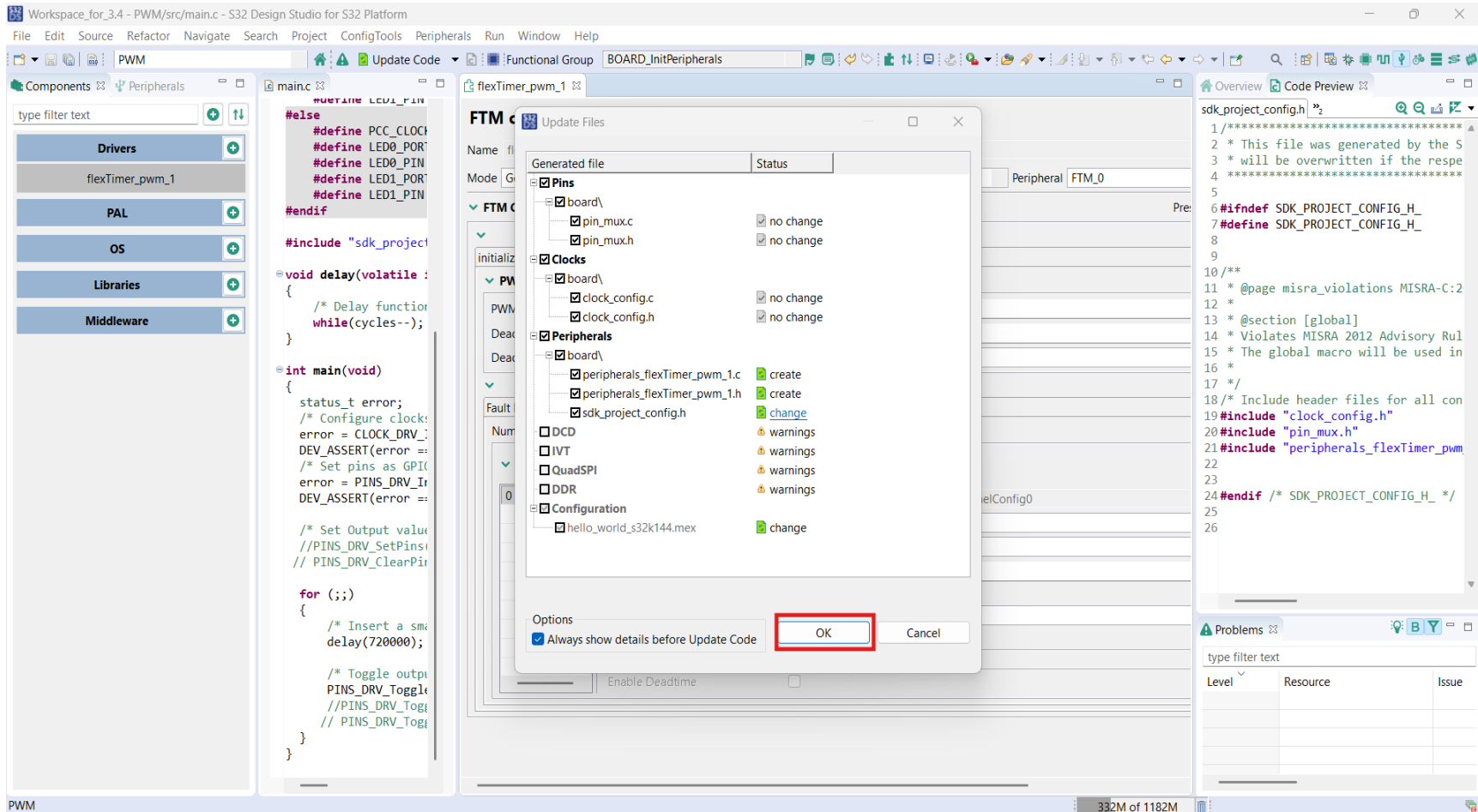
- Click on **Update Code**

The screenshot shows the S32 Design Studio IDE interface. The top toolbar has a red box highlighting the 'Update Code' button. The main workspace is divided into several panes:

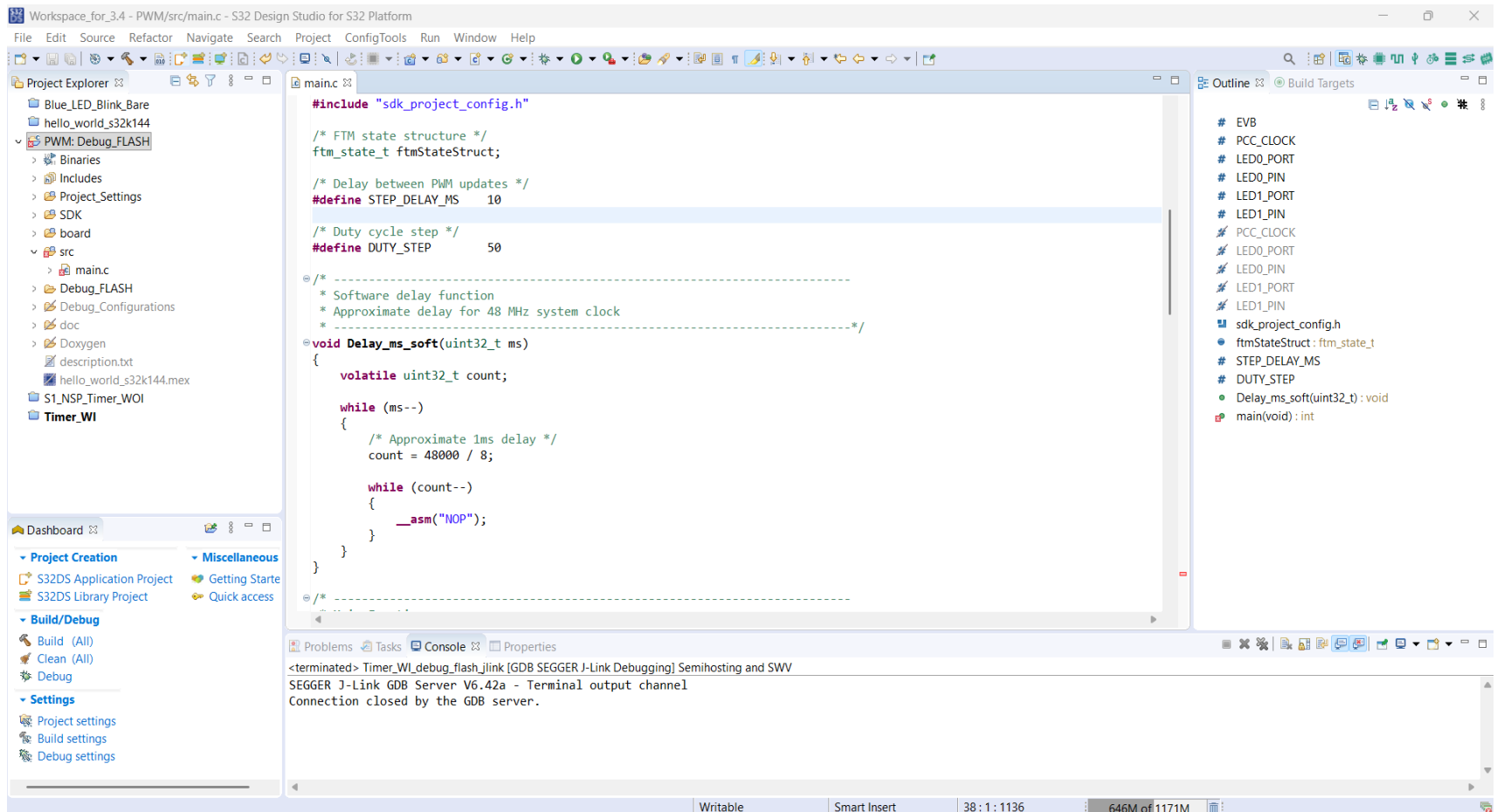
- Components:** A list of components including Drivers, flexTimer_pwm_1, PAL, OS, Libraries, and Middleware.
- Code Editor:** Displays the source code for `main.c`, showing the definition of `LED1_PIN`, the inclusion of `sdk_project`, and the implementation of the `Delay_ms_soft` function.
- FTM configuration [Drivers]:** A configuration pane for the `flexTimer_pwm_1` driver. It includes sections for FTM Configuration Set, PWM General Configuration, and Independent Channels Configuration List.
- Code Preview:** A pane showing the generated code for the `peripherals_flexTimer_pwm_1` driver, including comments and the `#include "peripherals_flexTimer_pwm_1"` statement.
- Problems:** A pane at the bottom right showing any errors or warnings.

The status bar at the bottom indicates the memory usage: 606M of 1164M.

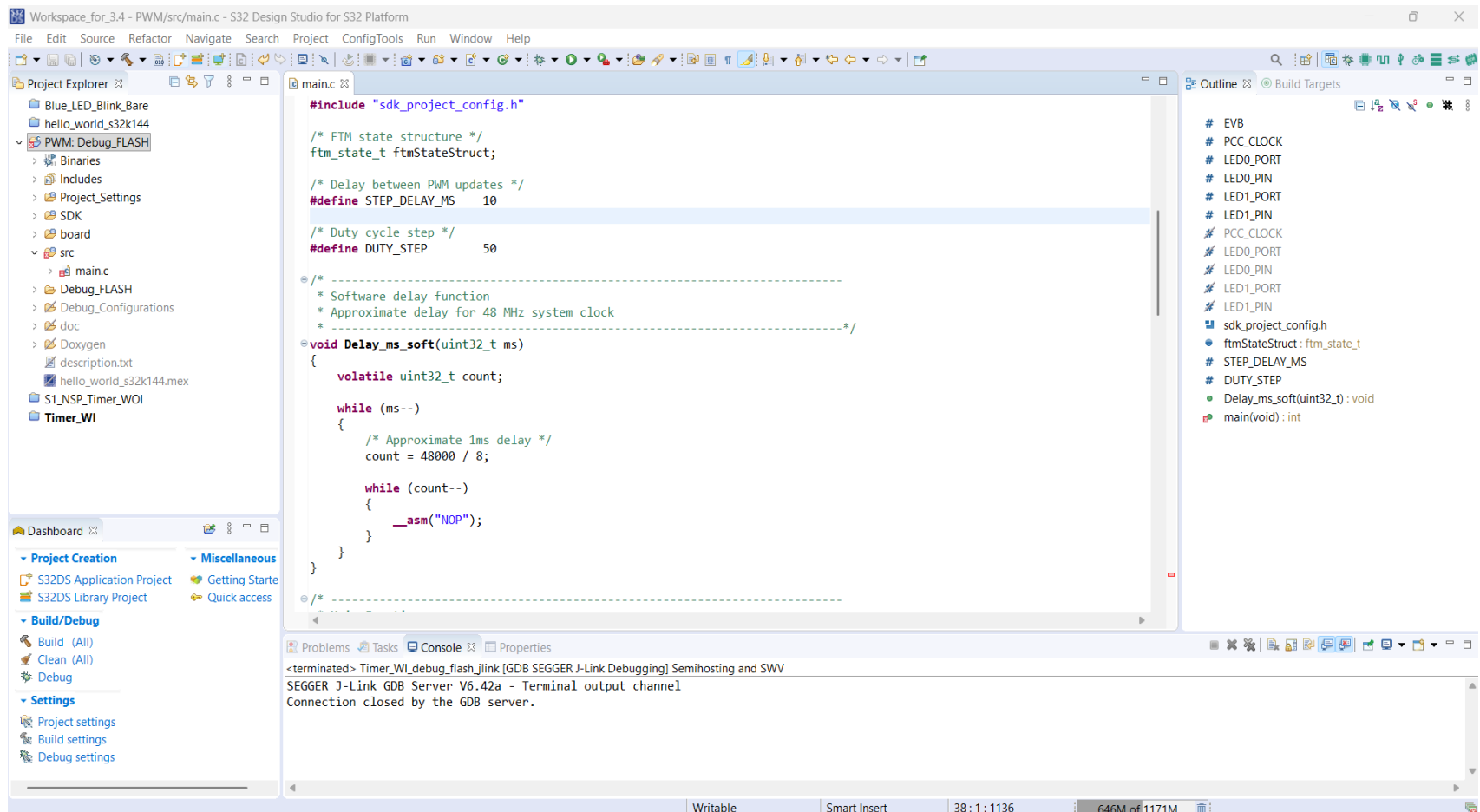
- Click on OK



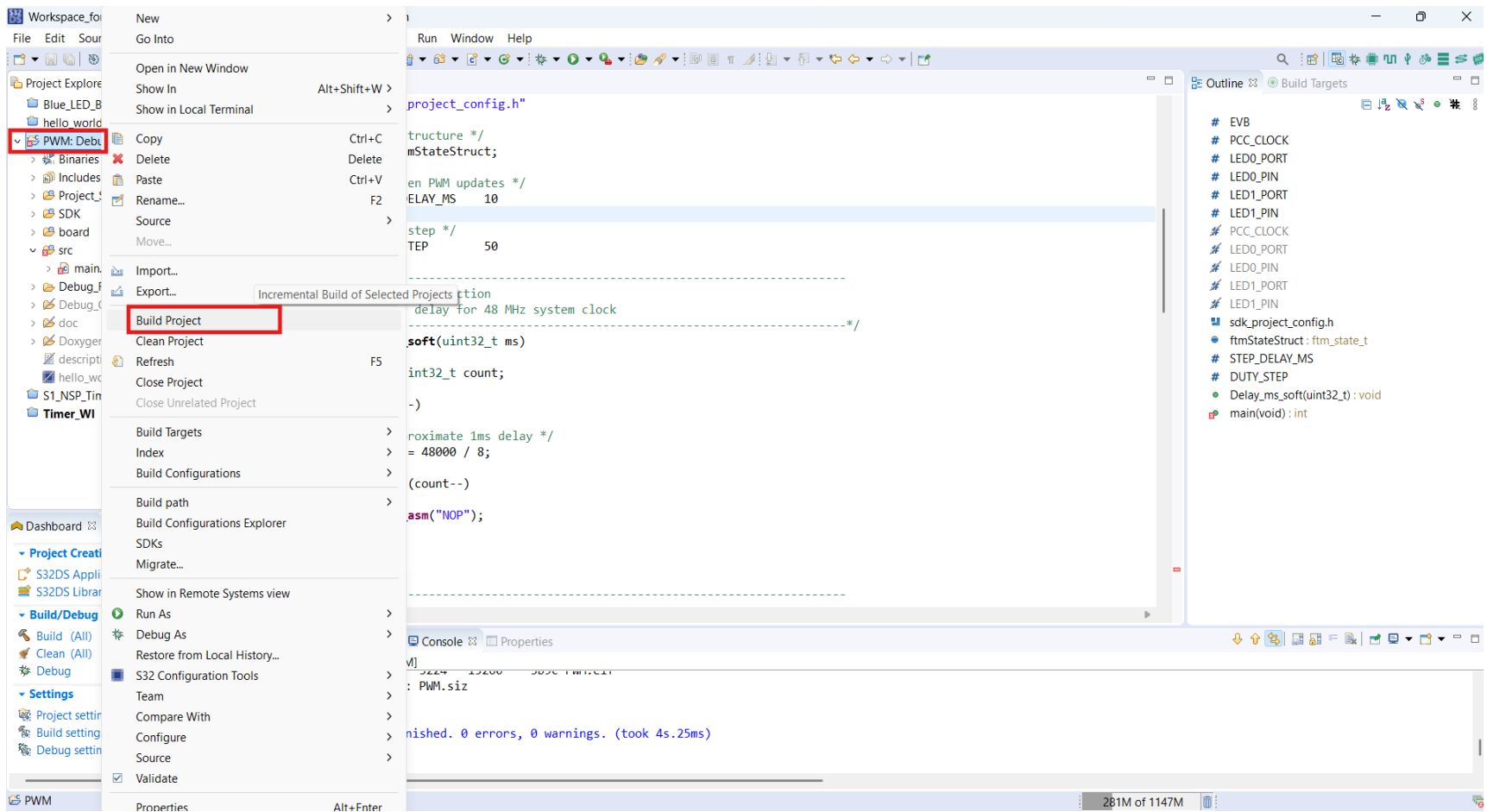
- After update click on C/C++ to change perspective



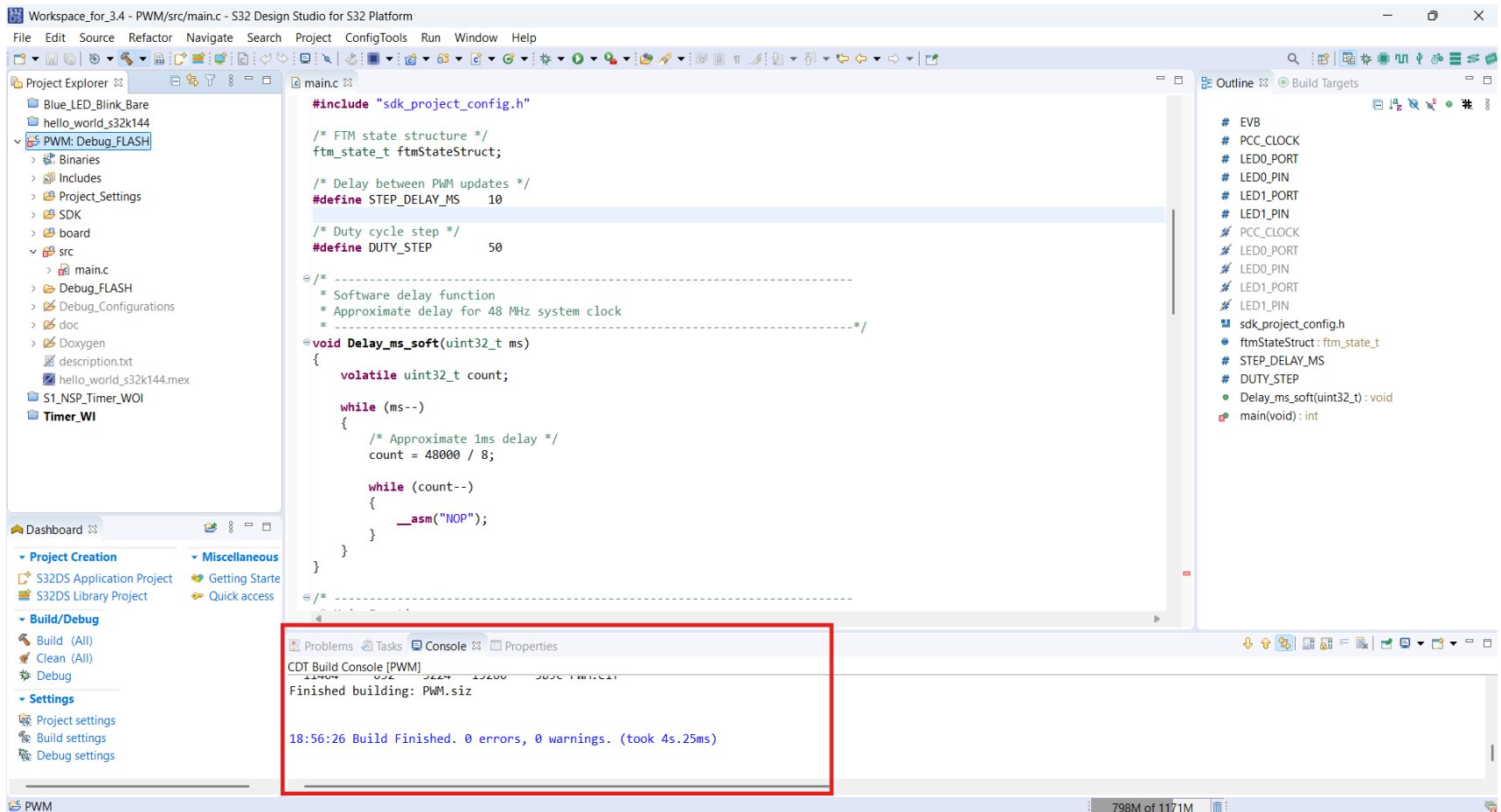
- C/C++ perspective and write c program



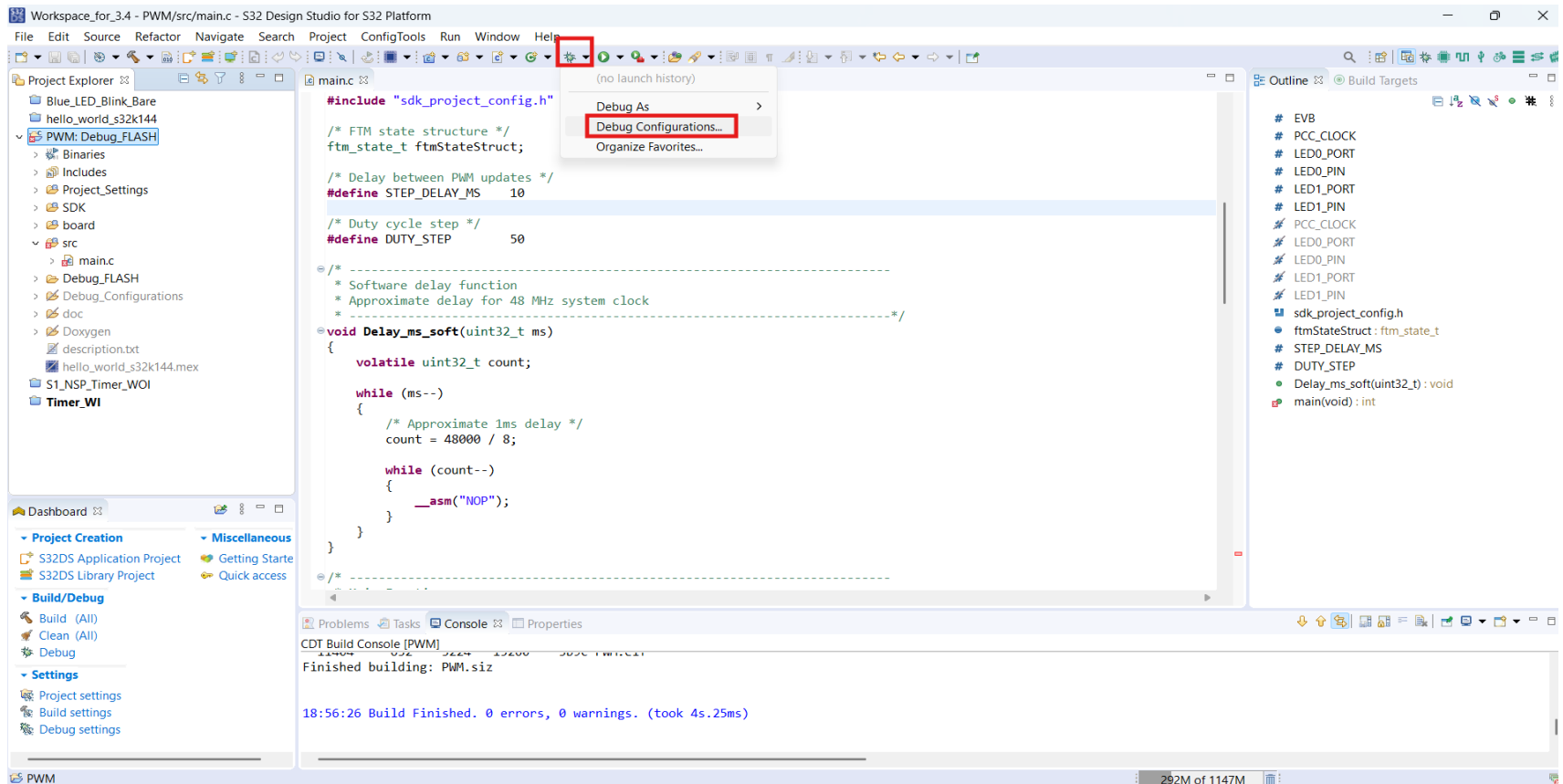
- Right click on project and click on Build Project



- Check for any errors



- Connect S32K144 development board to PC/Laptop
- Click on Debug→Debug Configurations



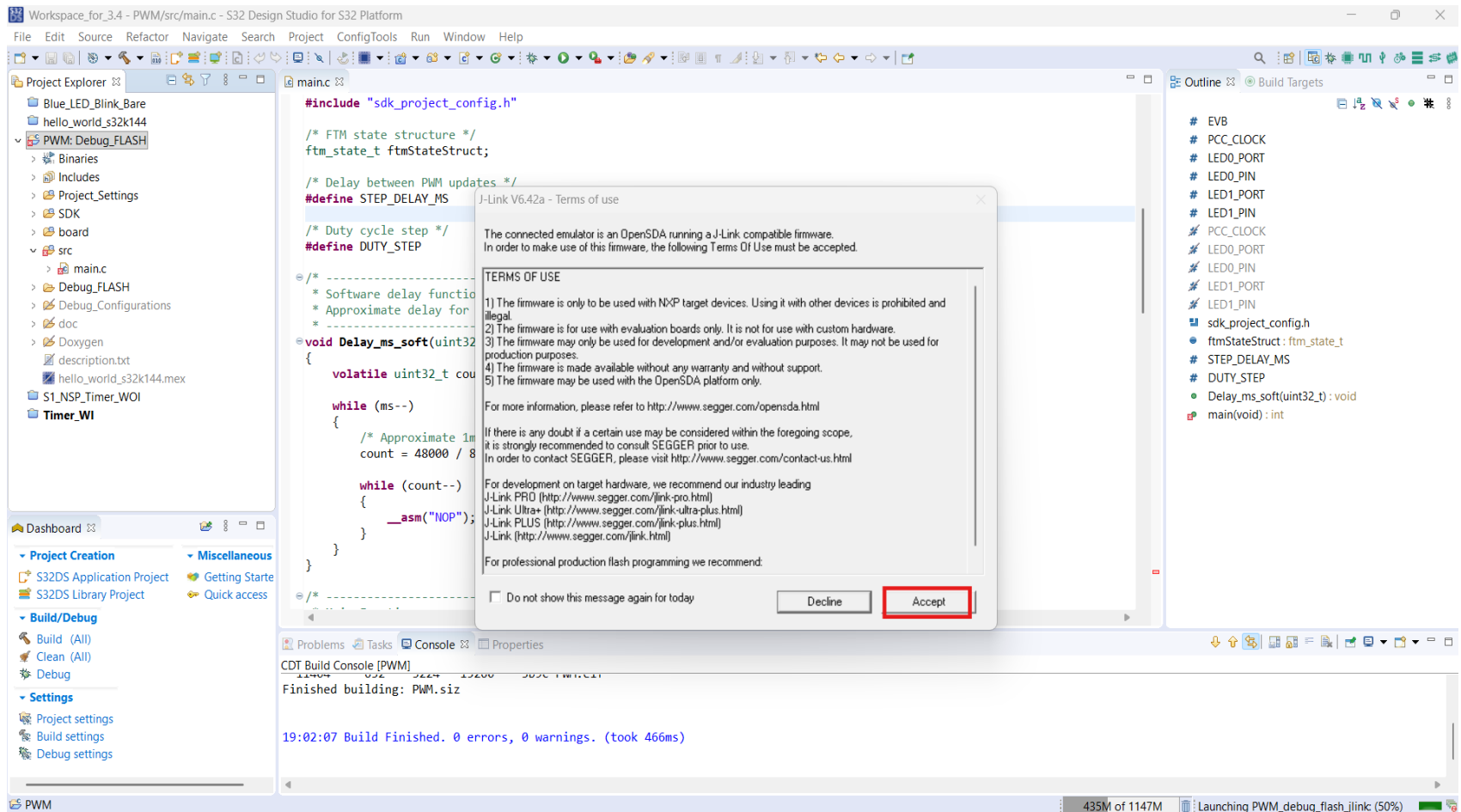
- Do configurations as below and Click on **Debug**

The screenshot displays the Eclipse IDE interface with the 'Debug Configurations' dialog box open. The 'Main' tab is selected, and the configuration 'PWM_debug_flash_jlink' is highlighted in the list on the left. The configuration details are as follows:

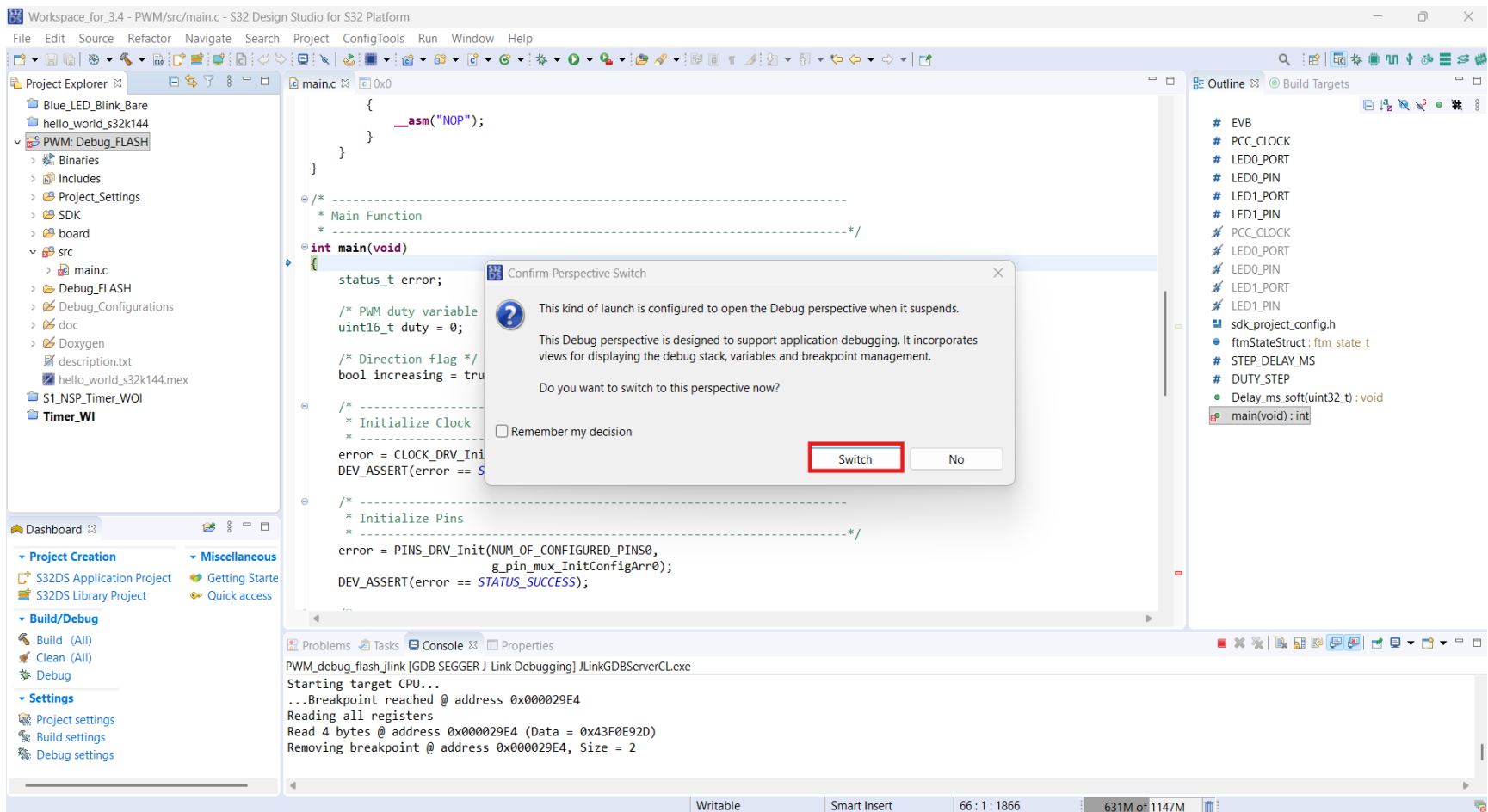
- Name:** PWM_debug_flash_jlink
- J-Link GDB Server Setup:**
 - ☒ Start the J-Link GDB server locally
 - Executable: \$(jlink_path)/\$(jlink_gdbserver)
 - Device name: S32K144
 - Endianness: ☒ Little
 - Connection: ☒ USB
 - Interface: ☒ JTAG
 - Initial speed: ☒ Fixed 1000 kHz
 - GDB port: 2331
 - SWO port: 2332
 - Telnet port: 2333
 - ☒ Verify downloads
 - ☒ Initialize registers on start
 - ☒ Local host only
 - ☐ Silent
- GDB Client Setup:**
 - Executable: \$(S32DS_ARM32_TOOLCHAIN_DIR)/bin/\$(arm32_cross_prefix)gdb
 - Other options: -singlerun -strict -timeout 0 -nogui
 - ☒ Allocate console for the GDB server
 - ☒ Allocate console for semihosting and SWO
 - Commands: set mem inaccessible-by-default off

The 'Debug' button at the bottom right of the dialog is highlighted with a red box. The status bar at the bottom shows '18:56:26 Build Finished. 0 errors, 0 warnings. (took 4s.25ms)'.

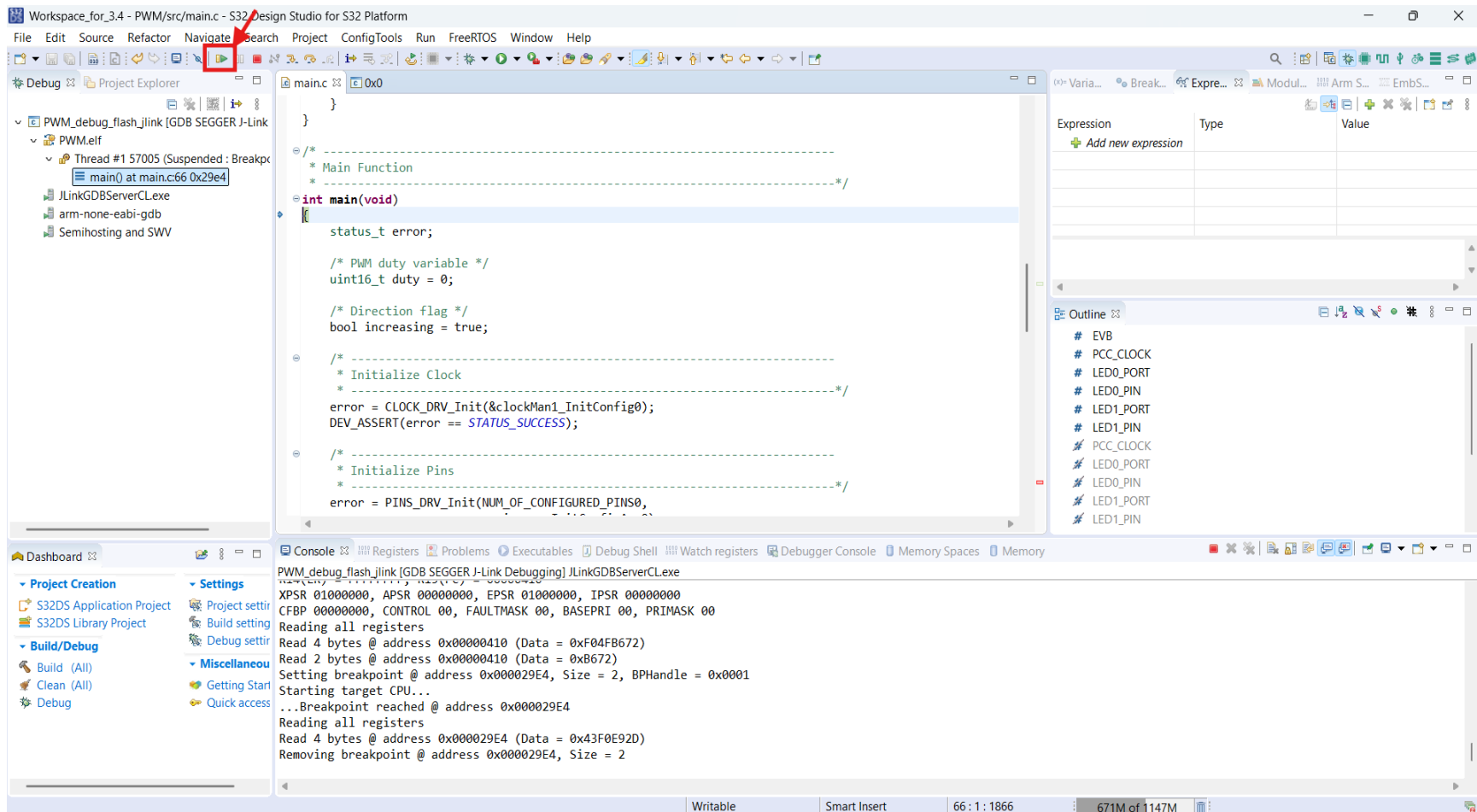
- Click on Accept



- Click on Switch, changes to Debug perspective

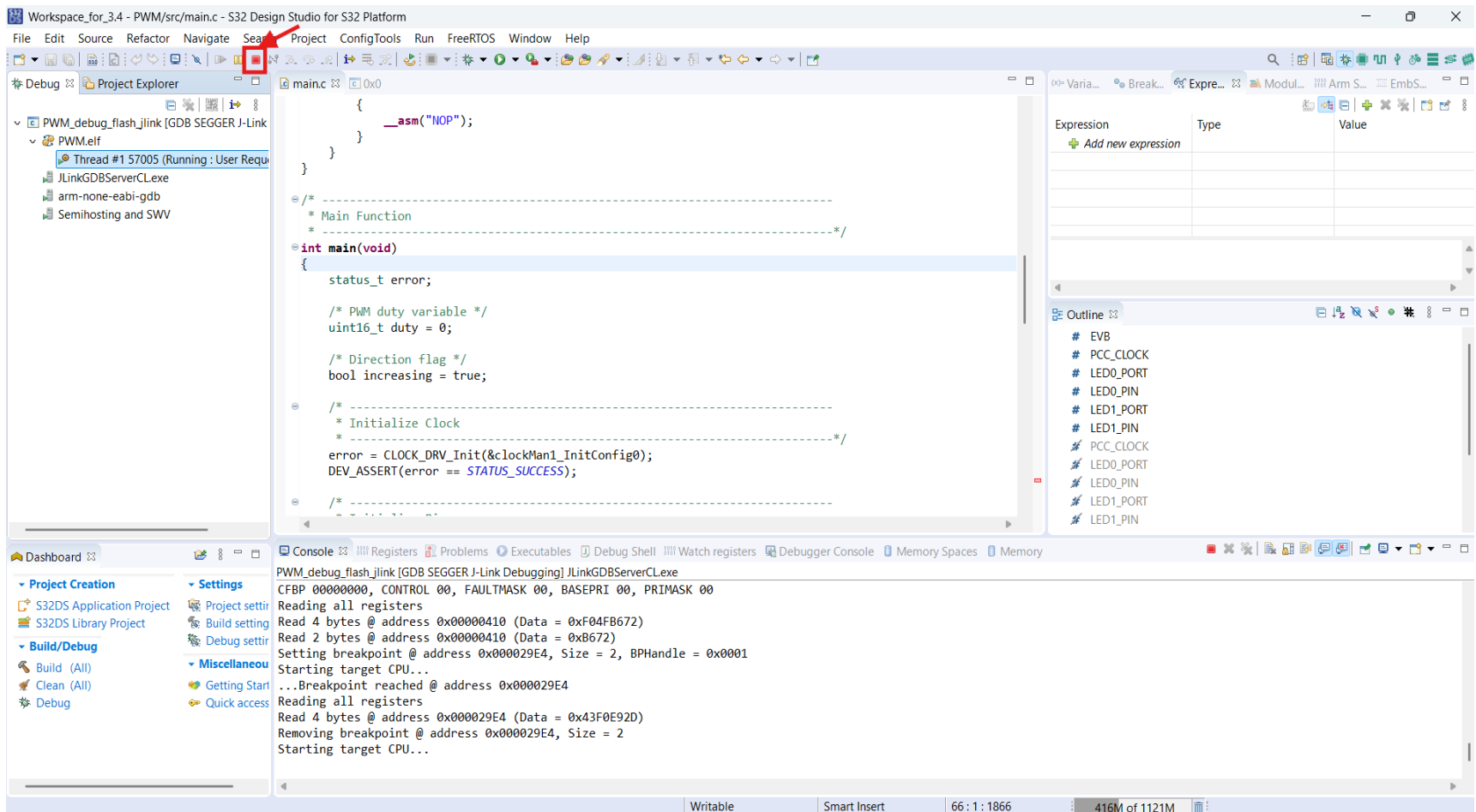


- To run code on board, click on Resume and see output on board



The screenshot shows the S32 Design Studio IDE interface. The top toolbar has a red box around the Resume button (a green play icon). The main editor displays the code for `main.c`, which includes comments for the Main Function and the `int main(void)` function. The function initializes variables, sets up the clock, and initializes pins. The Debug console at the bottom shows the output of the JLinkGDBServerCLex, including register values and breakpoint status. The Project Explorer on the left shows the project structure, including the `PWM_debug_flash_jlink` project and the `main.c` file.

- Click on Terminate to stop debugging



Workspace_for_3.4 - PWM/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help

Debug Project Explorer

PWM_debug_flash_jlink [GDB SEGGER J-Link]

PWM.elf

Thread #1 57005 (Running : User Request)

JLinkGDBServerCLExe

arm-none-eabi-gdb

Semihosting and SWV

main.c

```

0x0
{
    {
        __asm__("NOP");
    }
}

/* -----
 * Main Function
 * ----- */
int main(void)
{
    status_t error;

    /* PWM duty variable */
    uint16_t duty = 0;

    /* Direction flag */
    bool increasing = true;

    /* -----
     * Initialize Clock
     * ----- */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);

    /* -----
     * Main Function
     * ----- */
}

```

Expression Break... Expre... Modul... Arm S... EmbS...

Expression	Type	Value
Add new expression		

Outline

- # EVB
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN
- # PCC_CLOCK
- # LED0_PORT
- # LED0_PIN
- # LED1_PORT
- # LED1_PIN

Dashboard

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settir
 - Build setting
 - Debug settir
 - Miscellaneous
 - Getting Start
 - Quick access

Console

PWM_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCLExe

CFBP 00000000, CONTROL 00, FAULTMASK 00, BASEPRI 00, PRIMASK 00

Reading all registers

Read 4 bytes @ address 0x00000410 (Data = 0xF04FB672)

Read 2 bytes @ address 0x00000410 (Data = 0xB672)

Setting breakpoint @ address 0x000029E4, Size = 2, BPHandle = 0x0001

Starting target CPU...

...Breakpoint reached @ address 0x000029E4

Reading all registers

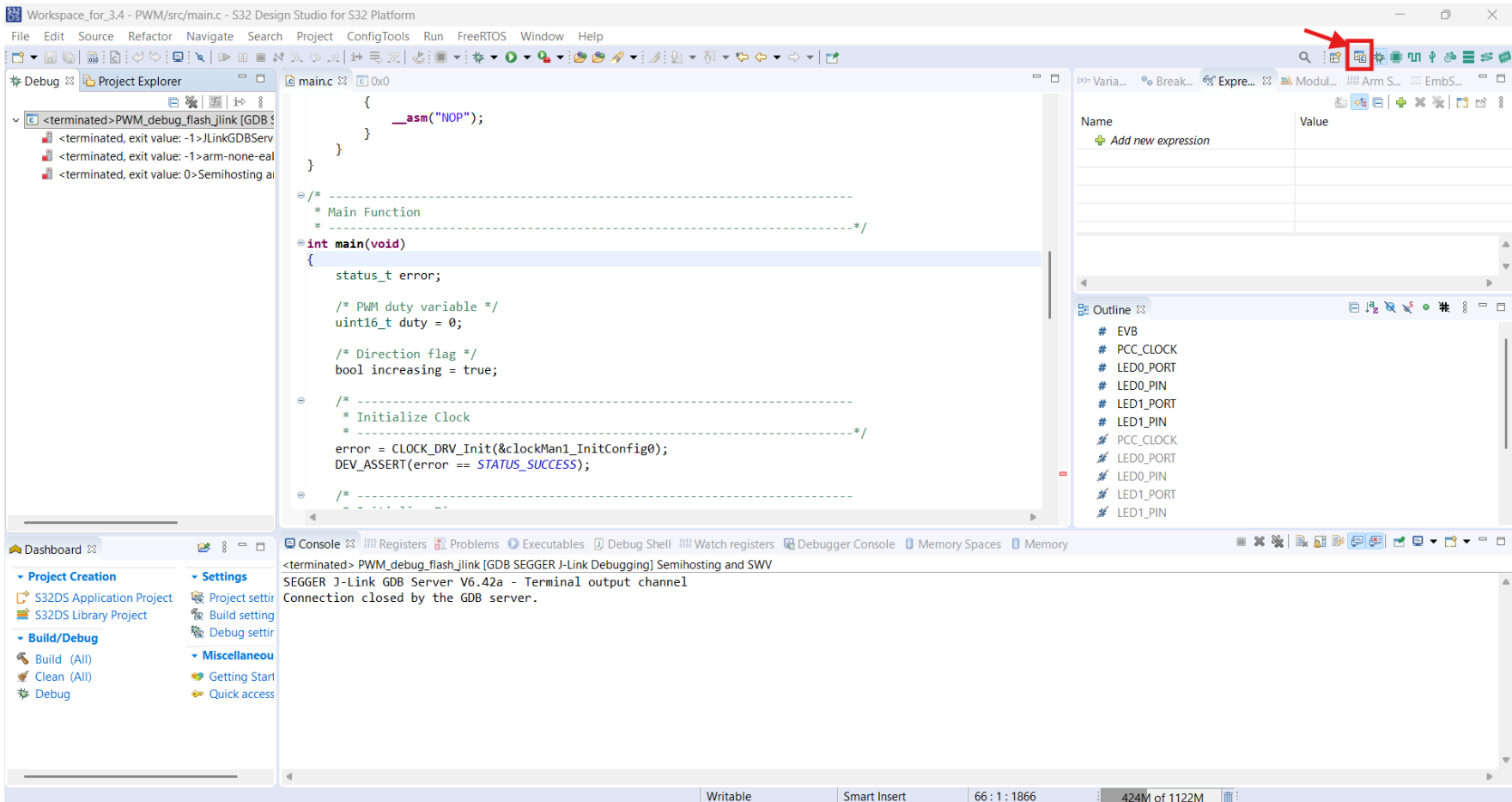
Read 4 bytes @ address 0x000029E4 (Data = 0x43F0E92D)

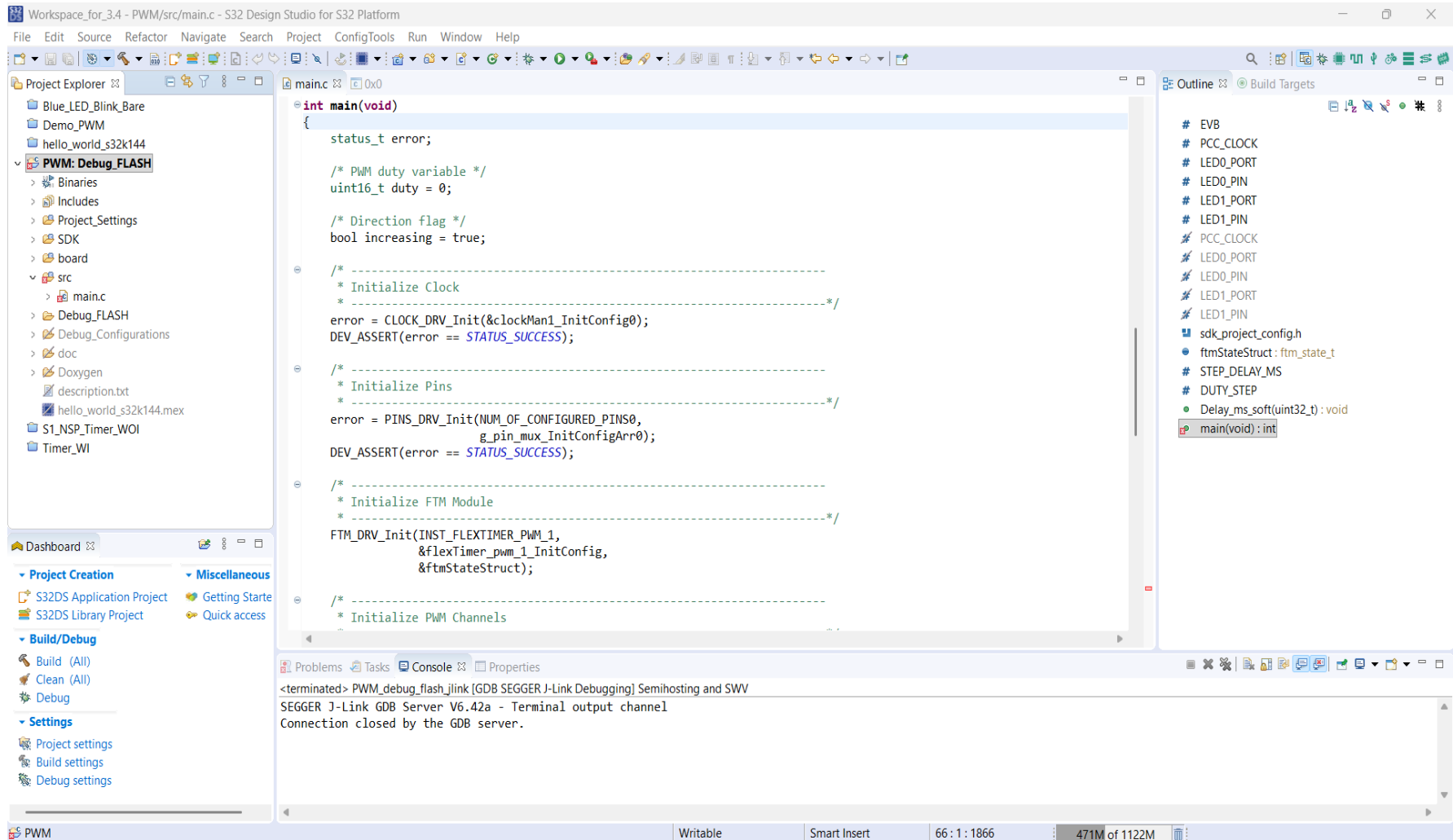
Removing breakpoint @ address 0x000029E4, Size = 2

Starting target CPU...

Writable Smart Insert 66:1:1866 416M of 1121M

- Click on C/C++ to change perspective







Expt.5. Analog Signal Acquisition Using ADC Module on the S32K144 Microcontroller with PWM Output Connected to ADC Input.

- **Objective**

- ✓ Develop an Embedded C application using the ADC module of the S32K144 microcontroller to acquire an analog signal derived from a PWM output.

- **Problem Description**

- ✓ Generate a PWM signal using the FTM module and feed the PWM signal to an ADC input through a suitable low-pass filter or signal-conditioning circuit. The ADC shall sample the resulting analog voltage and convert it into a digital value. The experiment demonstrates the relationship between PWM duty cycle and measured analog voltage.

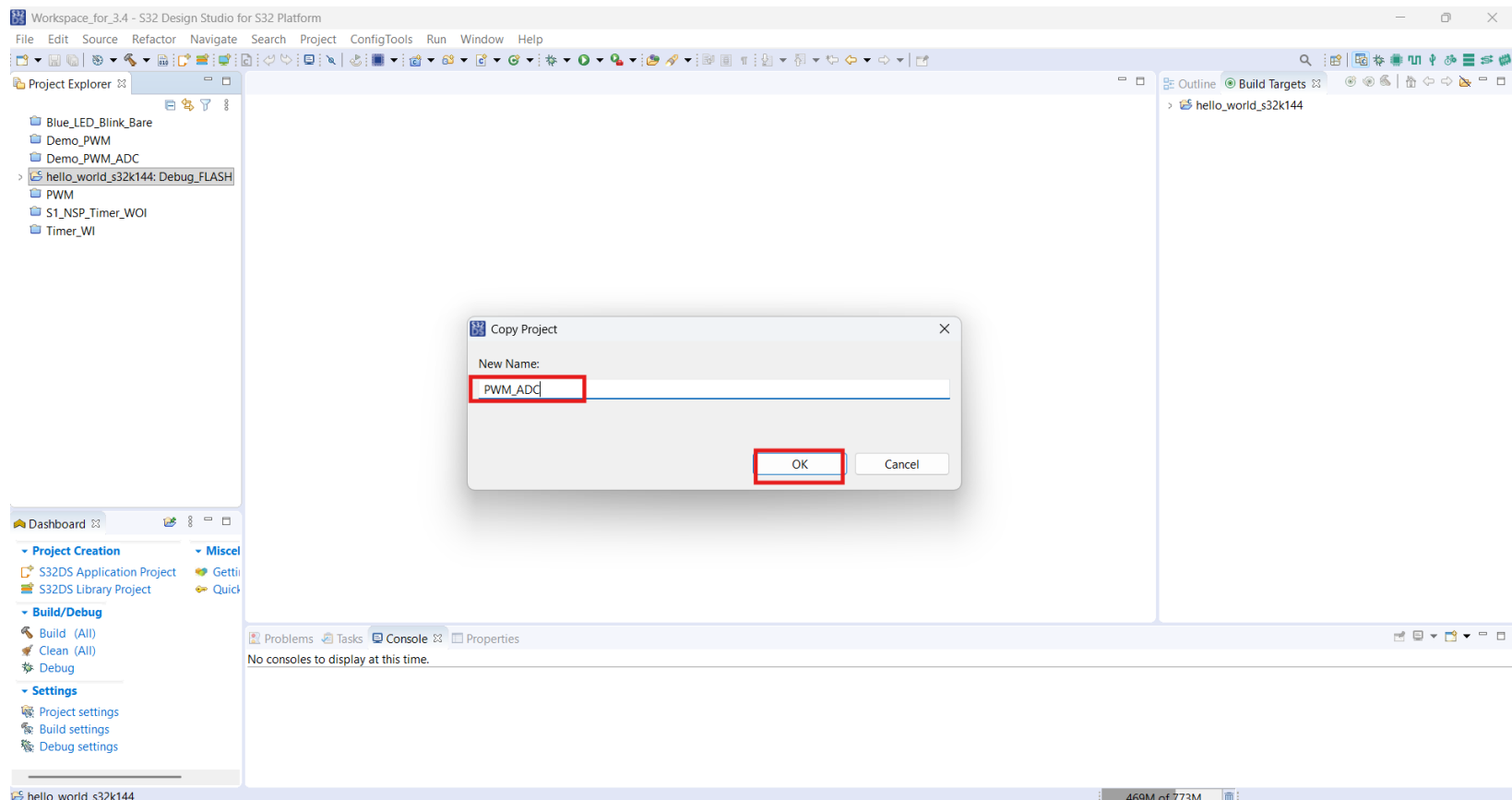
- **Functional Requirements**

- Configure FTM to generate a PWM signal at PTD13 (FTM2_CH4) at a fixed frequency.
- Vary the PWM duty cycle, for example 25%, 50%, and 75%.
- Connect the PWM output (PTD13) to ADC input PTB0 (ADC0-Channel-4) through a proper RC low-pass filter.
- Configure the ADC input PTB0 (ADC0-Channel-4) using S32 SDK.
- Start ADC conversion using software trigger or periodic trigger.
- Read ADC conversion result.
- Convert ADC result into voltage.

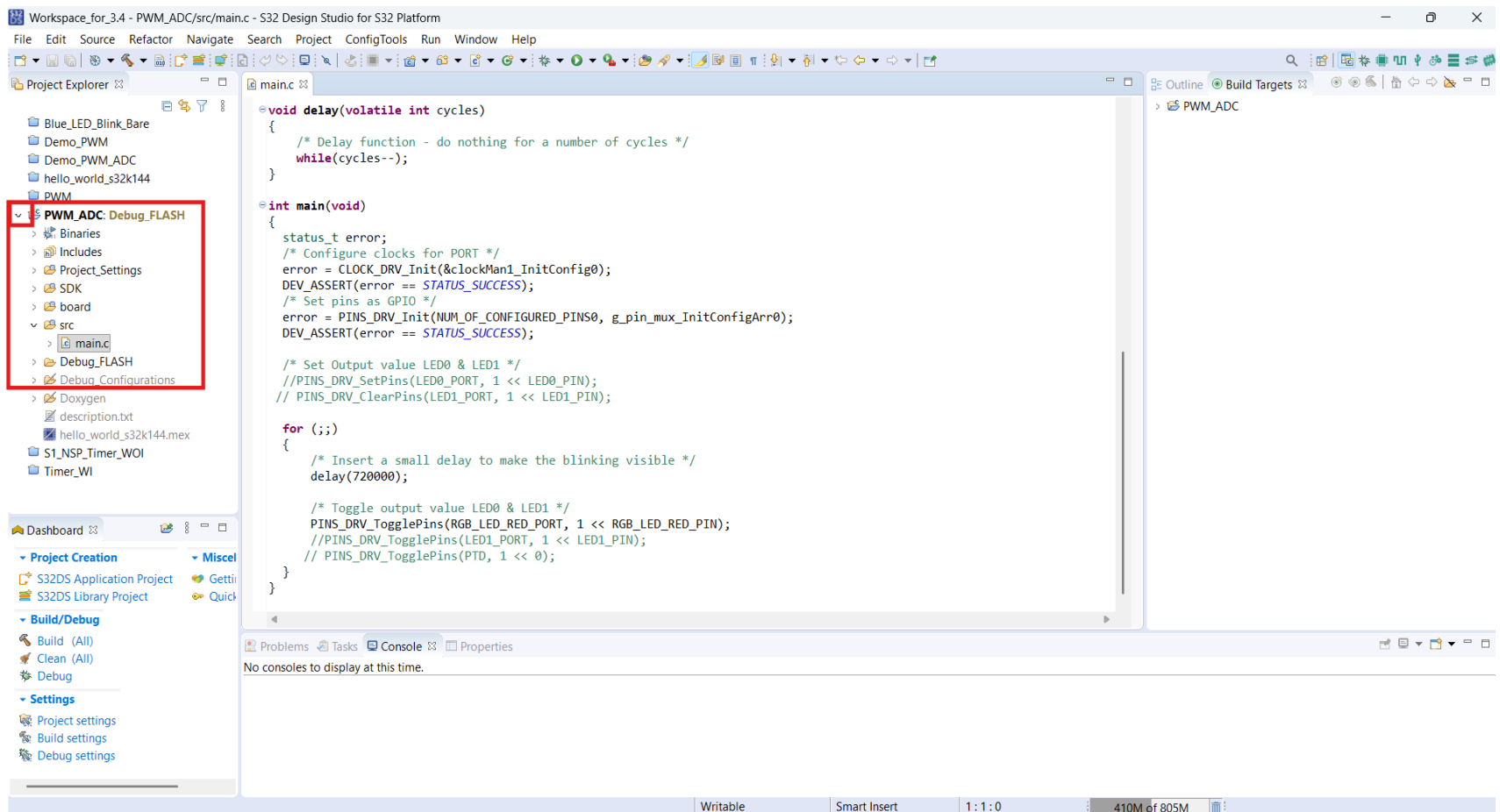
- **Expected Output**

- The ADC reading should change according to the PWM duty cycle. Higher duty cycle should produce a higher average voltage after filtering.

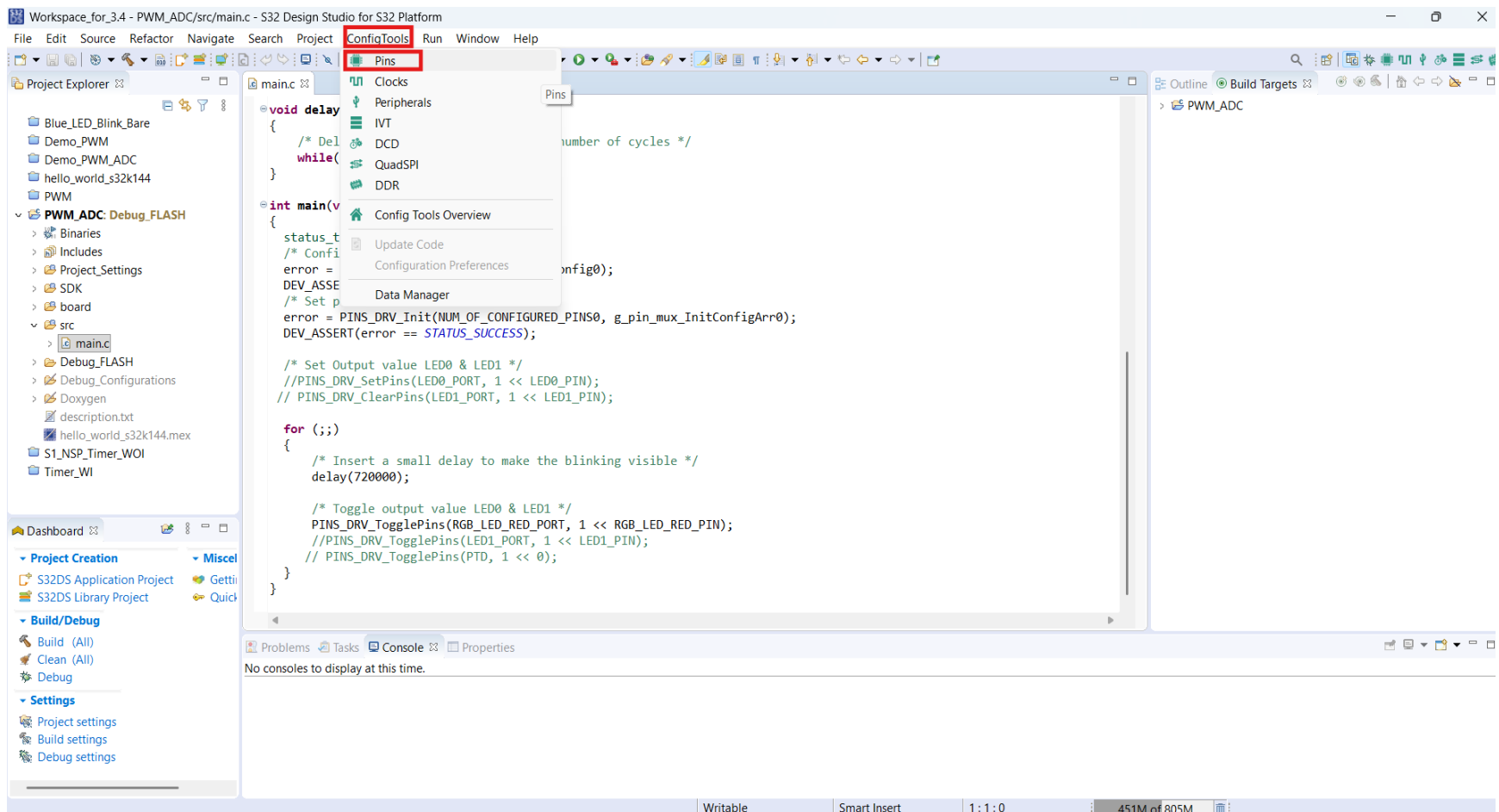
- Select hello_world_s32k144 project, copy and paste it and rename as PWM_ADC. Select PWM_ADC project and **Close Unrelated Projects**.



- Expand the project and open src→main.c file



- For Pin Routing Details: Click on ConfigTools→Pins



- Click on Pin 25, pop up window Pin[25] opens

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPi
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPi
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTB8			PTB8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPi
28	PTB4			PTB4	FTM0_CH4	LPSPi
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	
33	PTD5			PTD5	FTM2_CH3[...]	
34	PTD12			PTD12	FTM2_CH2	
35	PTD11			PTD11	FTM2_CH1[...]	

Pin [25] - X

All signals on pin [25]:

- ☐ FTM2_CH4 (FTM2,ch4)
- ☐ LPUART1_RX (LPUART1,rx)
- ☐ PTD13 (PORTD,port,13)
- ☐ RTC_CLKOUT (RTC,clkout)

Done

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 5

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi

PWM_ADC

565M of 814M

Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board/pin_mux.c

board/pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

- ## PWM ADC

Click on Done

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi
16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM0_CH1	LPSPi
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM0_CH0	LPSPi
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTB8			PTB8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPi
28	PTB4			PTB4	FTM0_CH4	LPSPi
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	
33	PTD5			PTD5	FTM2_CH3[...]	
34	PTD12			PTD12	FTM2_CH2	
35	PTD11			PTD11	FTM2_CH1	

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull :
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull

Pin [25]

All signals on pin [25]:

- ☒ FTM2_CH4 (FTM2, ch4)
- ☐ LPUART1_RX (LPUART1, rxd)
- ☐ PTD13 (PORTD, port, 13)
- ☐ RTC_CLKOUT (RTC, clkout)

Done

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

- ☒ Update code enabled
- board\pin_mux.c
- board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

PWM_ADC

378M of 776M

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code BOARD_InitPins

Pins Peripheral Signals Power Groups

Pin Pin name Label Identifier PORT FTM LPSPi

16	PTB6			PTB6		
17	PTE14			PTE14	FTM0_FLT1[...]	
18	PTE3			PTE3	FTM0_FLT0[...]	
19	PTE12			PTE12	FTM0_FLT3	
20	PTD17			PTD17	FTM0_FLT2	
21	PTD16	RGB_LED_GR...	RGB_LED_GR...	PTD16	FTM2_CH1	LPSPi
22	PTD15	RGB_LED_RED	RGB_LED_RED	PTD15	FTM2_CH0	LPSPi
23	PTE9			PTE9	FTM0_CH7	
24	PTD14			PTD14	FTM2_CH5	
25	PTD13			PTD13	FTM2_CH4	
26	PTE8			PTE8	FTM0_CH6	
27	PTB5			PTB5	FTM0_CH5	LPSPi
28	PTB4			PTB4	FTM0_CH4	LPSPi
29	PTC3			PTC3	FTM0_CH3	
30	PTC2			PTC2	FTM0_CH2	
31	PTD7			PTD7	FTM2_FLT3	
32	PTD6			PTD6	FTM2_FLT2	
33	PTD5			PTD5	FTM2_CH3[...]	
34	PTD12			PTD12	FTM2_CH2	
35	PTD11			PTD11	FTM2_CH11	

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull

S32K144_LQFP100 - LQFP 100 package

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board/pin_mux.c

board/pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue
-------	----------	-------

PWM_ADC

407M of 776M

- Click on Pin 54, pop up window Pin[54] opens

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin Pin name Label Identifier PORT FTM LPSPI

☐	46	PTC14		PTC14	FTM1_CH2	LPSPI
☐	47	PTB3		PTB3	FTM1_CH1[...]	LPSPI
☐	48	PTB2		PTB2	FTM1_CH0[...]	LPSPI
☐	49	PTC13		PTC13	FTM3_CH7[...]	
☐	50	PTC12		PTC12	FTM3_CH6[...]	
☐	51	PTC11		PTC11	FTM3_CH5	
☐	52	PTC10		PTC10	FTM3_CH4	
☐	53	PTB1		PTB1	TCLK0	LPSPI
☐	54	PTB0		PTB0		LPSPI
☐	55	PTC9		PTC9	FTM1_FLT1	
☐	56	PTC8		PTC8	FTM1_FLT0	
☐	57	PTA7		PTA7	FTM0_FLT2	
☐	58	PTA6		PTA6	FTM0_FLT1	LPSPI
☐	59	PTE7		PTE7	FTM0_CH7[...]	
☒	60	VSS_60				
☒	61	VDD_61				
☐	62	PTA17		PTA17	FTM0_CH6[...]	
☐	63	PTB17		PTB17	FTM0_CH5	LPSPI
☐	64	PTB16		PTB16	FTM0_CH4	LPSPI
☐	65	PTR15		PTR15	FTM0_CH3	LPSPI

Pin [54]

All signals on pin [54]:

- ☐ ADC0_SE4 (ADC0,se,4); routed by default
- ☐ ADC1_SE14 (ADC1,se,14); routed by default
- ☐ CAN0_RX (CAN0,rx,rx)
- ☐ LPSPI0_PCS0 (LPSPI0,pcs,0)
- ☐ LPTMR0_ALT3 (LPTMR0,alt,3)
- ☐ LPUART0_RX (LPUART0,rx)
- ☐ PTB0 (PORTB,port,0)

Done

Routing Details

Pins Signals type filter text

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull

PWM_ADC 446M of 776M

Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c
board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

- Select ADC0_SE4 and click on Done

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups Package

Pin name Label Identifier PORT FTM LPSPI

Pin [54]

All signals on pin [54]:

- ☒ ADC0_SE4 (ADC0,se,4); routed by default
- ☐ ADC1_SE14 (ADC1,se,14); routed by default
- ☐ CAN0_RX (CAN0,rx,rx)
- ☐ LPSPI0_PCS0 (LPSPI0,pcs,0)
- ☐ LPTMR0_ALT3 (LPTMR0,alt,3)
- ☐ LPUART0_RX (LPUART0,rx)
- ☐ PTB0 (PORTB,port,0)

Done

Routing Details

Pins Signals type filter text

Routing Details for BOARD... 7

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull :
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull

Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

type filter text

Level	Resource	Issue

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help

PWM_ADC Update Code Functional Group BOARD_InitPins

Pins Peripheral Signals Power Groups

type filter text

Pin	Pin name	Label	Identifier	PORT	FTM	LPSPi	ADC	LPUART
☐	46	PTC14		PTC14	FTM1_CH2	LPSPi2_PCS0	ADC0_SE12	
☐	47	PTB3		PTB3	FTM1_CH1[...]	LPSPi0_SIN	ADC0_SE7	
☐	48	PTB2		PTB2	FTM1_CH0[...]	LPSPi0_SCK	ADC0_SE6	
☐	49	PTC13		PTC13	FTM3_CH7[...]			LPUART2_RTS
☐	50	PTC12		PTC12	FTM3_CH6[...]			LPUART2_CTS
☐	51	PTC11		PTC11	FTM3_CH5			
☐	52	PTC10		PTC10	FTM3_CH4			
☐	53	PTB4		PTB4	FTM0	LPSPi0_SOUT	ADC0_SE5[...]	LPUART0_TX
☒	54	PTB0		PTB0	FTM0	LPSPi0_PCS0	ADC0_SE4[...]	LPUART0_RX
☐	55	PTC9		PTC9	FTM1_FLT1			LPUART1_TX[...]
☐	56	PTC8		PTC8	FTM1_FLT0			LPUART1_RX[...]
☐	57	PTA7		PTA7	FTM0_FLT2		ADC0_SE3	LPUART1_RTS
☐	58	PTA6		PTA6	FTM0_FLT1	LPSPi1_PCS1	ADC0_SE2	LPUART1_CTS
☐	59	PTE7		PTE7	FTM0_CH7[...]			
☒	60	VSS_60						
☒	61	VDD_61						
☐	62	PTA17		PTA17	FTM0_CH6[...]			
☐	63	PTB17		PTB17	FTM0_CH5	LPSPi1_PCS3		

Package

S32K144_LQFP100 - LQFP 100 package

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

☒ Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Routing Details

Pins Signals

Routing Details for BOARD... 7

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
54	ADC0	se, 4	->	[54] PTB0		n/a		Input	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi

PWM_ADC

340M of 797M

- For Peripheral Configuration: Select ConfigTools→Peripherals

Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project **ConfigTools** Pins Run Window Help

Pins Peripheral Signals Power Groups

Pin Pin name Label Identifier PORT

46 PTC14 PTC14

47 PTB3 PTB3

48 PTB2 PTB2

49 PTC13 PTC13

50 PTC12 PTC12

51 PTC11 PTC11

52 PTC10 PTC10

53 PTB1 PTB1

54 **PTB0** PTB0

55 PTC9 PTC9 FTM1_F

56 PTC8 PTC8 FTM1_F

57 PTA7 PTA7 FTM0_F

58 PTA6 PTA6 FTM0_F

59 PTE7 PTE7 FTM0_C

60 VSS_60

61 VDD_61

62 PTA17 PTA17 FTM0_C

63 PTB17 PTB17 FTM0_C

Peripherals

ADC0 ADC1 CAN0

CAN1 CAN2 CMP0

EWM FLEW0 FTM

FTM0 FTM1 FTM2

FTM0 JTAG LPSPI0

LPSPI1 LPSPI2

LPUART0 LPUART1

LPUART2 OSC PORTA

PORTB PORTC PORTD

ORTE Platform PowerAndGr

RTC SWD TRGMUX

S32K144_LQFP100 - LQFP 100 package

Routing Details

Pins Signals

Routing Details for BOARD... 7

#	Peripheral	Signal	Arrow	Routed pin/signal	Label	Identifier	Power group	Direction	Interrupt Status	Interrupt Configuration	Lock Register	Pull Enable	Pull Sel
40	PORTC	port, 0	->	[40] PTC0		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
39	PORTC	port, 1	->	[39] PTC1		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
22	PORTD	port, 15	->	[22] PTD15	RGB_LED_RED	RGB_LED_RED		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
21	PORTD	port, 16	->	[21] PTD16	RGB_LED_GREEN	RGB_LED_GREEN		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
4	PORTD	port, 0	->	[4] PTD0	RGB_LED_BLUE	RGB_LED_BLUE		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
25	FTM2	ch, 4	->	[25] PTD13		n/a		Output	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi
54	ADC0	se, 4	->	[54] PTB0		n/a		Input	Don't modify	Interrupt Status Flag (ISF) is disabled	Unlocked	Disabled	Pull Doi

Overview Code Preview Registers

Configuration - General Info

Configuration - HW Info

Processor: S32K144

Part number: S32K144_LQFP100

Core: Cortex-M4

SDK Version: s32sdk_s32k1xx_rtm_403

Project

Pins

Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.

Generated code

Update code enabled

board\pin_mux.c

board\pin_mux.h

Functional groups

BOARD_InitPins

Other tools

Problems

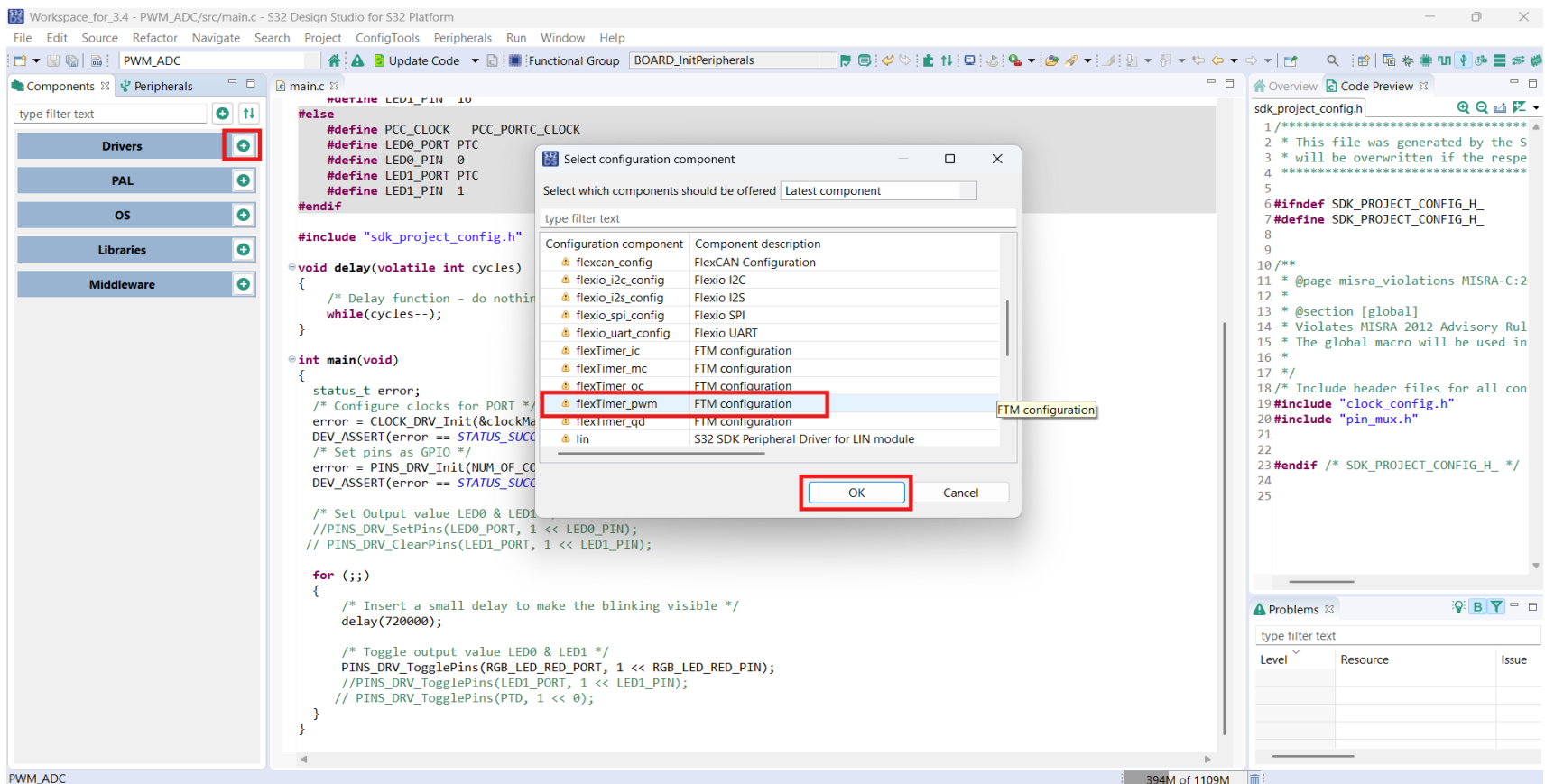
type filter text

Level Resource Issue

PWM_ADC

446M of 797M

- Click on Add Drivers, new pop-up window opens. Select FlexTimer_pwm and Click on OK



- Check settings as below for FTM for Initialization tab

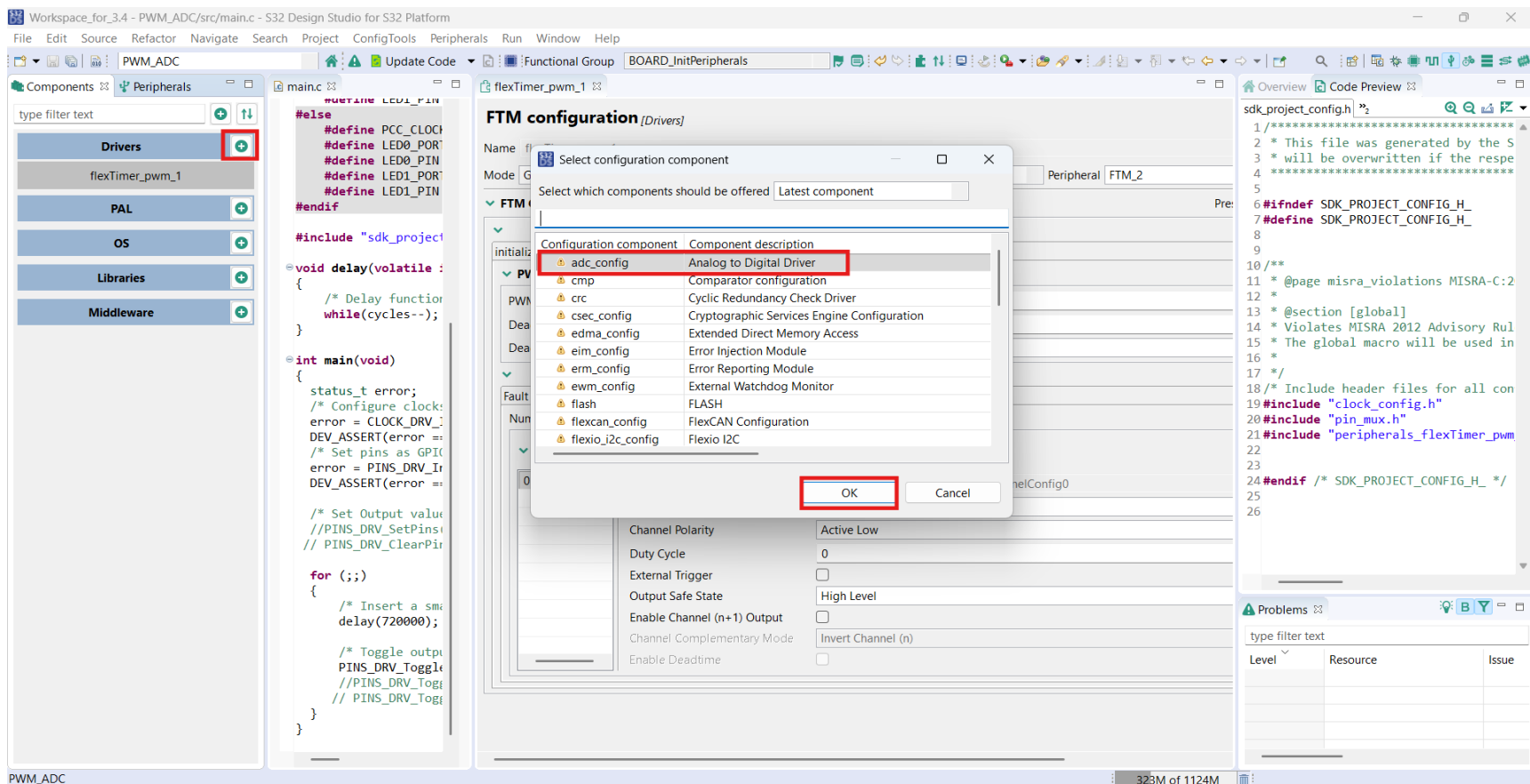
The screenshot displays the S32 Design Studio IDE interface for configuring the FTM (FlexTimer Module) for the flexTimer_pwm_1 peripheral. The 'Initialization' tab is active, showing the following configuration details:

- FTM configuration [Drivers]**
 - Name: flexTimer_pwm_1
 - Mode: General Mode
 - Peripheral: FTM_2
- FTM Configuration Set**
 - Initialization
- FTM Clock Configuration**
 - FTM Clock Source: System clock
 - Frequency of source clock (Hz): 48000000
 - Clock Source Prescaler: 1
 - Frequency of counter clock (Hz): 48000000
- FTM Global Configuration**
 - Mode Of Operation For FTM: Edge Aligned PWM
 - Debug Mode: Mode 3
 - Timer Overflow Interrupt: ☐
 - Initialization Trigger: ☐
- FTM Register Sync Configuration**
 - Software Trigger: ☒
 - Hardware Trigger 1: ☐
 - Hardware Trigger 2: ☐
 - Hardware Trigger 3: ☐
 - Sync Point: Immediate
 - Max Loading Point: ☐
 - Min Loading Point: ☐
 - Automatic Trigger Clear: ☐
 - Inverting Control Synchronization: PWM Synchronization
 - Software Output Control Synchronization: PWM Synchronization
 - Output Mask Synchronization: PWM Synchronization

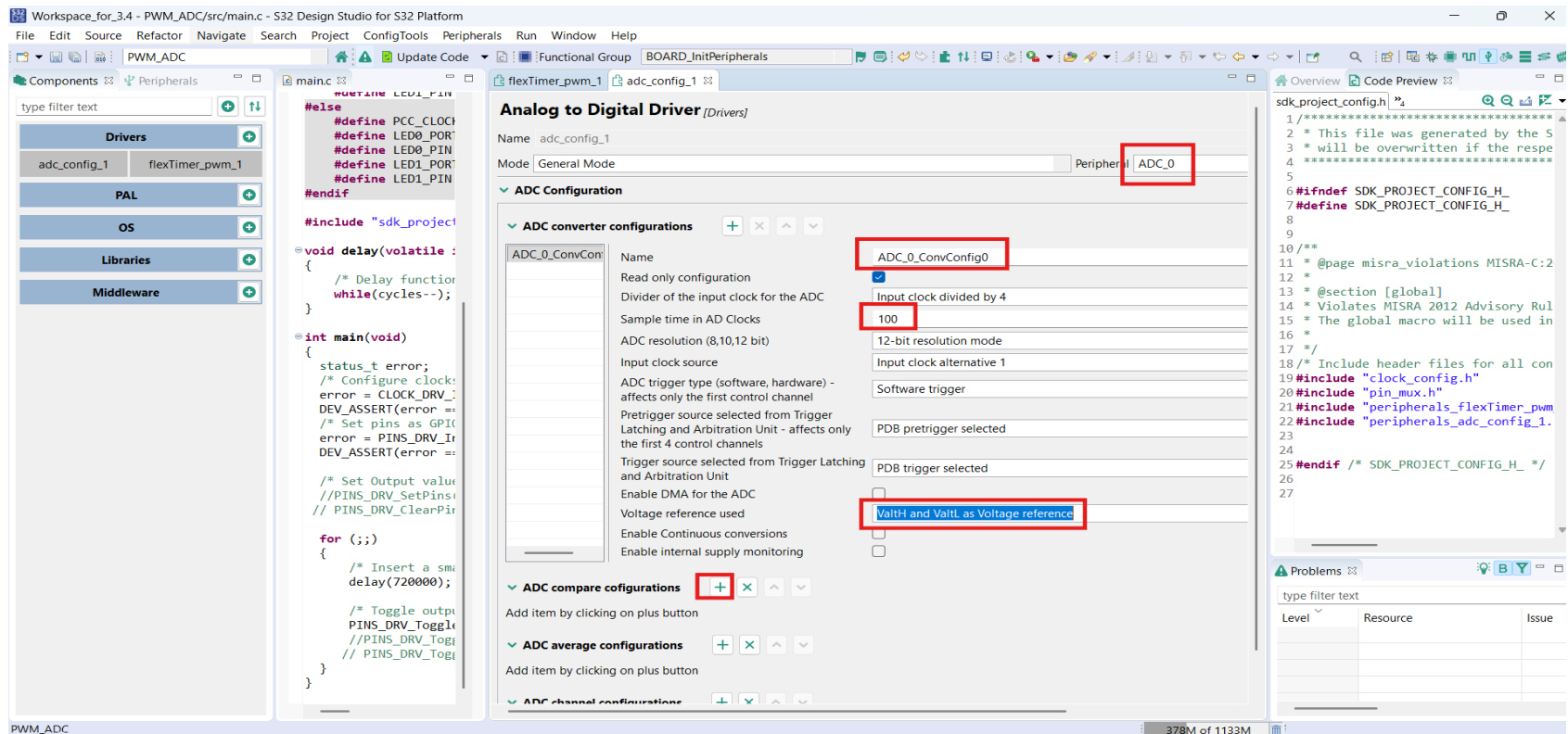
The left sidebar shows the project structure with 'main.c' open, displaying C code for PWM initialization. The right sidebar shows the 'Overview' and 'Code Preview' tabs, with 'sdk_project_config.h' open, showing project configuration macros.

-

- Click on Add Drivers, new pop-up window opens. Select adc_config and Click on OK



- Make changes as below for ADC converter configurations and Click on + (Add) button of ADC compare configurations, ADC average configurations and ADC channel configurations



- Make changes as below for ADC compare configurations, ADC average configurations and ADC channel configurations

The screenshot displays the S32 Design Studio IDE interface for a project named 'Workspace_for_3.4 - PWM_ADC/src/main.c'. The main window shows the C source code for 'main.c', which includes headers, defines, and a main function. The 'main' function configures the clock, sets pins as GPIO, and sets output values. The 'Components' pane on the left shows the project structure with 'adc_config_1' and 'flexTimer_pwm_1' selected. The 'ConfigTools' pane on the right shows the configuration for 'adc_config_1'. The 'ADC compare configurations' section shows 'ADC_0_HwCompConfig0' with 'Read only configuration' checked. The 'ADC average configurations' section shows 'ADC_0_HwAvgConfig0' with 'Read only configuration' checked and 'Selection for number of samples used for averaging' set to 'Hardware average of 32 samples'. The 'ADC channel configurations' section shows 'ADC_0_ChnConfig0' with 'Read only configuration' checked and 'Selection of input channel for measurement' set to 'External input channel 4'. The 'Overview' pane on the right shows the project configuration file 'sdk_project_config.h' with various macros defined.

main.c

```
#ifndef LED1_PIN
#define LED1_PIN
#endif

#include "sdk_project_config.h"

void delay(volatile unsigned int cycles) {
    while(cycles--);
}

int main(void) {
    status_t error;
    /* Configure clock */
    error = CLOCK_DRV_Init(CLOCK_DRV_InitConfig, NULL, 0);
    DEV_ASSERT(error == kStatus_Success);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(1, &PINS_DRV_InitConfig, NULL);
    DEV_ASSERT(error == kStatus_Success);

    /* Set Output value */
    //PINS_DRV_SetPins(PINS_DRV_SetPinsConfig);
    //PINS_DRV_ClearPins(PINS_DRV_ClearPinsConfig);

    for(;;) {
        /* Insert a small delay */
        delay(720000);

        /* Toggle output */
        PINS_DRV_Toggle(PINS_DRV_ToggleConfig);
        //PINS_DRV_Toggle(PINS_DRV_ToggleConfig);
    }
}
```

ADC compare configurations

Name	Read only configuration	Enable the compare feature	Enable Greater-Than functionality	Enable Range functionality	First Compare Value	Second Compare Value
ADC_0_HwCompConfig0	☒	☐	☐	☐	0	0

ADC average configurations

Name	Read only configuration	Enable averaging functionality	Selection for number of samples used for averaging
ADC_0_HwAvgConfig0	☒	☒	Hardware average of 32 samples

ADC channel configurations

Name	Read only configuration	Enable interrupts for this channel	Selection of input channel for measurement
ADC_0_ChnConfig0	☐	☐	External input channel 4

sdk_project_config.h

```
1/*****
2 * This file was generated by the S
3 * will be overwritten if the respo
4 *****/
5
6 #ifndef SDK_PROJECT_CONFIG_H
7 #define SDK_PROJECT_CONFIG_H
8
9
10 /**
11 * @page misra_violations MISRA-C:2
12 *
13 * @section [global]
14 * Violates MISRA 2012 Advisory Rul
15 * The global macro will be used in
16 *
17 */
18 /* Include header files for all con
19 #include "clock_config.h"
20 #include "pin_mux.h"
21 #include "peripherals_flexTimer_pwm
22 #include "peripherals_adc_config_1.
23
24
25 #endif /* SDK_PROJECT_CONFIG_H */
26
27
```

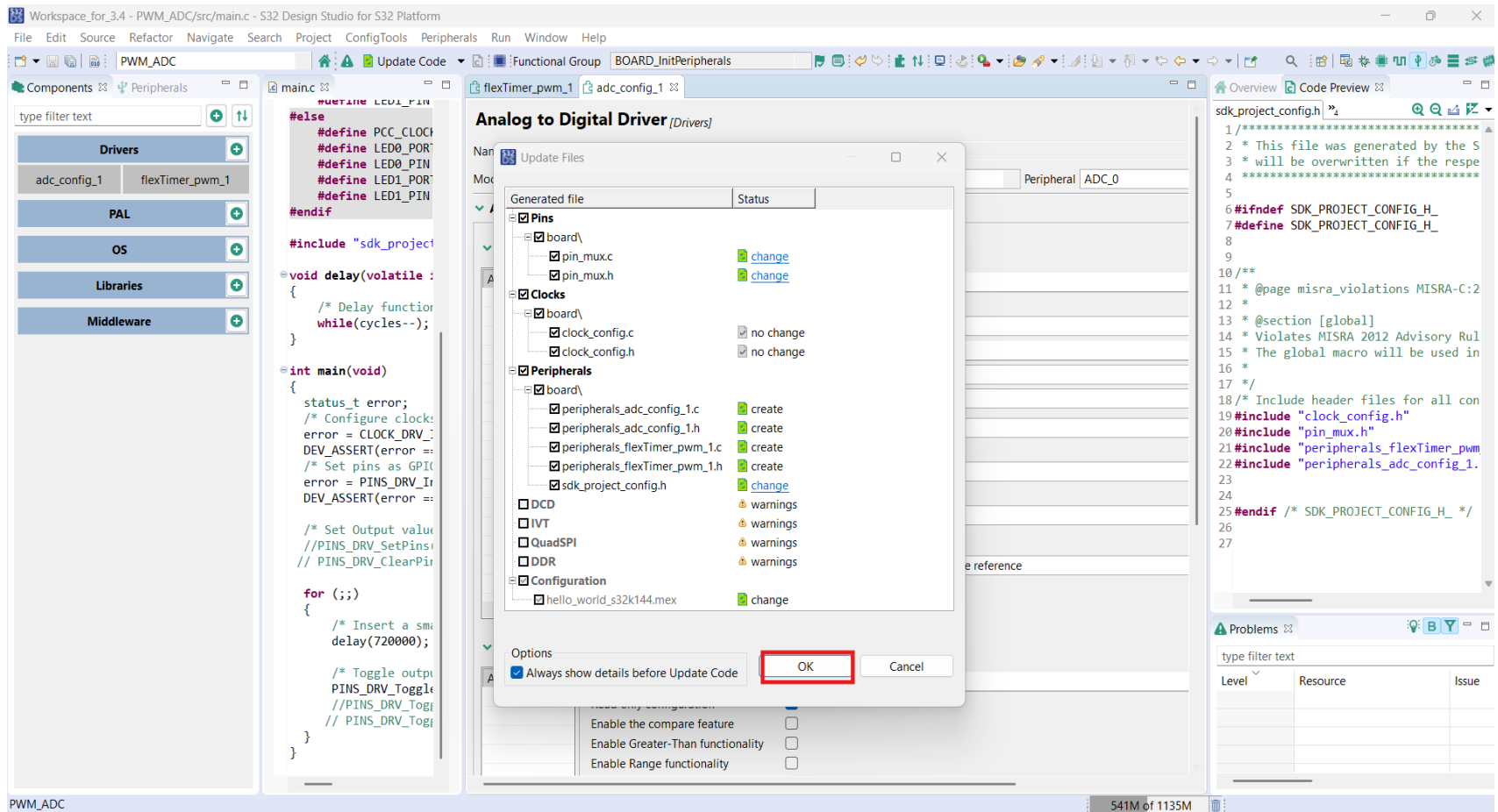
- Click on **Update Code**

The screenshot shows the S32 Design Studio for S32 Platform IDE. The top toolbar has the 'Update Code' button highlighted with a red box. The main workspace is divided into several panes:

- Components:** Shows a tree view with 'Drivers' containing 'adc_config_1' and 'flexTimer_pwm_1', and 'PAL'.
- main.c:** Displays C code for a delay function and a main function that configures pins and calls a delay.
- Analog to Digital Driver [Drivers]:** Shows the configuration for 'adc_config_1'. It includes sections for 'ADC converter configurations' and 'ADC compare configurations'. The 'ADC converter configurations' section is expanded, showing details for 'ADC_0_ConvConfig0' such as input clock division, sample time, resolution, and trigger settings.
- Code Preview:** Shows the 'sdk_project_config.h' file with various macros and includes.
- Problems:** A table at the bottom right showing any compilation errors or warnings.

The status bar at the bottom indicates 'PWM_ADC' and '484M of 1135M'.

- Click on OK



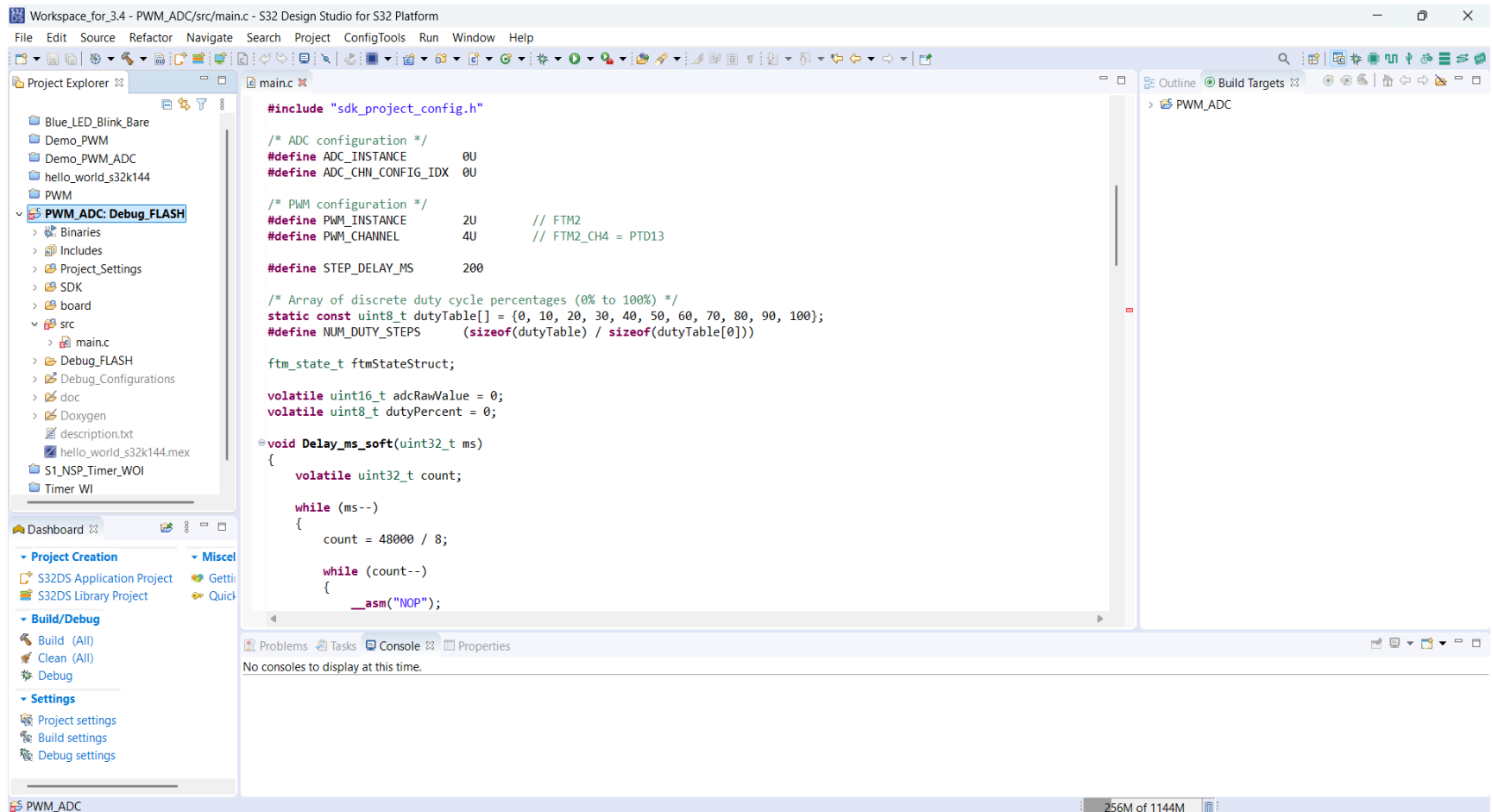
- After update click on C/C++ to change perspective

The screenshot displays the S32 Design Studio for S32 Platform IDE. The main window is divided into several panes:

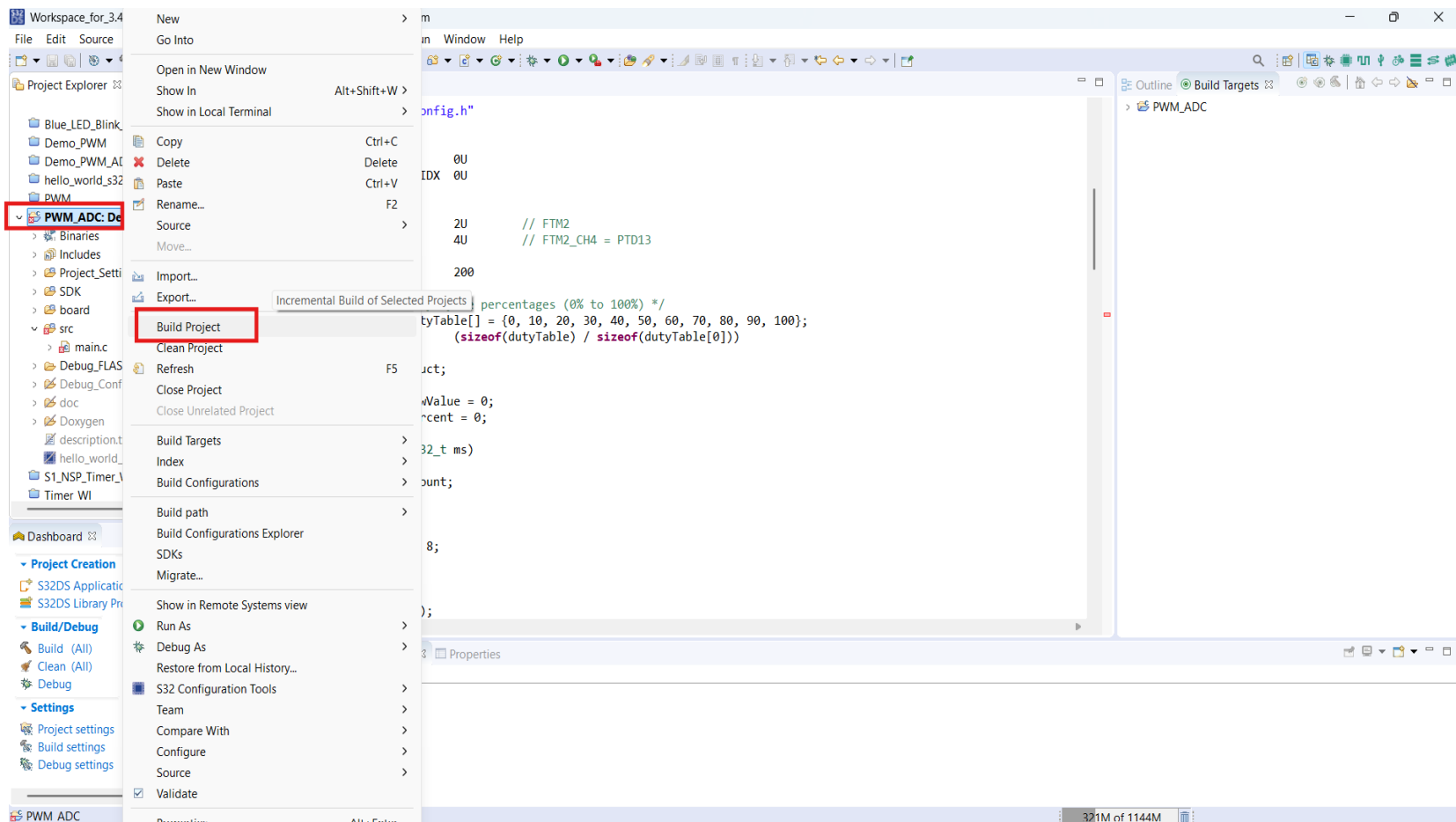
- Left Pane:** Contains a 'Components' view with a search filter 'type filter text'. Below it are categories: Drivers, PAL, OS, Libraries, and Middleware. The 'Drivers' category is expanded, showing 'adc_config_1' and 'flexTimer_pwm_1'.
- Top Pane:** Shows the 'main.c' file with C code. The code includes definitions for PCC_CLOCK, LED0_PORT, LED0_PIN, LED1_PORT, and LED1_PIN, followed by an include for 'sdk_project_config.h'. The code also contains a 'delay' function and a 'main' function that configures the clock, sets pins, and toggles the output.
- Right Pane:** Displays the 'Analog to Digital Driver' configuration for 'adc_config_1'. The configuration is set to 'General Mode' and 'Peripheral: ADC_0'. It shows 'ADC converter configurations' and 'ADC compare configurations'.
- Bottom Right Pane:** Shows the 'Code Preview' of 'sdk_project_config.h', which includes MISRA-C:2 compliance headers and project-specific includes like 'clock_config.h', 'pin_mux.h', 'peripherals_flexTimer_pwm', and 'peripherals_adc_config_1'.

A red arrow points to the 'C/C++' icon in the top right toolbar, indicating the action to switch to the C/C++ perspective.

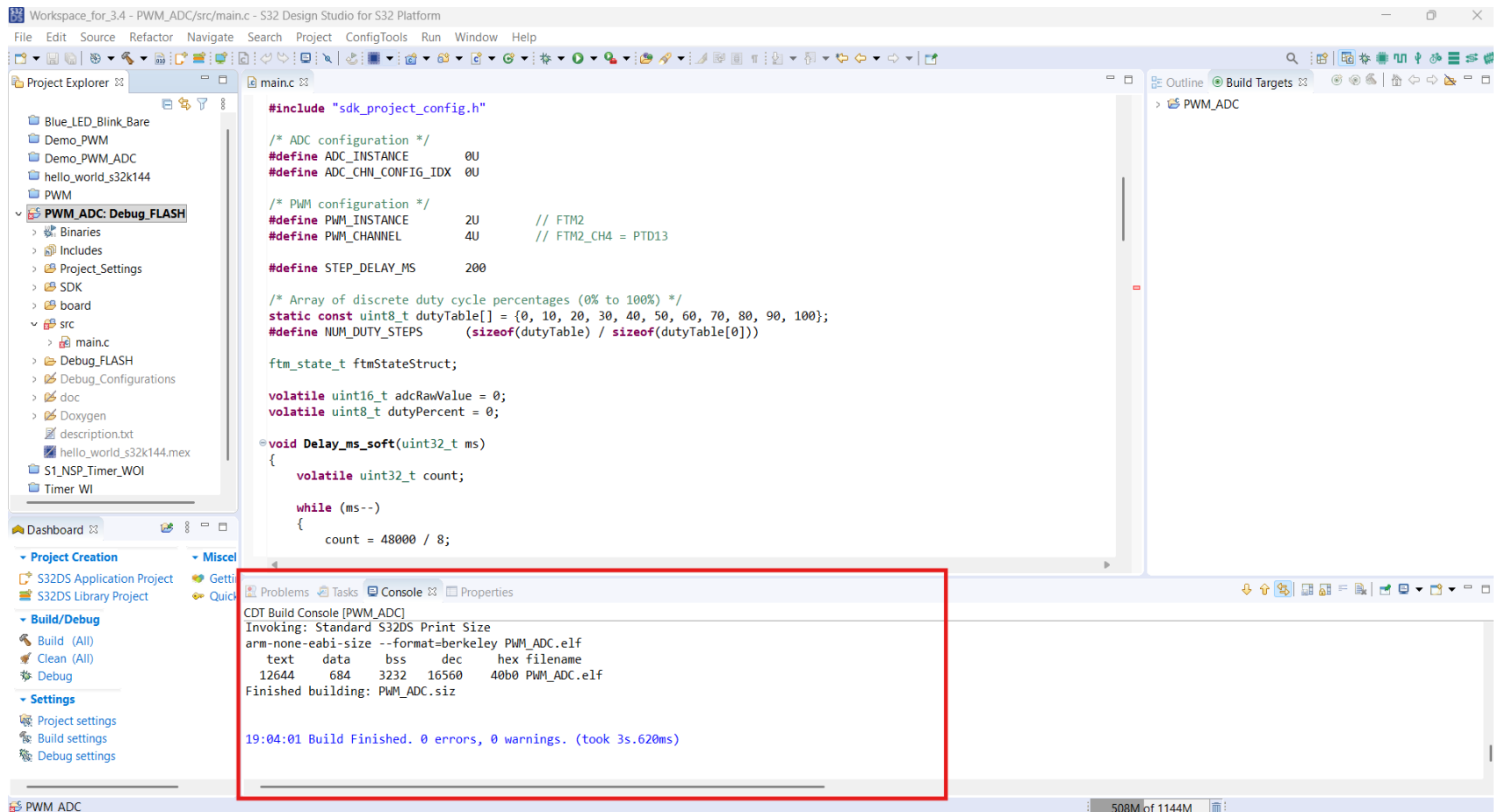
- C/C++ perspective and write c program



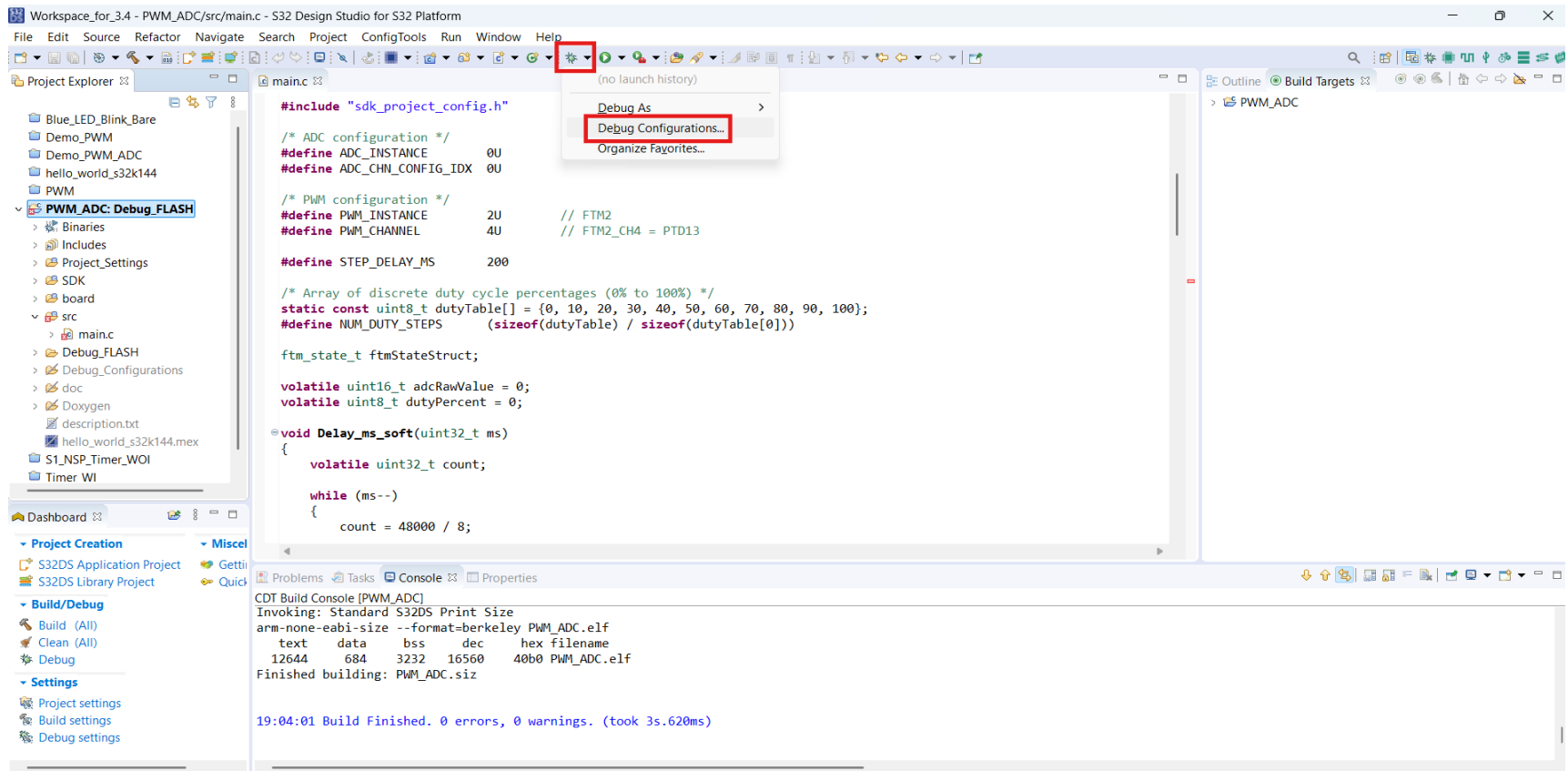
- Right click on project and click on Build Project



- Check for any errors



- Connect S32K144 development board to PC/Laptop
- Click on Debug→Debug Configurations



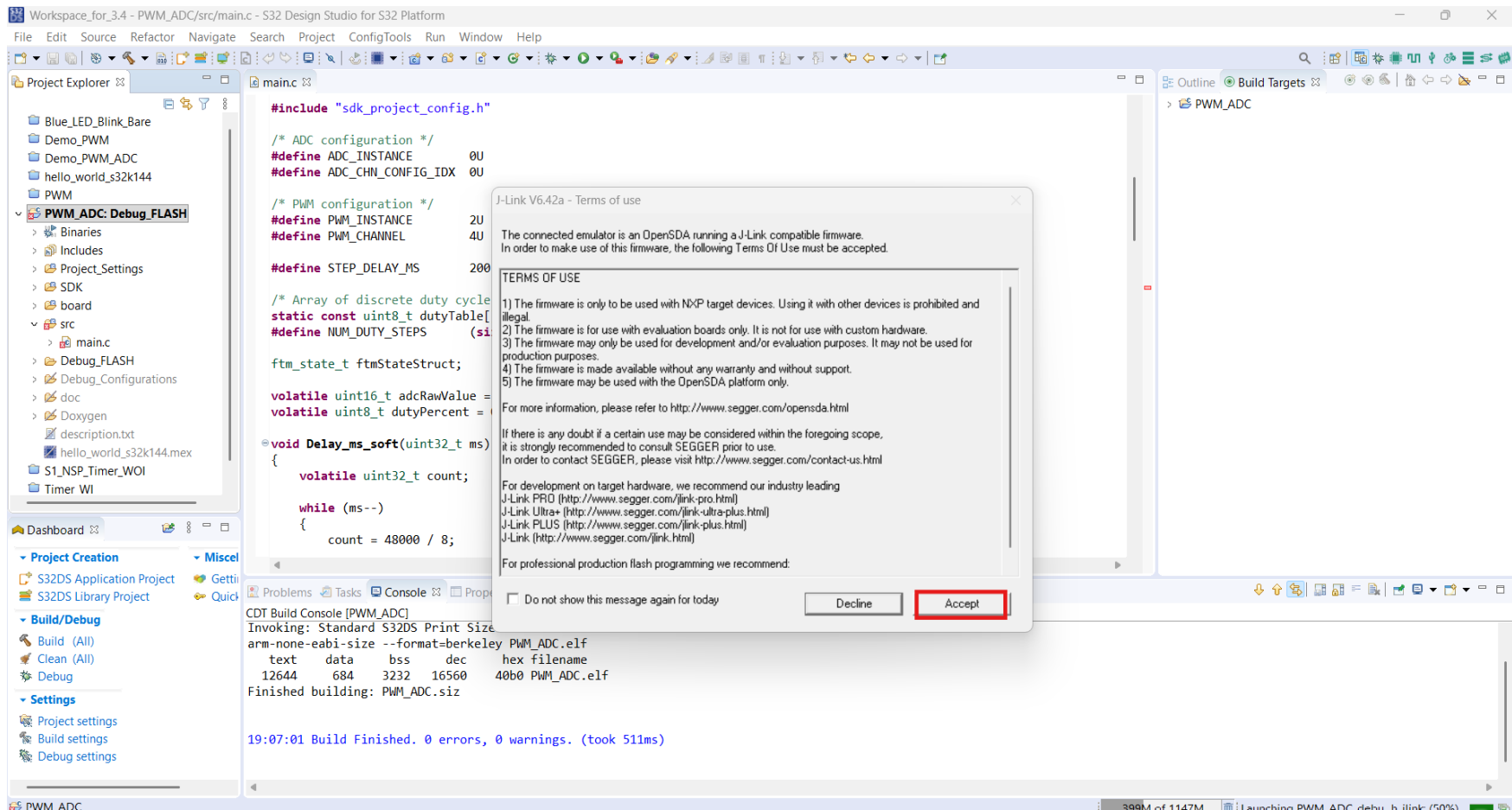
- Do configurations as below and Click on **Debug**

The screenshot shows the Eclipse IDE interface. On the left, the Project Explorer displays a project named 'PWM_ADC: Debug_FLASH'. The 'Debug Configurations' window is open, showing a list of configurations on the left and the configuration details on the right. The configuration 'PWM_ADC_debug_flash_jlink' is selected and highlighted with a red box. The configuration details on the right are as follows:

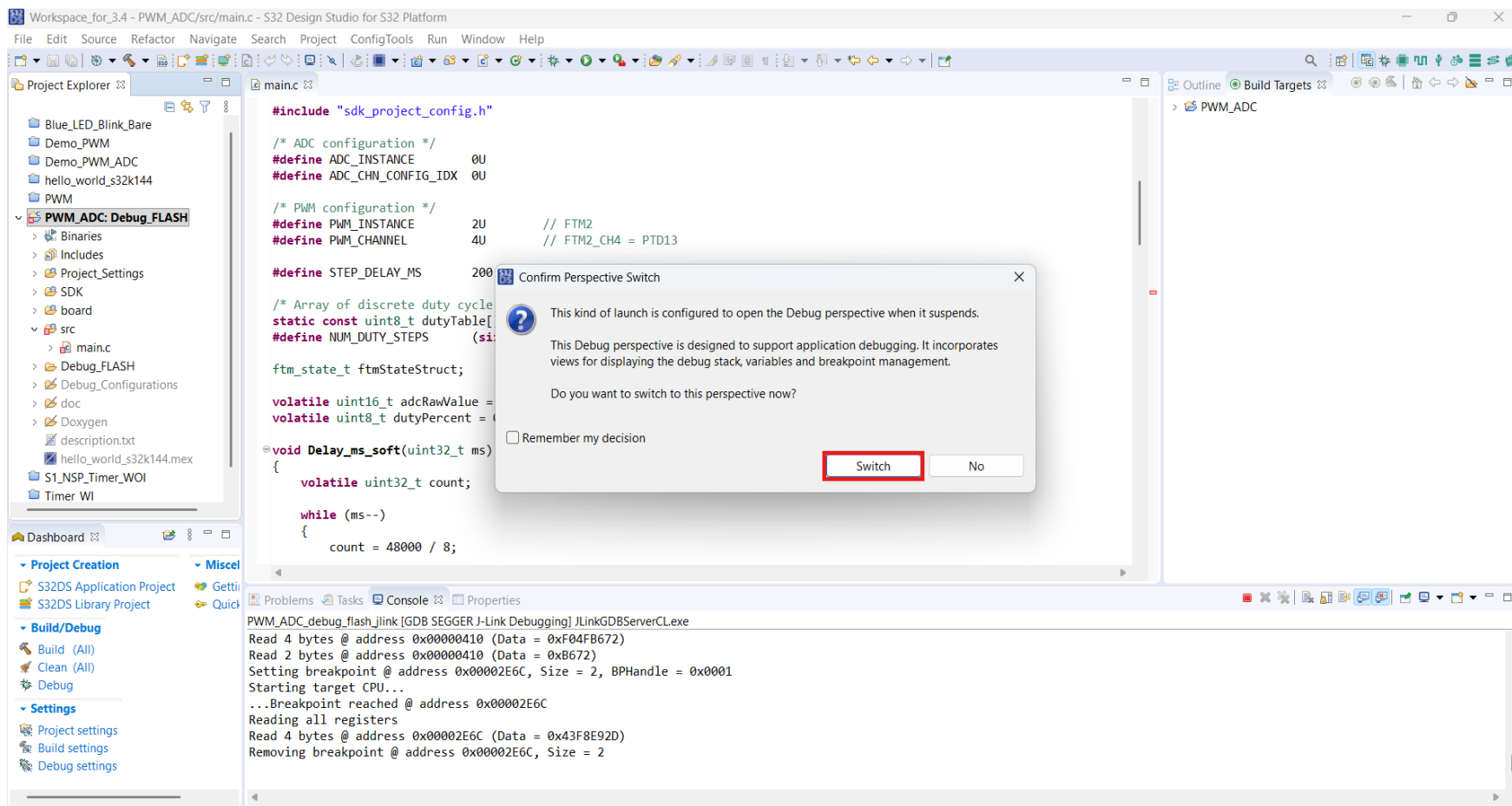
- Name: PWM_ADC_debug_flash_jlink
- Debugger: J-Link GDB Server Setup (highlighted with a red box)
- Start the J-Link GDB server locally: ☒ (Connect to running target: ☐)
- Executable: \${jlink_path}/\${jlink_gdbserver} (Browse... Variables...)
- Device name: S32K144 (Supported device names)
- Endianness: ☒ Little ☐ Big
- Connection: ☒ USB ☐ IP (USB serial or IP name/address)
- Interface: ☐ SWD ☒ JTAG (highlighted with a red box)
- Initial speed: ☐ Auto ☐ Adaptive ☒ Fixed 1000 kHz
- GDB port: 2331
- SWO port: 2332 (Verify downloads: ☒ Initialize registers on start: ☒)
- Telnet port: 2333 (Local host only: ☒ Silent: ☐)
- Log file: (Browse...)
- Other options: -singlesrun -strict -timeout 0 -nogui
- Allocate console for the GDB server: ☒ Allocate console for semihosting and SWO: ☒
- GDB Client Setup
- Executable: \${S32DS_ARM32_TOOLCHAIN_DIR}/bin/\${arm32_cross_prefix}gdb\${arm32_cross_prefix} (Browse... Variables...)
- Other options: (empty)
- Commands: set mem inaccessible-by-default off

At the bottom of the window, there are buttons for 'Revert', 'Apply', 'Debug' (highlighted with a red box), and 'Close'. The status bar at the bottom shows '19:04:01 Build Finished. 0 errors, 0 warnings. (took 3s.620ms)'.

- Click on Accept



- Click on Switch, changes to Debug perspective



- Add breakpoints as below and add variables in Expressions tab

The screenshot shows the S32 Design Studio IDE with a C program open in the main editor. The program is located at `Workspace_for_3.4 - PWM_ADC/src/main.c`. The code includes comments and function calls for PWM duty cycle control. Two breakpoints are set in the code: one at the line `if (adcRawValue > 2048)` and another at the `else` statement. The Expressions tab on the right shows a table with two entries: `dutyPercent` of type `volatile uint8_t` with value `0 \0'`, and `adcRawValue` of type `volatile uint16_t` with value `0`. The Outline tab on the right shows the project structure and a list of variables and functions. The Console tab at the bottom shows the output of the debugger, including register values and memory addresses.

main.c

```

dutyPercent = dutyTable[stepIndex];

/* Update PWM hardware channel */
PWM_SetDuty(dutyPercent);

/* Wait for RC low-pass filter to settle */
Delay_ms_soft(STEP_DELAY_MS);

/* Read filtered analog voltage */
adcRawValue = ADC_ReadValue();

/* Green LED threshold indicator (mid-scale > 2048) */
if (adcRawValue > 2048)
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      0); // LED ON (Active-Low)
}
else
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      1); // LED OFF (Active-Low)
}

/* Navigate through the steps: 0% -> 100% -> 0% */

```

Expressions

Expression	Type	Value
dutyPercent	volatile uint8_t	0 \0'
adcRawValue	volatile uint16_t	0
+ Add new expression		

Outline

- # PWM_CHANNEL
- # STEP_DELAY_MS
- du^ctyTable: const uint8_t[]
- # NUM_DUTY_STEPS
- ftmStateStruct: ftm_state_t
- ▼ adcRawValue: volatile uint16_t
- ▼ dutyPercent: volatile uint8_t
- Delay_ms_soft(uint32_t): void
- PWM_SetDuty(uint8_t): void
- ADC_ReadValue(void): uint16_t
- main(void): int

Console

PWM_ADC_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCL.exe

Reading all registers

Read 4 bytes @ address 0x00002E6C (Data = 0x43F8E92D)

Removing breakpoint @ address 0x00002E6C, Size = 2

Read 1 bytes @ address 0x20000022 (Data = 0x00)

Read 2 bytes @ address 0x20000020 (Data = 0x0000)

Read 2 bytes @ address 0x00002E70 (Data = 0x4827)

Read 4 bytes @ address 0x00002F10 (Data = 0x1FFF84A0)

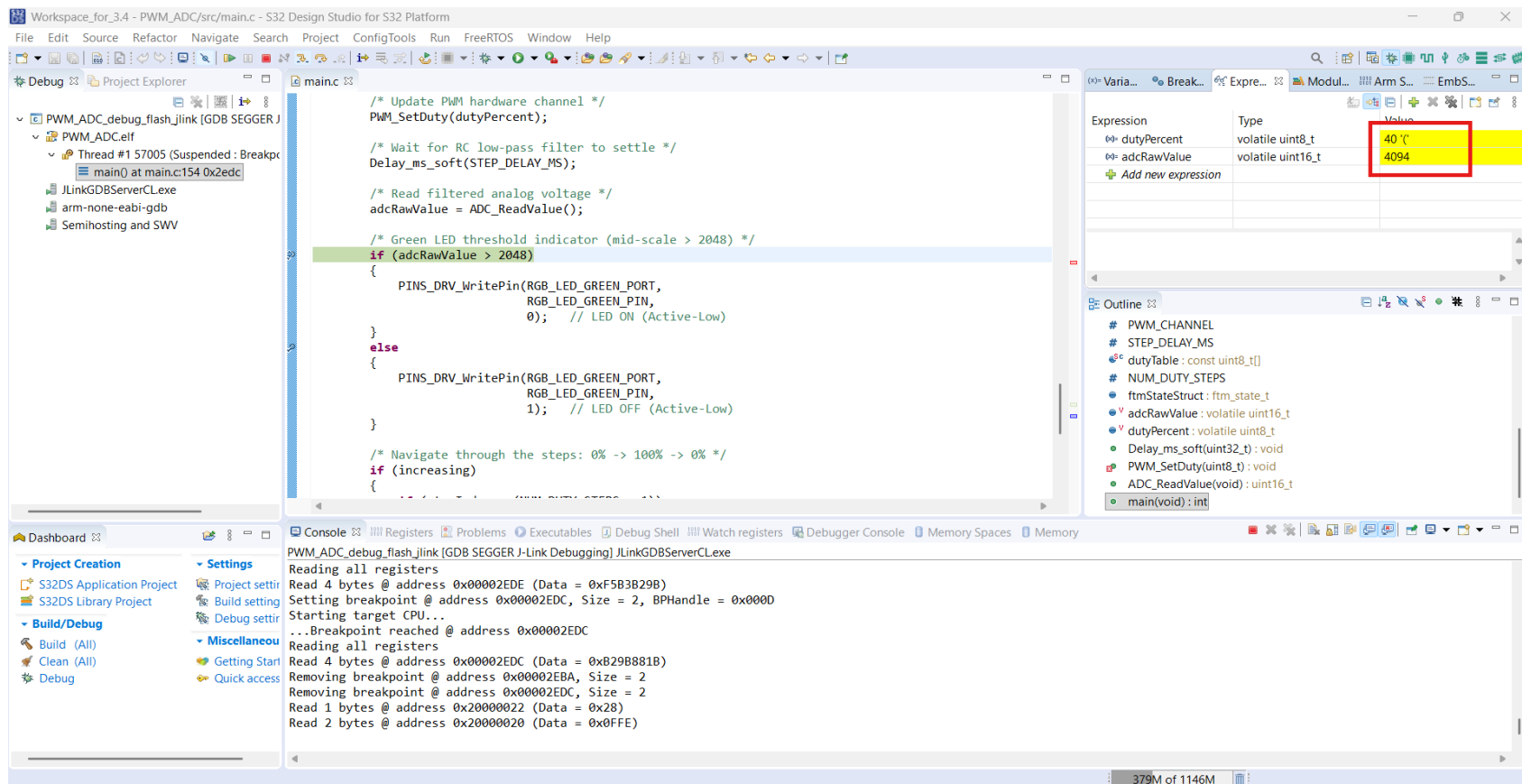
Read 2 bytes @ address 0x00002EDC (Data = 0x881B)

Read 2 bytes @ address 0x00002E70 (Data = 0x4827)

Read 4 bytes @ address 0x00002F10 (Data = 0x1FFF84A0)

Read 2 bytes @ address 0x00002EBA (Data = 0x4F1C)

- To run code on board, click on Resume and see variable values and output on board



Workspace_for_3.4 - PWM_ADC/src/main.c - S32 Design Studio for S32 Platform

File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help

Debug Project Explorer

PWM_ADC_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCLExe
 PWM_ADC.elf
 Thread #1 57005 (Suspended): Breakpoint at main.c:154 0x2edc
 JLinkGDBServerCLExe
 arm-none-eabi-gdb
 Semihosting and SWV

```

/* Update PWM hardware channel */
PWM_SetDuty(dutyPercent);

/* Wait for RC low-pass filter to settle */
Delay_ms_soft(STEP_DELAY_MS);

/* Read filtered analog voltage */
adcRawValue = ADC_ReadValue();

/* Green LED threshold indicator (mid-scale > 2048) */
if (adcRawValue > 2048)
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      0); // LED ON (Active-Low)
}
else
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      1); // LED OFF (Active-Low)
}

/* Navigate through the steps: 0% -> 100% -> 0% */
if (increasing)
{

```

Variable View:

Expression	Type	Value
dutyPercent	volatile uint8_t	40
adcRawValue	volatile uint16_t	4094

Outline:

- # PWM_CHANNEL
- # STEP_DELAY_MS
- dutyTable: const uint8_t[]
- # NUM_DUTY_STEPS
- ftmStateStruct: ftm_state_t
- adcRawValue: volatile uint16_t
- dutyPercent: volatile uint8_t
- Delay_ms_soft(uint32_t): void
- PWM_SetDuty(uint8_t): void
- ADC_ReadValue(void): uint16_t
- main(void): int

Dashboard:

- Project Creation
 - S32DS Application Project
 - S32DS Library Project
- Build/Debug
 - Build (All)
 - Clean (All)
 - Debug
- Settings
 - Project settir
 - Build settir
 - Debug settir
- Miscellaneous
 - Getting Start
 - Quick access

Console:

```

PWM_ADC_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCLExe
Reading all registers
Read 4 bytes @ address 0x00002EDE (Data = 0xF5B3B298)
Setting breakpoint @ address 0x00002EDC, Size = 2, BPHandle = 0x0000
Starting target CPU...
...Breakpoint reached @ address 0x00002EDC
Reading all registers
Read 4 bytes @ address 0x00002EDC (Data = 0xB29B881B)
Removing breakpoint @ address 0x00002EBA, Size = 2
Removing breakpoint @ address 0x00002EDC, Size = 2
Read 1 bytes @ address 0x20000022 (Data = 0x28)
Read 2 bytes @ address 0x20000020 (Data = 0x0FFE)

```

379M of 1146M

- To run code on board, click on Resume and see variable values and output on board

The screenshot displays the S32 Design Studio IDE interface for a project named 'Workspace_for_3.4 - PWM_ADC/src/main.c'. The code in the editor is as follows:

```

/* Update PWM hardware channel */
PWM_SetDuty(dutyPercent);

/* Wait for RC low-pass filter to settle */
Delay_ms_soft(STEP_DELAY_MS);

/* Read filtered analog voltage */
adcRawValue = ADC_ReadValue();

/* Green LED threshold indicator (mid-scale > 2048) */
if (adcRawValue > 2048)
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      0); // LED ON (Active-Low)
}
else
{
    PINS_DRV_WritePin(RGB_LED_GREEN_PORT,
                      RGB_LED_GREEN_PIN,
                      1); // LED OFF (Active-Low)
}

/* Navigate through the steps: 0% -> 100% -> 0% */
if (increasing)
{

```

The debugger is set to a breakpoint at address 0x2edc in the `main()` function. The 'Variables' pane on the right shows the following variables:

Expression	Type	Value
dutyPercent	volatile uint8_t	90
adcRawValue	volatile uint16_t	0

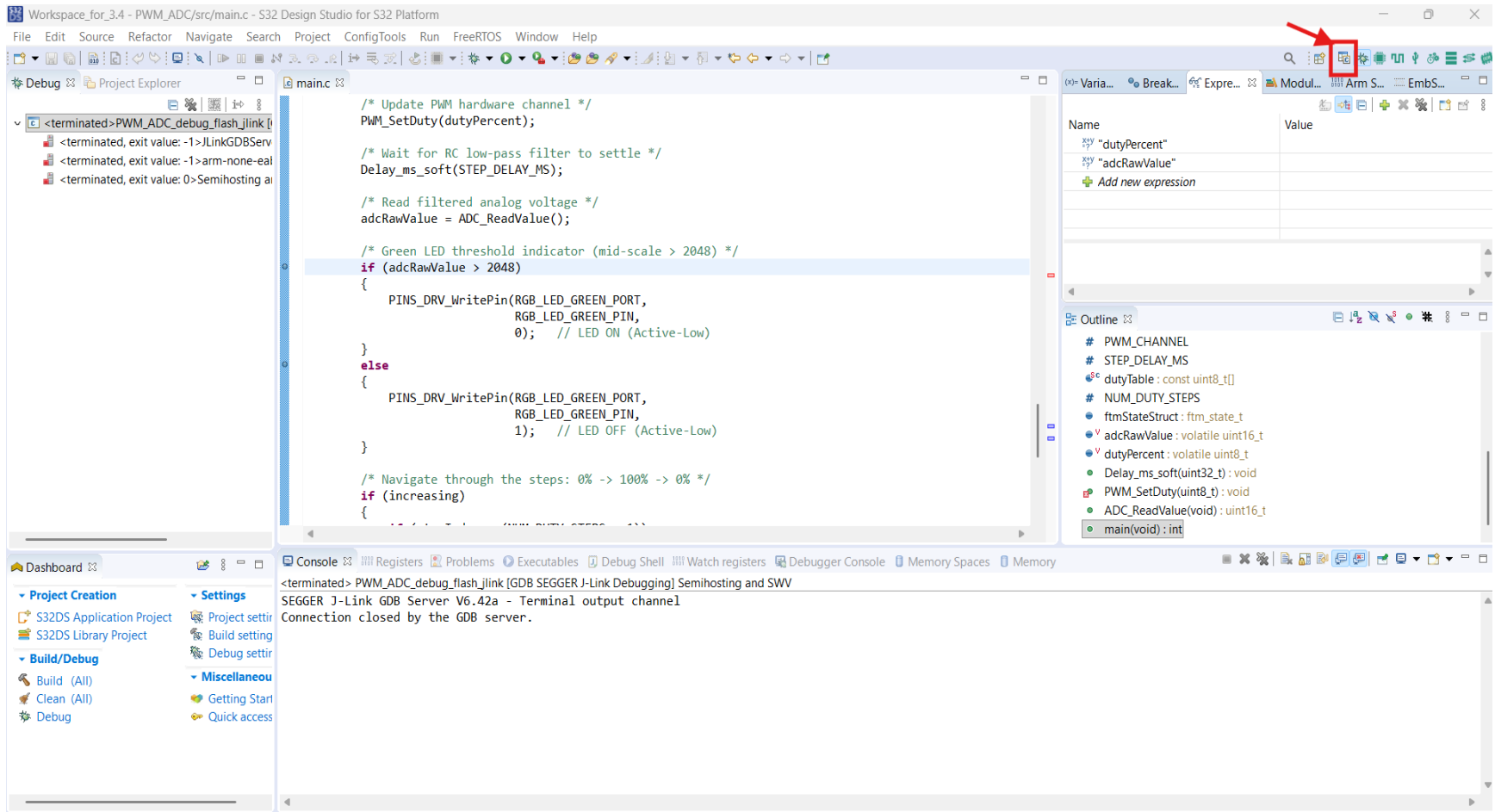
The 'Console' pane at the bottom shows the following output:

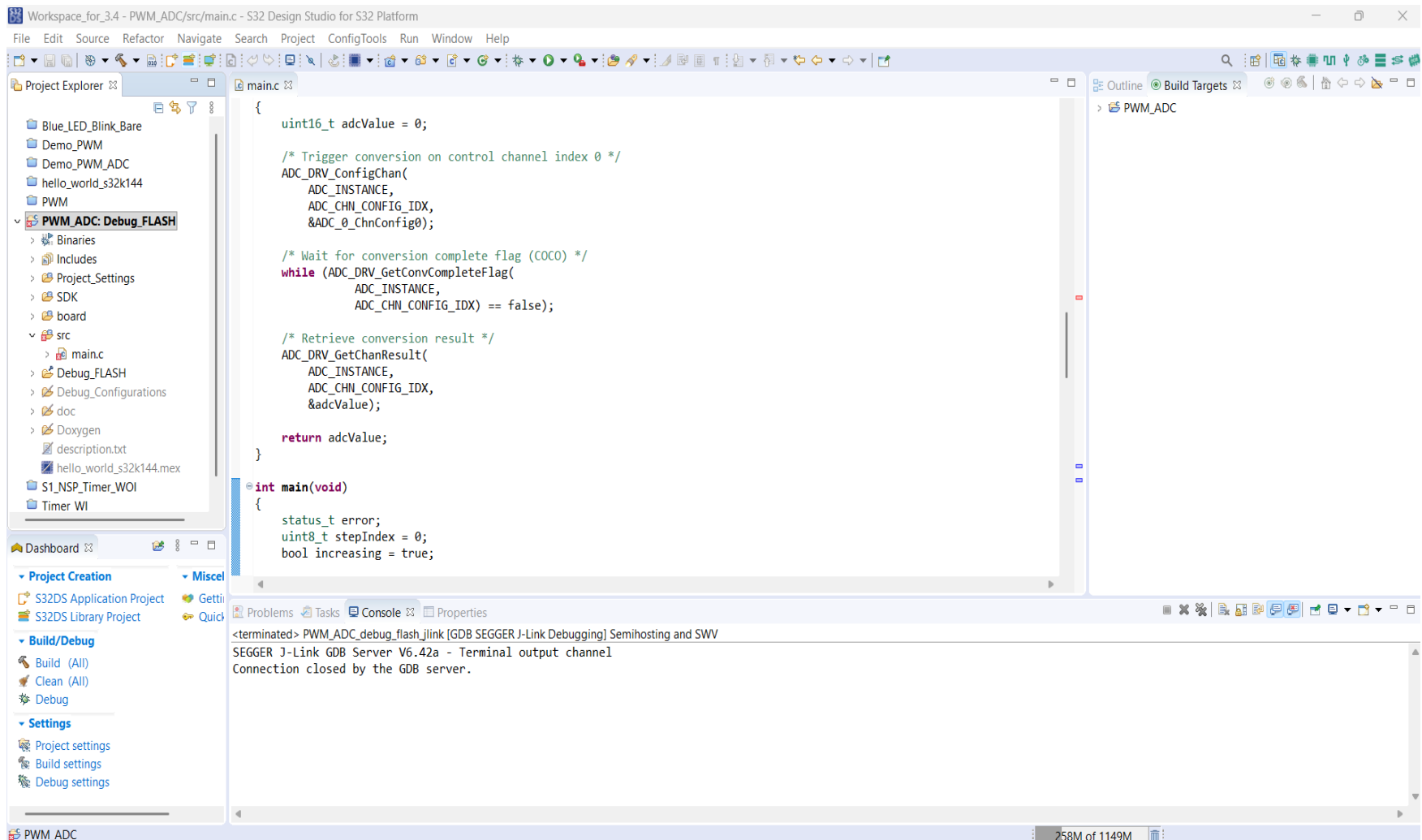
```

PWM_ADC_debug_flash_jlink [GDB SEGGER J-Link Debugging] JLinkGDBServerCL.exe
Reading all registers
Read 4 bytes @ address 0x00002EDE (Data = 0xF5B3B29B)
Setting breakpoint @ address 0x00002EDC, Size = 2, BPHandle = 0x0017
Starting target CPU...
...Breakpoint reached @ address 0x00002EDC
Reading all registers
Read 4 bytes @ address 0x00002EDC (Data = 0xB29B881B)
Removing breakpoint @ address 0x00002EBA, Size = 2
Removing breakpoint @ address 0x00002EDC, Size = 2
Read 1 bytes @ address 0x20000022 (Data = 0x5A)
Read 2 bytes @ address 0x20000020 (Data = 0x0000)

```


- Click on C/C++ to change perspective





Kind Courtesy of reference books/papers/Links:

1. S32K1xx Series Reference Manual
2. S32SDK User Manual - S32K1xx RTM 4.0.3
3. <https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-s32-platform:S32DS-S32PLATFORM>

Thank you